Exam : FCSS_NST_SE-7.6

Title : FCSS - Network Security 7.6 Support Engineer

Vendor : Fortinet

Version : V13.65

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QUESTION NO: 1
Consider the scenario where the server name indication (SNI) does not match either the common name (CN) or any of the subject alternative names (SAN) in the server certificate. Which action will FortiGate take when using the default settings for SSL certificate inspection?
A. FortiGate uses the SNI from the user's web browser.
B. FortiGate closes the connection because this represents an invalid SSL/TLS configuration.
C. FortiGate uses the first entry listed in the SAN field in the server certificate.
D. FortiGate uses the CN information from the Subject field in the server certificate.
Answer: D
Explanation:
When FortiGate performs SSL certificate inspection with default settings, it checks if the Server Name Indication (SNI) matches either the Common Name (CN) or any Subject Alternative Name (SAN) in the server certificate. If there is no match, FortiGate does not block the connection; instead, it uses the CN value from the certificate's subject field to continue web filtering and categorization.
This behavior is described in the official Fortinet 7.6.4 Administration Guide:
"Check the SNI in the hello message with the CN or SAN field in the returned server certificate: Enable: If it is mismatched, use the CN in the server certificate." This is the default (Enable) mode, which differs from the Strict mode that would block the mismatched connection.
By default, this policy ensures service continuity and prevents disruptions due to certificate mismatches, allowing FortiGate to log and inspect based on the CN even when the requested SNI does not match. It provides a balance between connection reliability and the accuracy of filtering by certificate identity, allowing security policies to remain functional without unnecessary blocks. This approach is recommended by Fortinet to maintain usability for end-users while still supporting granular inspection.
References:
FortiGate 7.6.4 Administration Guide: Certificate Inspection
SSL/SSH Inspection Profile Configuration

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QUESTION NO: 2
In which two slates is a given session categorized as ephemeral? (Choose two.)
A. A UDP session with only one packet received
B. A UOP session with packets sent and received
C. A TCP session waiting for the SYN ACK
D. A TCP session waiting for FIN ACK
Answer: A,C
Explanation:
The study guide states:
"FortiGate categorizes an entry in the session table as an ephemeral session when it is a TCP session that is not fully established (three-way handshake not completed), or when it is a UDP session with only one packet received." This directly proves:
A is correct because a UDP session with only one packet received is ephemeral.

C is correct because a TCP session waiting for the SYN/ACK is not fully established, so it is ephemeral. The study guide's TCP state table shows that the handshake is only completed when the session reaches ESTABLISHED Why the other options are wrong:
B is wrong because once UDP traffic has been seen in both directions, it is no longer the "single packet received" condition described for ephemeral sessions. The study guide says for UDP: 00 = one way, 01 = both ways D is wrong because a TCP session waiting for FIN/ACK is already in the closing stage after establishment, not in the "not fully established" stage. The study guide explains that after both sides close the session, FortiGate can keep it briefly in the table in state value 5 for out-of-order packets after FIN/ACK

QUESTION NO: 3
Refer to the exhibit.

```
# diagnose debug application fssod -1
# diagnose debug enable
[fsso_svr.c:save_result:579] event_id=4768, logon=bobby, domain=FSSO
workstation=, ip=10.124.2.90 port=49215, time=1372061722
```

Partial output of the fssod daemon real-time debug command is shown. Which two conclusions can you draw from the output? (Choose two answers)
A. FSSO cannot verify if the user is still logged in.
B. Fortinet Single Sign-On (FSSO) is using DC Agent mode to detect logon events.
C. FortiGate is frequently polling the workstation in case the user has logged out.
D. FSSO is using agentless polling mode to detect logon events.
E. FortiGate polled this event through TCP port 8000.
Answer: C,D
Explanation:
The correct answers are C and D.
The key clue is the command itself:
diagnose debug application fssod -1
The study guide explicitly states: "There is a specific FortiGate daemon that handles the polling mode. It is the fssod daemon. To enable agentless polling mode real-time debug use the command: diagnose debug application fssod -1." That directly proves D. FSSO is using agentless polling mode to detect logon events.
The study guide also states: "In agentless polling mode, FortiGate frequently polls all workstations (as a standalone collector agent does) to check which users are still logged in. You can sniffer this traffic on port 445." That directly proves C. FortiGate is frequently polling the workstation in case the user has logged out.
Why the other options are wrong:
A is wrong because the "cannot verify if the user is still logged in" / Not Verified condition is described for the collector agent workstation status, not as a conclusion from this FortiGate fssod debug line. The study guide says: "A user goes to not verified status when they log out, or when there is a problem in the polling done by the collector agent to the workstation." B is wrong because DC Agent mode is part of agent-based FSSO, where DC agents send events to a collector agent. This output is from the fssod daemon, which the study guide ties to agentless polling mode, not DC Agent mode.

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E is wrong because TCP port 8000 is used for communication between the collector agent and FortiGate, while in agentless polling mode FortiGate polls workstations and that traffic can be sniffed on TCP port 445.
So the verified answers are: C, D.

QUESTION NO: 4
Which two protocol states indicate that traffic is bidirectional? (Choose two.)
A. proto_state=01 for a TCP session.
B. proto_state=01 for a UDP session.
C. proto_state=05 for a TCP session.
D. proto_state=00 for an ICMP session.
Answer: A,B
Explanation:
The correct answers are A and B.
For UDP, the study guide states this directly: "For UDP, the session state can have only two values: 00 when traffic is only one way, and 01 when traffic is two ways. For ICMP, the protocol state is always 00." That makes B correct and D incorrect.
For TCP, the study guide explains that the protocol state is a two-digit number, where the first digit is the server-side state and the second digit is the client-side state. It also states that the first digit is 0 when the session is not subject to any inspection, and the TCP state table shows that value 1 = ESTABLISHED So, for a normal non-proxied/non-inspected TCP session, proto_state=01 means the TCP session is in the ESTABLISHED state. An established TCP session means the three-way handshake has completed, which requires traffic in both directions. That is why A is correct.
The study guide also says: "proto_state=11 means that the TCP three-way handshake for both server-side and client-side is completed (ESTABLISHED)." This confirms that TCP state value 1 represents an established state.
Why C is not selected: the study guide defines value 5 as TIME_WAIT and says: "When a session is closed by both the sender and receiver, FortiGate keeps that session in the session table for a few seconds... This is the state value 5." So proto_state=05 represents a closing/closed TCP session in TIME_WAIT, not the normal bidirectional state the question is testing.
Therefore, the verified answers are A and B.

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QUESTION NO: 5
Refer to the exhibit.

```
# diagnose sys  session stat
misc info:    session_count=1523 setup_rate=0 exp_count=0 reflect_count=0 clash=113
   memory_tension_drop=0 ephemeral=0/131072 removeable=0 extreme_low_mem=0
   npu_session_count=0
delete=0, flush=1, dev_down=0/0
session walkers: active=0, vf-42, dev-0, saddr-0, npu-0
TCP sessions:
    562 in ESTABLISHED state
    15 in SYN_RECV state
    27 in CLOSE state
```

The partial output of diagnose sys session stat command is shown.

Which statement about the output shown in the exhibit is correct?

A. 113 sessions have been dropped because of memory page exhaustion.

B. There have been 131072 recorded ephemeral sessions but there are no current ones.

C. 562 TCP sessions have their proto_state set to 01 if there is no inspection.

D. 27 sessions have expired but are still in the session table in case any out-of-order packets arrive.

Answer: C

Explanation:

The correct answer is C.

The exhibit shows:

562 in ESTABLISHED state

27 in CLOSE state

memory_tension_drop=0

ephemeral=0/131072

According to the study guide, for TCP sessions: "The protocol state in the session table is a two-digit number. For TCP, the first number (from left to right) is related to the server-side state and is 0 when the session is not subject to any inspection (flow or proxy)... The second digit is the client-side state." The same page also shows that value 1 = ESTABLISHED So, if a TCP session is in ESTABLISHED state and there is no inspection, its proto_state is 01:

first digit 0 = no inspection

second digit 1 = ESTABLISHED

That makes C correct. This is also consistent with FortiOS examples showing established TCP sessions with proto=6 proto_state=01 Why the other options are wrong:

A is wrong because the field that indicates sessions dropped due to low free memory is memory_tension_drop, and in the exhibit it is 0, not 113. The study guide states: "If there is a lack of free memory, the kernel deletes the oldest sessions. The command shown on this slide displays the number of sessions the kernel deleted because of this mechanism." So 113 is the clash value, not memory-tension drops.

B is wrong because ephemeral=0/131072 does not mean 131072 ephemeral sessions were recorded. The study guide explains that FortiGate "sets a hard limit on the maximum number of ephemeral sessions that can exist at the same time in the session table." Therefore:

0 = current ephemeral sessions

131072 = maximum allowed ephemeral sessions for that model/context

D is wrong because the study guide says the temporary retention for possible out-of-order packets happens in state value 5 (TIME_WAIT): "When a session is closed by both the sender and receiver, FortiGate keeps that session in the session table for a few seconds, to allow for any out-of-order packets that might arrive after the FIN/ACK packet. This is the state value 5." But the exhibit shows 27 in CLOSE state, and the same table shows CLOSE = 6, not TIME_WAIT So the verified answer is C.

QUESTION NO: 6

Refer to the exhibit.

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Partial output of FortiOS kernel slabs

```
---omitted---
ip6_session            5      0   1472    5    2 : tunables   24   12    8 : slabdata   1   1
tcp_session            3      5   1500    5    2 : tunables   24   12    8 : slabdata   1   1
ip_session            10     10   1408    5    2 : tunables   24   12    8 : slabdata   2   2
ext4_groupinfo_4k     32     56    144   28    1 : tunables  120   60    8 : slabdata   2   2
RAWv6                  7      7   1152    7    2 : tunables   24   12    8 : slabdata   1   1
UDPv6                 15     10   1408    5    2 : tunables   24   12    8 : slabdata   4   4
tw_sock_TCPv6          0      0    248   16    1 : tunables  120   60    8 : slabdata   0   0
request_sock_TCPv6     0      0    312   13    1 : tunables   54   27    8 : slabdata   0   0
TCPv6                 47     48   2304    3    2 : tunables   24   12    8 : slabdata  16  16
---omitted---
```

The partial output of FortiOS kernel slabs is shown. Which statement about total slab size is true?

A. The total slab size of the ip_session slab is 14080 kB and is associated with the user space.

B. The total slab size of the tcp_session slab is 7500 kB and is associated with the kernel.

C. The total slab size of the ip6_session slab is 1472 kB and is associated with the kernel.

D. The total slab size of the UDPv6 slab is 14080 kB and is associated with the user space.

Answer: B

Explanation:

The correct answer is B.

The study guide explicitly states that slabs are used by the kernel: "The kernel memory slabs are collections of objects with a common purpose. The kernel uses them to store information in memory." It also gives the exact calculation method: "Total slab size = available objects x object size" and explains that in the diagnose hardware sysinfo slab output, the columns are active objects, available objects, and object size From the exhibit:

tcp_session 3 5 1500 ...

available objects = 5

object size = 1500

So:

Total slab size = 5 × 1500 = 7500

That matches option B.

Why the other options are wrong:

A: ip_session 10 10 1408 ... gives 10 × 1408 = 14080, but slabs are associated with the kernel, not user space C: ip6_session 5 0 1472 ... gives 0 × 1472 = 0, not 1472 D: UDPv6 15 10 1408 ... gives 10 × 1408 = 14080, but again slabs are associated with the kernel, not user space So the verified answer is B.

QUESTION NO: 7

Refer to the exhibit.

```
---omitted---
[ 7608] wksta_check: failed to connect to workstation:10.0.155.23 (10.0.155.23)
[ 7608] wksta_check_wmi: cannot get UserName, error code:0x0
---omitted---
```

The output from a collector agent log is shown. The collector agent is showing the status of a

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workstation as Not Verified. What are two common causes for this message? (Choose two.)

A. The workstation has come out of hibernate mode.

B. The workstation remote registry service is not running.

C. Traffic to ports 139 and 445 is blocked.

D. DNS cannot resolve the workstation name.

Answer: B,C

Explanation:

The correct answers are B and C.

The study guide has a section titled "Not Verified Status on the Collector Agent" and states: "The collector agent cannot verify if the user is still logged in" and lists these common causes:

"A firewall is blocking traffic to port 139 and 445"

"The workstation remote registry service is not running"

The guide also explains the verification method:

"For WMI polling mode, the collector agent checks the WMI service. For all the other modes, the collector agent checks the HKEY_USERS hive through remote registry services." If the workstation does not respond to these checks, the status can become not verified An additional requirements slide in the same study guide confirms:

"TCP ports 139 and 445 must be open between the collector agent and all workstations"

"Remote registry service must be up and running on each workstation"

Why the other options are wrong:

A is wrong because the study guide mentions a workstation coming out of hibernate mode under a different problem: "No Internet After IP Address Change", not as a common cause of Not Verified status D is wrong because DNS resolution issues are also discussed under the IP address change scenario, where the collector agent uses DNS to resolve the workstation name after an IP change. That is separate from the Not Verified causes listed for this log message So the verified answers are: B, C.

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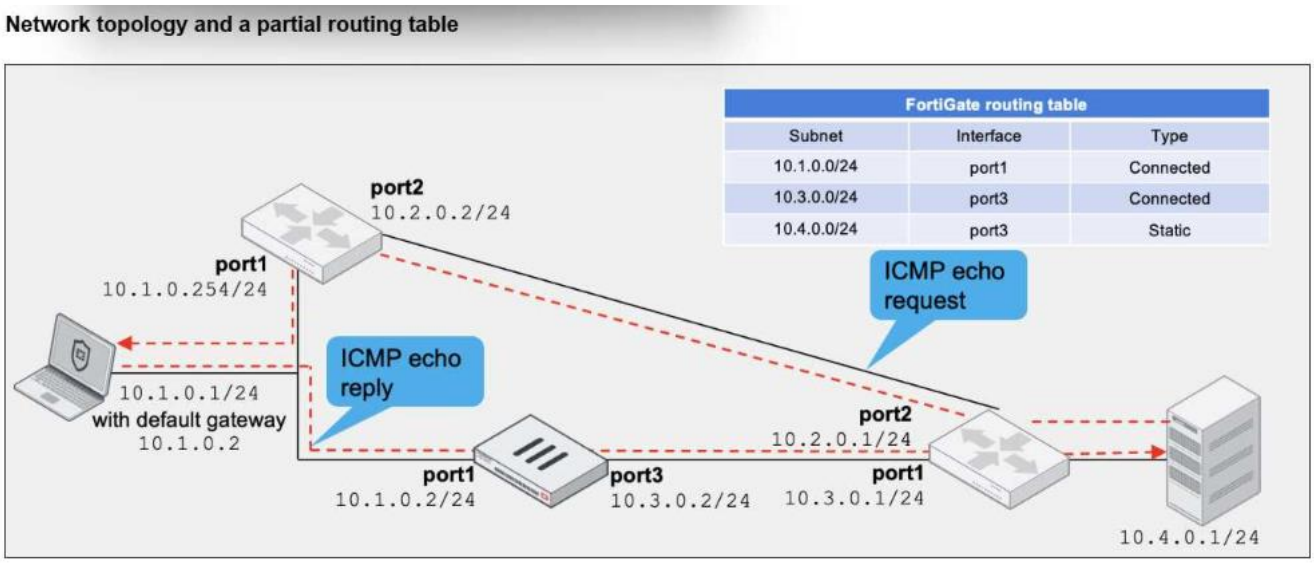
QUESTION NO: 8

Refer to the exhibit.



A network topology and a partial routing table are shown.

FortiGate has already been configured with a firewall policy that allows all ICMP traffic to flow from port1 to port3.
Which two changes can the administrator perform to ensure the server at 10.4.0.1/24 receives the ICMP echo reply from the laptop at 10.1.0.1/24? (Choose two.)

A. Enable asymmetric routing under config system settings.

B. Change the FortiGate configuration from strict RPF check mode to feasible RPF check mode.

C. Modify the default gateway on the laptop from 10.1.0.2 to 10.1.0.254.

D. Add a default static route on FortiGate to forward all traffic to port3.

Answer: A,C

Explanation:
The correct answers are A and C.
The study guide describes this exact asymmetric ICMP scenario. It states:
"The server sends an echo request to the PC through port2 of the local router, effectively bypassing FortiGate. When it receives the echo request, the PC responds with an echo reply through its default gateway, 10.1.0.2, which is port1 on FortiGate. Because there is no existing session, the echo reply is dropped. All subsequent echo replies are blocked." That means the current problem exists because:
the ICMP request bypasses FortiGate
the ICMP reply goes through FortiGate
FortiGate has no matching session, so it drops the reply
The study guide then shows the exact corrective option:
"Allowing asymmetric routing:"
```
config system settings
set asymroute enable
end
```
It further explains:
"After the packet passes through the FortiGate CPU, FortiGate forwards the packet using the FIB, even though there are no session matches. FortiGate forwards all subsequent echo replies using the FIB." So A is correct.
The other valid fix is to make the traffic symmetric by changing the laptop's default gateway so the reply no longer goes through FortiGate. In the exhibit, the alternate gateway is 10.1.0.254, which is the local router on the same subnet. If the laptop uses 10.1.0.254 instead of 10.1.0.2, the ICMP echo reply follows the same bypass path as the echo request, so the server receives it without involving FortiGate session validation. This makes C correct.
Why the other options are wrong:
B is wrong because this is not an RPF problem. The study guide explains RPF as a reverse path lookup used to validate whether a packet arrived on a legitimate interface, mainly for spoofing protection. The issue in this scenario is a missing session due to asymmetric routing, not a strict-versus-feasible RPF failure D is wrong because FortiGate already has the specific route 10.4.0.0/24 through port3 in the routing table shown in the exhibit, so adding a default static route to port3 is unnecessary and not the reason the echo reply is being dropped So the verified answers are: A, C.

QUESTION NO: 9

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Refer to the exhibit.

Debug output

```
# diagnose vpn tunnel list
name=VPN_0 ver=2 serial=644 94.186.234.212:4500->78.34.86.60:4500 tun_id=172.31.0.236 tun_id6=::1
0.0.6.65 dst_mtu=1500 dpd-link=on weight=1
bound_if=32 lgwy=static/1 tun=intf mode=dial_inst/3 encap=none/66488 options[103b8]=npu create_dev rgwy-
chg rport-chg frag-rfc  role=sync-primary accept_traffic=1 overlay_id=0

parent=VPN index=0
proxyid_num=1 child_num=0 refcnt=5 ilast=0 olast=0 ad=/0
stat: rxp=2144005 txp=1132253 rxb=158609260 txb=1091224036
dpd: mode=on-idle on=1 idle=5000ms retry=2 count=0 seqno=1
natt: mode=silent draft=0 interval=10 remote_port=4500
fec: egress=0 ingress=0
proxyid=VPN proto=0 sa=1 ref=17 serial=1
  src: 0:0.0.0.0-255.255.255.255:0
  dst: 0:0.0.0.0-255.255.255.255:0
  SA:  ref=4 options=227 type=00 soft=0 mtu=1406 expire=419/0B replaywin=2048
       seqno=1ed4 esn=0 replaywin_lastseq=00003e00 qat=0 rekey=0 hash_search_len=1
  life: type=01 bytes=0/0 timeout=1786/1800
  dec: spi=6e0f6228 esp=aes key=32 e170300e48adf71905f0b1eed6b935c077f71ea97c51e2a5ae400a5014726293
       ah=sha512 key=64 c958832024cb1e8609385b0aa022cc9c0381f73e23d5ae92a8409d999fa5c4fb65d8bf9f3d516e5b
3cf230f90b4ad238ee463bd44da484790bbfd508ba13a80b
  enc: spi=104b1b89 esp=aes key=32 4fafb61b0afb1afa8b18b4985d7e97bc4657befa8dc494c13244639581620a91
       ah=sha512 key=64 a568609290a97b682149048d00e10c8cd628f74ec8d0dcf4d80960bbd6e9047e9f5a2af75a5c58ba
60bde32666239c708eb4c04aff215d6e1207fb811d6642b1
  dec:pkts/bytes=15872/2112552, enc:pkts/bytes=15776/3136559
  npu_flag=02 npu_rgwy=78.34.86.60 npu_lgwy=94.186.234.212 npu_selid=643 dec_npuid=1 enc_npuid=0
```

The output of the command diagnose vpn tunnel list is shown.
Reviewing the debug command, what is the current status of the traffic flowing through the tunnel?

A. The outbound IPsec SA was copied to the NPU.

B. NP6 is handling the offloading.

C. The inbound and outbound IPsec SAs were copied to the NPU.

D. The inbound IPsec SA was copied to the NPU.

Answer: D

Explanation:
The correct answer is D. The inbound IPsec SA was copied to the NPU.
The exhibit shows:
npu_flag=02
dec_npuid=1
enc_npuid=0
The study guide gives the exact meaning of the npu_flag field:
npu_flag=00 = Both IPsec SAs loaded to the kernel
npu_flag=01 = Outbound IPsec SA copied to NPU
npu_flag=02 = Inbound IPsec SA copied to NPU

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npu_flag=03 = Both outbound and inbound IPsec SAs copied to NPU
It also explains: "If the first IPsec packet is inbound and can be offloaded, the inbound SA is copied to the NPU and the npu_flag changes to 02. After both SAs are copied to the NPU, the npu_flag changes to 03." So with npu_flag=02, only the inbound SA has been copied to the NPU. That makes D correct.
Why the other options are wrong:
A is wrong because outbound-only offload would be npu_flag=01, not 02
C is wrong because both directions offloaded would be npu_flag=03, not 02 B is wrong because dec_npuid=1 identifies the NPU ID used for decryption, but it does not state that the processor is specifically NP6. The study guide only maps the offload state through the npu_flag values in this context, not the NPU model from this field alone So the verified answer is: D.

QUESTION NO: 10
Refer to the exhibit.

Output of a session

```
session info: proto=6 proto_state=01 duration=157 expire=3559 timeout=3600 flags=00000000 socktype=0 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=0 policy_dir=0 tunnel=/ vlan_cos=0/255
user=User1 state=log may_dirty authed f00 acct-ext
statistic(bytes/packets/allow_err): org=2137/14/1 reply=1663/12/1 tuples=2
tx speed(Bps/kbps): 1/0 rx speed(Bps/kbps): 1/0
orgin->sink: org pre->post, reply pre->post dev=5->3/3->5 gwy=10.1.0.254/10.1.10.1
hook=pre dir=org act=noop 10.1.10.1:34830->35.241.9.150:443(0.0.0.0:0)
hook=post dir=reply act=noop 35.241.9.150:443->10.1.10.1:34830(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 pol_uuid_idx=14735 auth_info=2 chk_client_info=0 vd=0
serial=0000352e tos=ff/ff app_list=0 app=0 url_cat=0
rpdb_link_id=00000000 ngfwid=n/anpu_state=0x000100
no_ofld_reason:  npu-flag-off
```

The exhibit shows the output of a session. Which two statements are correct? (Choose two.)

A. The session did not match a firewall policy.

B. The gateway to the destination is 10.1.10.1.

C. The session was initiated from an authenticated user.

D. The TCP session has been successfully established.

Answer: C,D
Explanation:
The correct answers are C and D.
For D, the session output shows proto=6, which means TCP, and proto_state=01. The study guide explains that for TCP sessions, "the first number (from left to right) is related to the server-side state and is 0 when the session is not subject to any inspection (flow or proxy). ... The second digit is the client-side state." It also shows that value 1 = ESTABLISHED So proto_state=01 means:
first digit 0 = no inspection
second digit 1 = ESTABLISHED
That confirms the TCP session has been successfully established.
For C, the study guide section "Session for an Authenticated User" states: "Any session for traffic coming from an authenticated user contains the authed flag. Additionally, the username is added to the session information." In the exhibit, the session contains user=User1 and an authentication state flag (authd), which indicates the session is associated with an authenticated user.

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Why the other options are wrong:
A is wrong because the session did match a firewall policy. The study guide says the session output includes "The ID of the matching policy" In the exhibit, the session shows policy_id=1, so a firewall policy was matched.
B is wrong because the study guide explains the gwy field as:
first value = gateway to destination
second value = gateway to source
The exhibit shows gwy=10.1.0.254/10.1.10.1, so:
gateway to destination = 10.1.0.254
gateway to source = 10.1.10.1
So the verified answers are: C, D.

QUESTION NO: 11
Refer to the exhibit.

VPN tunnel details

```
Hub # get vpn ipsec tunnel details
gateway
    name: 'Hub2Spoke1'
    type: route-based
    local-gatewav: 10.10.1.1:0 (static)
    remote-gateway: 10.10.2.2:0 (static)
    mode: ike-vl
    interface: 'wan2' (6)
    rx packets: 1025 bytes: 524402 errors: 0
    tx packets: 641 bytes: 93 errors: 0
    dpd: on-demand/negotiated idle: 20000ms retry: 3 count: 0
    selectors
      name: 'Hub2Spoke1'
      auto-negotiate: disable
      mode: tunnel
      src: 0:192.168.1.0/0.0.0.0:0
      dst:0:10.10.20.0/0.0.0.0:0
      SA
        lifetime/rekey: 43200/32137
        mtu: 1438
        tx-esp-seq: 2ce
        replay: enabled
        inbound
          spi: 01e54b14
          enc: aes-cb 914dc5d092667ed436ea7f6efb867976
          auth:  shal a81b019d4cdfda32ce51e6b01d0b1ea42a74adce
        outbound
          spi: 3dd3545f
          enc:  aes-cb 017b8ff6c4ba21eac99b22380b7de74d
          auth: shal   edd8141f4956140eef703d9042621d3dbf5cd961
        NPU acceleration: encryption (outbound) decryption (inbound)
```

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Partial output of the get vpn ipsec tunnel details command is shown. Based on the output, which two statements are correct? (Choose two.)

A. The npu_flag for this tunnel is 02.

B. Different SPI values are a result of auto-negotiation being disabled for phase2 selectors.
C. The npu_flag for this tunnel is 03.
D. Anti-replay is enabled.
Answer: C,D
Explanation:
The correct answers are C and D.
The study guide's get vpn ipsec tunnel details example shows:
replay: enabled
inbound and outbound sections with separate SPIs
NPU acceleration: encryption(outbound) decryption(inbound)
and it labels these as "Phase 2 SAs for each direction" and "Hardware acceleration" This directly proves D. Anti-replay is enabled, because the output explicitly says replay: enabled
For the NPU status, the study guide explains the exact npu_flag meanings:
npu_flag=00 = both IPsec SAs loaded to the kernel
npu_flag=01 = outbound IPsec SA copied to NPU
npu_flag=02 = inbound IPsec SA copied to NPU
npu_flag=03 = both outbound and inbound IPsec SAs copied to NPU
Because the exhibit shows hardware acceleration in both directions - encryption(outbound) and decryption(inbound) - the matching npu_flag is 03, not 02. That makes C correct and A incorrect.
Why B is wrong:
The same study guide output labels the tunnel as having Phase 2 SAs for each direction, so different inbound and outbound SPIs are normal for the two SAs. Also, the FortiOS administration guide explains that auto-negotiate controls whether phase 2 SA negotiation is initiated automatically, not whether inbound and outbound SPIs are different: "By default the phase 2 security association (SA) is not negotiated until a peer attempts to send data... Auto-negotiate initiates the phase 2 SA negotiation automatically..." So the verified answers are: C, D.

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QUESTION NO: 12
What can cause an IKEv2 tunnel to go down after it was initially brought up successfully?
A. Mismatched traffic selectors (phase 2 / "quick-mode selectors") were detected during the CREATE_CHILD_SA exchange.
B. A mismatched proposal was detected during the IKE_AUTH exchange.
C. A mismatched pre-shared key was detected during the IKE_AUTH exchange.
D. A mismatched Diffie-Hellman group was detected during the IKE_SA_INIT exchange.
Answer: A
Explanation:
The correct answer is A.
The study guide explains the IKEv2 exchange order very clearly:
"The initial exchanges are: IKE_SA_INIT and IKE_AUTH."
"Create_Child_SA exchange: Creates a new child SA or rekeys an existing child SA." It also states:
"After successful IKE_SA_INIT and IKE_AUTH exchanges, the CHILD_SA exchange takes place. In this exchange, the peers negotiate the CHILD_SA and the traffic selectors - traffic

selector responder (TSr) and traffic selector initiator (TSi)." That is why A is correct: if the tunnel was initially brought up successfully, then the initial exchanges already succeeded. A later problem during CREATE_CHILD_SA, especially with traffic selectors/phase 2 selectors, can cause the tunnel to fail during rekey or child-SA renegotiation.
Why the other options are wrong:
B is wrong because proposal mismatch for the IKE SA is handled during IKE_SA_INIT, not after the tunnel is already up. The study guide says IKE_SA_INIT negotiates the security settings to protect the IKE traffic C is wrong because a pre-shared key mismatch is part of authentication and would prevent successful initial establishment during IKE_AUTH. The study guide shows that after IKE_AUTH, "authentication succeeded" and "established IKE SA" when it works D is wrong because a Diffie-Hellman mismatch belongs to IKE_SA_INIT, which happens before the tunnel comes up. The study guide also states: "By IKEv2 design, no Diffie-Hellman public key is exchanged during an IKE_AUTH exchange." So the verified answer is: A.

QUESTION NO: 13
Refer to the exhibit, which shows the output of get router info bgp summary.

```
get router info bgp summary

VRF 0 BGP router identifier 172.16.1.254, local AS number 65100
BGP table version is 3
2 BGP AS-PATH entries
0 BGP community entries

Neighbor        V         AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.64.1.254    4        100      18      20        3    0    0 00:02:55        1
100.64.2.254    4        100      0        0        0    0    0 never        Active

Total number of neighbors 2
```

Which two statements are true? (Choose two.)
A. The local ForliGate has received one prefix from BGP neighbor 100.64.1.254.
B. The TCP connection with BGP neighbor 100.64.2.254 was successful.
C. The local FortiGate has received 18 packets from a BGP neighbor.
D. The local FortiGate is still calculating the prefixes received from BGP neighbor 100.64.2.264
Answer: A,C
Explanation:
The get router info bgp summary output lists BGP neighbor status:
Prefix Reception: The "State/PfxRcd" column shows the number of prefixes received from the neighbor-neighbor 100.64.1.254 has "1", confirming option A.
Received Message Count: Under "MsgRcvd", 18 packets have been received from neighbor 100.64.1.254. This matches option C.
The second neighbor 100.64.2.254 is in "Active" state and has received/sent 0 packets, indicating that its TCP connection is NOT established, disproving option B.
There is no indication anywhere that the router is "still calculating" prefixes; "Active" just means no session is established, so option D is incorrect.
References:
FortiOS BGP Command Reference: BGP Neighbor States, PfxRcd, and Counters

QUESTION NO: 14
Refer to the exhibit.

```
# get router info bgp summary

VRF 0 BGP router identifier 172.16.23.58, local AS number 65100
BGP table version is 2
1 BGP AS-PATH entries
0 BGP community entries

Neighbor        V         AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.16.23.153   4        100     218    3220        0    0    0 never     Connect

Total number of neighbors 1
```

The output of a BGP debug command is shown.
Why has the local router at 172.16.23.58 been unable to establish adjacency with its only neighbor?
A. The neighbor router has become unreachable, which is evident by the low ratio of messages received to messages sent.
B. The local router has not received an OPEN message from the neighbor.
C. The local router has not received a SYN/ACK packet from the neighbor.
D. There is no active route to the BGP neighbor.
Answer: C
Explanation:
The correct answer is C.
The exhibit shows the neighbor state as Connect in the State/PfxRcd column. The study guide explains the BGP states exactly as follows:
"Connect: Waiting for a successful three-way TCP connection"
"OpenSent: Waiting for an OPEN message from the peer"
"Established: Peers have successfully exchanged OPEN and keepalive messages" Because the router is still in Connect state, the TCP three-way handshake has not completed yet. In practical terms, the local router has sent the TCP SYN but has not successfully received the SYN/ACK needed to complete the handshake. That is why C is correct.
Why the other options are wrong:
A is wrong because the message counters alone do not prove that the neighbor is unreachable. The study guide says the State/PfxRcd field shows the BGP state when the session is not established, and here that state is specifically Connect B is wrong because waiting for an OPEN message happens in OpenSent, not Connect D is not the best answer for this output. The study guide ties the displayed state directly to the protocol phase: Connect means the device is still waiting for a successful TCP handshake So the verified answer is: C.

QUESTION NO: 15
During the SAML negotiation process, in which section does the Identity Provider (IdP) provide the SAML attributes used in the authentication process to the Service Provider (SP)?
A. Bindings HTTP post
B. Assertion dump

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C. Authentication request

D. Authentication response

Answer: B

Explanation:

The correct answer is B. Assertion dump.

The study guide states: "SAML attributes are pieces of information about a user that are exchanged between IdPs and SPs during the SAML authentication process. These attributes are included in the SAML assertion, which is built by the IdP as part of the authentication process." The same study guide page for real-time SAML troubleshooting shows the section labeled **** Assertion Dump ****, and inside that assertion it displays the actual user attributes, such as:

<saml:Attribute Name="username">

<saml:Attribute Name="groups">

It also explicitly marks this part as "Attributes sent by IdP"

Why the other options are wrong:

A . Bindings HTTP post is incorrect because bindings define how SAML messages are transported, not the section that contains the attributes. The study guide says: "Bindings: Define how SAML protocol messages are transmitted over different communication channels." C . Authentication request is incorrect because that is built by the SP and sent toward the IdP, not where the IdP's user attributes are shown. The study guide's flow says the SP "Builds auth request" and the IdP later "Builds auth response." D . Authentication response is broader than the exact section being asked. The exact section in the study guide where the IdP-provided attributes are shown is the Assertion dump.

So the verified answer is: B.

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QUESTION NO: 16

What are two functions of automation stitches? (Choose two.)

A. You can configure automation stitches to run diagnostic commands and attach the results to an email message when CPU or memory usage exceeds specified thresholds.

B. You can configure automation stitches to modify packet headers and payloads if specific traffic triggers an anomaly IPS event.

C. You can configure automation stitches to insert a delay between actions if the automation stitches are set to execute actions in parallel.

D. You can configure automation stitches to take parameters from previous actions as input for the next action if the automation stitches are set to execute actions in sequence.

Answer: A,D

Explanation:

The correct answers are A and D.

The Network Security Support Engineer 7.6 Study Guide explains that automation stitches consist of a trigger and one or more actions, and that they can detect events such as high CPU and conserve mode across the Security Fabric The FortiOS administration guide then gives the exact practical example for A:

"Automation stitches can be created to run a CLI script and send an email message when memory or CPU usage exceeds specified thresholds." It also shows examples where the email body contains the script output using:

%%results%%
That directly confirms A.
For D, the FortiOS administration guide states:
"The URI and HTTP body can use parameters from logs or previous action results." It also explains the execution model:
"The stitch Action execution can be set to either Sequential or Parallel. In sequential execution actions will execute one after another with a delay (if specified). ... In parallel execution all actions will execute immediately when the stitch is triggered." A concrete example shows a second action using the output of the first action:
"This string for the body text includes the results from the preceding CLI script action." with Body=%%results%%...
That confirms D.
Why the other options are wrong:
B is wrong because automation stitches can be triggered by IPS events, but the documentation does not describe them as modifying packet headers or payloads. Instead, they perform actions such as email, CLI script, webhook, quarantine, and similar automated responses C is wrong because delay is associated with sequential execution, not parallel execution. The guide explicitly says: "A delay can be added before an action if Sequential action execution is used." So the verified answers are: A, D.

QUESTION NO: 17
Refer to the exhibit.

```
config system interface
    edit "port1"
        set preserve-session-route enable
    next
end

# diagnose sys session list
session info: proto=1 proto_state=00 duration=4 expire=55 timeout=0 refresh_dir=both flags=00000000 socktype=0 sockport=0 av_idx=0 use=3
state=log may_dirty npu f00 route_preserve
orgin->sink: org pre->post, reply pre->post dev=7->19/19->7 gwy=100.64.1.1/10.0.1.101

# diagnose netlink interface list | grep index=19
if=port1 family=00 type=768 index=19 mtu=1420 link=0 master=0
```

The port1 interface configuration on FortiGate and partial session information for ICMP traffic are shown.
Which two things happen to the session information if a routing change occurs that affects this session? (Choose two answers)
A. This session will be unaffected by routing changes. The routing changes will apply only to new sessions.
B. The session will be flagged as dirty but no route lookups will be performed.
C. The session information will not change unless the current route has been removed from the routing table.
D. The session information will not change even when the active route has been removed from the routing table.
Answer: A,C
Explanation:
The correct answers are A and C.
The exhibit shows that preserve-session-route is enabled on port1:
config system interface
edit "port1"

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set preserve-session-route enable
next
end
The study guide explains the effect of this setting exactly:
"enable: FortiGate marks existing session routing information as persistent, and applies only the modified routes to new sessions" It also states:
"The current route must still be present in the FIB. Otherwise, FortiGate flags the session as dirty and reevaluates it" And the same page further clarifies:
"If you enable this setting, sessions passing through that interface continue to pass without being affected by the routing changes. The routing changes apply only to new sessions. If the route is removed from the FIB, then FortiGate must flag the session as dirty, flush its gateway information, and reevaluate the session." This proves:
A is correct because with preserve-session-route enable, existing sessions are normally preserved and routing changes apply only to new sessions.
C is correct because the session remains unchanged unless the current route is removed from the FIB/routing table, in which case FortiGate dirties and reevaluates the session.
Why the other options are wrong:
B is wrong because when the active route is removed, FortiGate does not simply mark the session dirty and stop there. The study guide says it "flags the session as dirty and reevaluates it", which means route lookup happens again.
D is wrong because the session does change if the active route is removed. FortiGate flushes gateway information and reevaluates the session.
So the verified answers are: A, C.

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QUESTION NO: 18
Refer to the exhibit.

Diagnose output

```
-=- Server List (Mon May  6 03:47:52 2024) -=-
IP                                      Weight    RTT Flags   TZ   FortiGuard-requests  Curr Lost Total Lost            Updated Time
64.26.151.37                                10     45         -5                262432          0        846 Mon May  6 03:47:43 2024
64.26.151.35                                10     46         -5                329072          0       6806 Mon May  6 03:47:43 2024
66.117.56.37                                10     75         -5                 71638          0        275 Mon May  6 03:47:43 2024
65.210.95.240                               20     71         -8                 36875          0         92 Mon May  6 03:47:43 2024
209.22.147.36                               20    103 DI      -8                 34784          0       1070 Mon May  6 03:47:43 2024
208.91.112.194                              20    107 D       -8                 35170          0       1533 Mon May  6 03:47:43 2024
96.45.33.65                                 60    144          0                 33728          0        120 Mon May  6 03:47:43 2024
80.85.69.41                                 71    226          1                 33797          0        192 Mon May  6 03:47:43 2024
62.209.40.74                               150     97          9                 33754          0        145 Mon May  6 03:47:43 2024
121.111.236.179                             45     44  F       -5                 26410      26226      26227 Mon May  6 03:47:43 2024
```

Partial output of command diagnose debug rating is shown. Which FDS server will the FortiGate algorithm choose?

A. 96.45.33.65

B. 208.91.112.194

C. 64.26.151.37

D. 209.22.147.36

Answer: C

Explanation:
The correct answer is C. 64.26.151.37.
The study guide explains the FortiGuard flags shown by diagnose debug rating:
D = Default
I = Initial
T = Timing

F = Failed
and specifically: "F = The server is down"
So even though 121.111.236.179 has the lowest RTT in the exhibit, it has the F flag, meaning FortiGate considers that server failed/down, so it will not be chosen.
To determine which active server is selected, the FortiOS administration guide states:
"The server list is sorted first by weight. The server with the smallest RTT appears at the top of the list regardless of weight. ... Therefore the top position in the list is selected based on RTT while the other positions are based on weight." Among the valid, non-failed choices in the exhibit:
64.26.151.37 → RTT 45
209.22.147.36 → RTT 103
96.45.33.65 → RTT 144
208.91.112.194 → RTT 107
The active server with the lowest RTT is 64.26.151.37, so that is the server FortiGate will choose.
So the verified answer is: C.

QUESTION NO: 19
Refer to the exhibit.

```
id=65308 trace_id=21 func=print_pkt_detail line=5998 msg="vd-root:0 received a packet(proto=6, 10.0.11.50:51426->100.65.1.111:211)
tun_id=0.0.0.0 from port4. flag [S], seq 449078039, ack 0, win 64240"
id=65308 trace_id=21 func=init_ip_session_common line=6198 msg="allocate a new session-0000040d"
id=65308 trace_id=21 func=__vf_ip_route_input_rcu line=2116 msg="find a route: flag=00000000 gw-100.65.0.254 via port2"
id=65308 trace_id=21 func=__iprope_tree_check line=535 msg="gnum-100004, use addr/intf hash, len=2"
id=65308 trace_id=21 func=get_new_addr line=1303 msg="find SNAT: IP-100.65.0.101(from IPPOOL), port-51426"
id=65308 trace_id=21 func=fw_forward_handler line=1007 msg="Allowed by Policy-1: SNAT"
id=65308 trace_id=21 func=ip_session_confirm_final line=3203 msg="npu_state=0x1100, hook=4"
id=65308 trace_id=21 func=ids_receive line=466 msg="send to ips"
id=65308 trace_id=21 func=__ip_session_run_tuple line=3525 msg="SNAT 10.0.11.50->100.65.0.101:51426"
id=65308 trace_id=21 func=__ip_session_run_tuple line=3579 msg="run helper-ftp(dir=original)"
```

An administrator has configured a firewall policy to use proxy-based inspection mode. What could explain the messages observed in the debug flow output?

A. At least one protocol port under Protocol Options has been mapped to Any.

B. SSL deep inspection is not configured.

C. The FTP protocol has not yet been mapped to port 211 under Protocol Options.

D. FortiGate does not have enough free memory to perform proxy-based inspections.

Answer: A
Explanation:
The correct answer is A.
The debug flow shows:
traffic is going to TCP port 211
FortiGate logs run helper-ftp(dir=original)
The study guide explains exactly what that message means:
"In this example, the run helper-ftp message indicates that the FTP session helper is being used." Under normal proxy-based inspection, protocol handling is controlled by Protocol Options. The FortiOS administration guide states:
"Protocol port mapping only works with proxy-based inspection." and "The ports can be modified to inspect any port with flowing traffic." So if the policy is configured for proxy-based inspection but the debug still shows the FTP session helper on port 211, the most likely explanation is that the FTP protocol mapping in Protocol Options is broad enough to match unexpectedly, such as being mapped to Any. That would cause FortiGate to identify the traffic as FTP and invoke the helper.

Why the other options are wrong:
B is wrong because SSL deep inspection is unrelated to this debug. The traffic shown is plain TCP/211, and the key message is about the FTP helper, not SSL decryption.
C is wrong because if FTP had not been mapped to port 211, FortiGate would be less likely to treat this traffic as FTP. The observed run helper-ftp indicates FTP handling is being triggered.
D is wrong because low-memory conserve behavior would typically cause inspection bypass or blocking behavior, not specifically the run helper-ftp message. The study guide's helper example ties this message to session-helper use, not memory shortage.
So the verified answer is: A.

QUESTION NO: 20
What can cause an IKEv2 tunnel to go down after it was initially brought up successfully?
A. A mismatched proposal was detected during the IKE_AUTH exchange.
B. A mismatched Diffie-Hellman group was detected during the IKE_SA_INIT exchange.
C. A mismatched pre-shared key was detected during the IKE_AUTH exchange.
D. Mismatched quick-mode selectors were detected during the CREATE_CHILD_SA exchange.
Answer: D
Explanation:
The correct answer is D.
The study guide explains that IKEv2 has two initial exchanges:
IKE_SA_INIT
IKE_AUTH
and then later exchanges such as:
CREATE_CHILD_SA
It also states the roles of those exchanges:
IKE_SA_INIT negotiates the security settings for IKE traffic
IKE_AUTH performs mutual authentication and sets up the piggyback child SA
CREATE_CHILD_SA creates a new child SA or rekeys an existing child SA Most importantly, the study guide explicitly says:
"By IKEv2 design, no Diffie-Hellman public key is exchanged during an IKE_AUTH exchange. Consequently, any phase 2 Diffie-Hellman group configuration mismatch between FortiGate and the peer is experienced only during the first rekey (CREATE_CHILD_SA exchange) of the child SA created during IKE_AUTH." This proves the key idea behind the question: an IKEv2 tunnel can come up successfully first, then fail later during a CREATE_CHILD_SA rekey/renegotiation event because of a phase 2 mismatch. Among the provided options, the matching later-stage cause is mismatched quick-mode selectors during CREATE_CHILD_SA.
Why the other options are wrong:
A is wrong because if the proposal mismatch were in the initial negotiation path, the tunnel would fail during establishment, not after it was already up. The study guide places initial tunnel establishment in IKE_SA_INIT and IKE_AUTH B is wrong because a mismatch in IKE_SA_INIT affects the initial establishment stage, not a tunnel that was already brought up successfully C is wrong because a pre-shared key mismatch is part of authentication during

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IKE_AUTH, so the tunnel would not come up successfully in the first place

QUESTION NO: 21
A VPN tunnel is up. To monitor traffic flow, the administrator enters the following CLI commands on an SSH session on FortiGate:
diagnose debug enable
diagnose sniffer packet any 'udp and port 500' 4
However, the sniffer does not show any output. Assuming default configuration values, what are two possible reasons there is no output? (Choose two answers)
A. The filter should be modified to also capture packets for TCP port 443 or UDP port 4500.
B. NAT Traversal is enabled.
C. The sniffer must be restricted to the remote peer IP address.
D. The sniffer output will be ignored because running diagnose debug enable shows only application real-time debugs.
Answer: A,B
Explanation:
The correct answers are A and B.
The study guide says:
"If NAT-T is enabled, and there is a FortiGate located in the middle that is running NAT, the sniffer command must use a different filter. In this case, IKE traffic uses UDP port 500, but switches to UDP port 4500 during the tunnel negotiation. Additionally, ESP traffic is encapsulated inside the UDP 4500 channel." It also says:
"In some networks, UDP is blocked by firewalls or ISPs. In those cases, you can configure your VPN tunnel to use IKE over TCP in the phase 1 configuration. The default IKE TCP port is 443..." And the study guide gives the correct capture examples:
No NAT: host <remote-gw> and udp port 500
With NAT and NAT-T: host <remote-gw> and (udp port 500 or udp port 4500) So:
B is correct because with NAT Traversal enabled, the tunnel may no longer be using only UDP 500. It can move to UDP 4500, so the current filter may miss the traffic.
A is correct because the filter may need to be expanded to include UDP 4500 for NAT-T, or TCP 443 when IKE over TCP is used.
Why the other options are wrong:
C is wrong because restricting the filter to the remote peer IP can make the capture more precise, but it is not required for the sniffer to display output. The problem here is the port/protocol choice, not the lack of a host filter. The study guide examples use host filtering as an aid, not as a requirement.
D is wrong because diagnose debug enable is used to enable real-time debug output for applications, but it does not suppress or invalidate sniffer output. Sniffer capture is a separate command path. Fortinet documentation separately documents diagnose sniffer packet ... for packet capture and diagnose debug enable for debug features.
So the verified answers are: A, B.

QUESTION NO: 22
Refer to the exhibit.

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```
# get router info bgp summary

VRF 0 BGP router identifier 172.16.1.254, local AS number 65100
BGP table version is 3
2 BGP AS-PATH entries
0 BGP community entries

Neighbor        V         AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.64.1.254    4        100      18      20        3    0    0 00:02:55        1
100.64.2.254    4        100       0       0        0    0    0 never         Active

Total number of neighbors 2
```

The output of the get router info bgp summary command is shown.
Which statement regarding adjacencies between the local router and its neighbors is correct?

A. The local router and neighbor 100.64.2.254 are unable to establish adjacency until the adjacency with neighbor 100.64.1.254 ceases.

B. The local router and neighbor 100.64.2.254 are unable to establish adjacency because the TCP session could not be established.

C. The local router and neighbor 100.64.1.254 established adjacency because the priority of 100.64.1.254 is higher than that of 100.64.2.254.

D. The local router and neighbor 100.64.2.254 are unable to establish adjacency because AS 100 is already used by neighbor 100.64.1.254.

Answer: B

Explanation:
The correct answer is B.
In the exhibit:
Neighbor 100.64.1.254 shows State/PfxRcd = 1, which means the session is established and the local FortiGate has received 1 prefix Neighbor 100.64.2.254 shows State/PfxRcd = Active
The study guide explains the BGP states exactly:
Connect: Waiting for a successful three-way TCP connection
Active: Unable to establish the TCP session
OpenSent: Waiting for an OPEN message from the peer
OpenConfirm: Waiting for the keepalive message from the peer
Established: Peers have successfully exchanged OPEN and keepalive messages It also explains how to read the State/PfxRcd column:
"If the state is not established, this column displays the BGP state. If the state is established, this column displays the number of prefixes that the local FortiGate received from that neighbor." Therefore, because neighbor 100.64.2.254 is in Active state, the correct conclusion is that the BGP adjacency cannot form because the TCP session could not be established.
Why the other options are wrong:
A is wrong because BGP can establish adjacencies with multiple neighbors independently; one established neighbor does not block another C is wrong because BGP adjacency is not established based on neighbor "priority"; the output shows adjacency is established because the session completed and prefixes were exchanged D is wrong because having two neighbors in the same remote AS is valid in BGP and does not prevent adjacency formation
So the verified answer is: B.

QUESTION NO: 23
Refer to the exhibit.

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```
**** SP Login Dump ****<lasso:Login
xmlns:lasso="http://www.entrouvert.org/namespaces/lasso/0.0"
xmlns:samlp="urn:oasis:names:tc:SAML:2.0:protocol"
xmlns:saml="urn:oasis:names:tc:SAML:2.0:assertion"
LoginDumpVersion="2"><lasso:Request><samlp:AuthnRequest
ID="_EEC718A47FB37B472B205B11153ED409" Version="2.0" IssueInstant="2024-02-
21T00:58:44Z" Destination="https://10.1.10.2/saml-idp/nst/login/"
SignType="0" SignMethod="0" ForceAuthn="false" IsPassive="false"
ProtocolBinding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST"
AssertionConsumerServiceURL="https://10.1.10.254:1003/remote/saml/login/"><sa
ml:Issuer>https://10.1.10.254:1003/remote/saml/metadata/</saml:Issuer><samlp:
NameIDPolicy Format="urn:oasis:names:tc:SAML:1.1:nameid-format:unspecified"
AllowCreate="true"/></samlp:AuthnRequest></lasso:Request><lasso:RemoteProvide
rID>http://10.1.10.2/samlidp/nst/metadata/</lasso:RemoteProviderID><lasso:Msg
Url>https://10.1.10.2/saml-
idp/nst/login/?SAMLRequest=jZJfT8IwFMW%2FytL30W5sAZtBwhhEEtQF0AdfTN0u0GRrZ2%2
Fnn29vGWIwUeJLk97eX%2B85p01Q1FXDJ63dqxW8tIDWe68rhbw7GJHWKK4FSuRK1IDcFnw9uVnys
Md4Y7TVha7IGXKZEIhgrNSKeItsRJ5ms%4</lasso:HttpRequestMethod><lasso:RequestID>
_EEC718A47FB37B472B205B11153ED409</lasso:RequestID></lasso:Login>
```

The output from using the command diagnose debug application samld -1 to diagnose a SAML connection is shown. Based on this output, which two conclusions can you draw? (Choose two answers)

A. The IdP IP address is 10.1.10.254.

B. The SP IP address is 10.1.10.254.

C. The SP IP address is 10.1.10.2.

D. The IdP IP address is 10.1.10.2.

Answer: B,D

Explanation:

The correct answers are B and D.

The study guide explains that in the SP Login Dump section, FortiGate is acting as the service provider (SP), and that you should read these fields:

"The IdP SSO URL, from the setting idp-single-sign-on-url in the FortiGate configuration"

"The SP SSO URL, from the setting single-sign-on-url in the FortiGate configuration"

"The IdP Entity ID, from the setting id-entity-id in the FortiGate configuration"

"The SP Entity ID, from the setting entity-id setting in the FortiGate configuration" In the exhibit:

Destination="https://10.1.10.2/saml-idp/nst/login/" → this is the IdP SSO URL

<lasso:RemoteProviderID>http://10.1.10.2/samlidp/nst/metadata/</lasso:RemoteProviderID> → this is the IdP Entity ID

AssertionConsumerServiceURL="https://10.1.10.254:1003/remote/saml/login/" → this is the SP SSO URL

<saml:Issuer>https://10.1.10.254:1003/remote/saml/metadata/</saml:Issuer> → this is the SP Entity ID The same study-guide example shows this exact mapping pattern, where:

Destination points to the IdP

AssertionConsumerServiceURL and Issuer point to the SP

Therefore:

10.1.10.2 is the IdP → D

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10.1.10.254 is the SP → B
So the verified answers are: B, D.

QUESTION NO: 24
Which exchange lakes care of DoS protection in IKEv2?
A. Create_CHILD_SA
B. IKE_Auth
C. IKE_Req_INIT
D. IKE_SA_NIT
Answer: C
Explanation:
The IKE_SA_INIT exchange in IKEv2 is responsible for DoS protection measures. During IKE_SA_INIT, before authentication and further exchange, the responder can use cookie challenges (per RFC 7296 and Fortinet VPN documentation). If a DoS attack is suspected (many requests from the same source), the responder replies with a cookie. Only after the initiator returns the correct cookie does the exchange proceed, protecting the responder from state exhaustion and certain forms of DoS traffic at the handshake stage.
References:
FortiOS VPN Manual: IKEv2 Exchange Process and DoS Protections
IKEv2 RFC 7296: Description of IKE_SA_INIT and DoS Cookie Mechanism

QUESTION NO: 25
Refer to the exhibit.

High Availability configuration status

```
NGFW-1 # get sys ha status
HA Health Status: OK
Model: FortiGate-VM64
Mode: HA A-P
Group ID: 12
Debug: 0
Cluster Uptime: 0 days 0h:15m:25s
Cluster state change time: 20xx-04-18 12:07:47
Primary selected using:
    <20xx/04/18 12:07:47> FGVM010000077649 is selected as the primary because its override priority is
larger than peer member FGVM010000077650.
ses_pickup: disable
override: disable
Configuration Status:
     FGVM010000077649(updated 5 seconds ago): in-sync
     FGVM010000077649 chksum dump: a4 36 c1 0f 19 35 dc 6c 51 c3 c7 bc d5 58 cd d3
     FGVM010000077650(updated 1 seconds ago): in-sync
     FGVM010000077650 chksum dump: a4 36 c1 0f 19 35 dc 6c 51 c3 c7 bc d5 58 cd d3
System Usage stats:
    FGVM010000077649 (updated 4 seconds ago) :
        sessions=166, average-cpu-user/nice/system/idle=19/08/08/99%, memory=458
    FGVM010000077650 (updated 1 seconds ago) :
        sessions=3, average-cpu-user/nice/system/idle=08/08/08/100%, memory=448
HBDEV stats:
    FGVM010000077649 (updated 4 seconds ago) :
        port7: physical/1000auto, up, rx-bytes/packets/dropped/errors=167663/567/0/0, tx=262623/656/0/0
    FGVM010000077650 (updated 1 seconds ago) :
        port7: physical/1000auto, up, rx-bytes/packets/dropped/errors=271373/680/0/0, tx=176013/592/0/0
---omitted---
```

Which two statements about the output are true, considering NGFW-1 and NGFW-2 have been up for a week? (Choose two.)
A. If FGVM...649 is rebooted, FGVM...650 will become the primary FortiGate and retain that role, even after FGVM...649 rejoins the cluster.
B. If port7 becomes disconnected on the secondary FortiGate, both FortiGate devices will elect themselves as primary.
C. If a configuration change is made to the secondary FortiGate, the Configuration Status will not change.

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D. If a configuration change is made to the primary FortiGate at this time, the secondary will initiate a synchronization reset.

Answer: A,B

Explanation:

The correct answers are A and B.

The exhibit shows:

override: disable

both members are currently in-sync

only port7 appears under HBDEV stats, so it is the active heartbeat interface the cluster is in HA A-P mode Why A is correct:

With override disabled, after a failover the new primary keeps that role when the old primary comes back. The FortiOS administration guide states:

"When the primary FortiGate rejoins the cluster the secondary FortiGate continues to operate as the primary FortiGate." So if FGVM...649 reboots and FGVM...650 becomes primary, FGVM...650 will remain primary after FGVM...649 rejoins.

Why B is correct:

The study guide states:

"When FortiGate devices configured in an HA cluster lose communication with each other on the heartbeat interface, each FortiGate assumes the role of the primary device." The exhibit shows only port7 as the heartbeat device in HBDEV stats So if port7 is disconnected and heartbeat communication is lost, the cluster can enter a split-brain condition, where both units believe they are primary. The FortiOS administration guide confirms the same behavior: loss of heartbeat communication causes each member to think it is the primary Why the other options are wrong:

C is wrong because configuration synchronization status is specifically used to detect whether secondary members remain synchronized with the primary. If members are no longer synchronized, the status changes from in-sync to out-of-sync D is wrong because the study guide explains that during a configuration change, checksums may differ briefly while changes are copied, but it does not describe this as the secondary initiating a "synchronization reset" So the verified answers are: A, B.

QUESTION NO: 26

Refer to the exhibit, which shows the output of a diagnose command. What can you conclude from the RTT value?

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```
FGT # diagnose debug rating
Locale       : english

Service      : Web-filter
Status       : Enable
License      : Contract

Service      : Antispam
Status       : Disable

Service      : Virus Outbreak Prevention
Status       : Disable

Num. of servers : 1
Protocol        : https
Port            : 443
Anycast         : Enable
Default servers : Included

-=- Server List (Mon May  1 03:47:52 2023) -=-

IP                                        Weight    RTT Flags   TZ   FortiGuard-requests  Curr Lost Total Lost            Updated Time
64.26.151.37                                  10     45         -5                262432          0        846 Mon May  1 03:47:43 2023
64.26.151.35                                  10     46         -5                329072          0       6806 Mon May  1 03:47:43 2023
66.117.56.37                                  10     75         -5                 71638          0        275 Mon May  1 03:47:43 2023
65.210.95.240                                 20     71         -8                 36875          0         92 Mon May  1 03:47:43 2023
209.22.147.36                                 20    103 DI      -8                 34784          0       1070 Mon May  1 03:47:43 2023
208.91.112.194                                20    107 D       -8                 35170          0       1533 Mon May  1 03:47:43 2023
96.45.33.65                                   60    144          0                 33728          0        120 Mon May  1 03:47:43 2023
80.85.69.41                                   71    226          1                 33797          0        192 Mon May  1 03:47:43 2023
62.209.40.74                                 150     97          9                 33754          0        145 Mon May  1 03:47:43 2023
121.111.236.179                               45     44  F       -5                 26410      26226      26227 Mon May  1 03:47:43 2023
```

A. Its value represents the time it takes to receive a response after a rating request is sent to a particular server.

B. Its value is incremented with each packet lost.

C. It determines which FortiGuard server is used for license validation.

D. Its initial value is statically set to 10.

Answer: A

Explanation:

The correct answer is A.

The study guide explicitly explains the diagnose debug rating table and says that for each server IP, the output shows "The round trip delay" That means the RTT value represents the time it takes for FortiGate to send a request and receive the reply from that FortiGuard server.

The FortiOS administration guide also confirms this by stating:

"Each server is probed for Round Trip Time (RTT) every two minutes."

Why the other options are wrong:

B is wrong because packet loss is shown separately by Curr Lost and Total Lost, while RTT is the round-trip delay C is wrong because license-validation behavior is indicated by flags such as I = Initial, not by the RTT value itself D is wrong because the documents do not say RTT starts at a fixed value of 10; it is measured dynamically as round-trip delay So the verified answer is: A.

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QUESTION NO: 27

Which of the following regarding protocol states is true? (Choose one answer)

A. proto_state=00 indicates that UDP traffic flows in both directions.

B. proto_state=01 indicates an established TCP session.

C. proto_state=10 indicates an established TCP session.

D. proto_state=01 indicates one-way ICMP traffic.

Answer: B

Explanation:

The correct answer is B.

The study guide states that for TCP, the protocol state is a two-digit number. The first digit is the server-side state and is 0 when the session is not subject to inspection. The second digit is the client-side state. It also shows that value 1 = ESTABLISHED So, for a normal TCP session with no inspection, proto_state=01 means:
first digit 0 = no inspection
second digit 1 = ESTABLISHED
That makes B correct.
Why the other options are wrong:
A is wrong because for UDP, the study guide says 00 = one-way traffic and 01 = two-way traffic C is wrong because 10 does not represent the normal established TCP session described in the study guide. The established example shown is based on state value 1, and the guide explicitly highlights proto_state=11 when both server-side and client-side TCP handshakes are completed D is wrong because for ICMP, the study guide says "the protocol state is always 00"
====

QUESTION NO: 28
Which statement about IKEv2 is true?
A. Both IKEv1 and IKEv2 share the feature of asymmetric authentication.
B. IKEv1 and IKEv2 have enough of the header format in common that both versions can run over the same UDP port.
C. IKEv1 and IKEv2 use the same TCP port but run on different UDP ports.
D. IKEv1 and IKEv2 share the concept of phase1 and phase2.
Answer: B
Explanation:
The correct answer is B.
The study guide explicitly states: "IKE version 2 does not interoperate with IKE version 1, but they share enough of the header format that both versions can unambiguously operate over the same UDP port." That directly proves B.
Why the other options are wrong:
A is wrong because the study guide shows authentication methods as Asymmetric for IKEv2 and Symmetric for IKEv1 C is wrong because the study guide does not say they use the same TCP port; instead, it specifically says they can operate over the same UDP port D is wrong because the study guide states: "IKEv2 does not use the concept of phase 1 or phase 2", even though FortiOS CLI/GUI still uses those terms for configuration purposes

QUESTION NO: 29
Refer to the exhibit, which shows partial outputs from two routing debug commands.

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```
FortiGate # get router info routing-table database

Routing table for VRF=0
S           0.0.0.0/0 [20/0] via 100.64.2.254, port2, [10/0]
S        *> 0.0.0.0/0 [10/0] via 100.64.1.254, port1

FortiGate # get router info routing-table all

Routing table for VRF=0
S*          0.0.0.0/0 [10/0] via 100.64.1.254, port1
```

Why is the port2 default route not in the second command output?

A. The port2 interface is disabled in the FortiGate configuration.

B. The port1 default route has a higher priority value than the default route using port2.

C. The port1 default route has a lower priority value than the default route using port2.

D. The port1 default route has a lower distance than the default route using port2.

Answer: D

Explanation:

The correct answer is D.

In the exhibit, get router info routing-table database shows both static default routes:

0.0.0.0/0 [20/0] via 100.64.2.254, port2

0.0.0.0/0 [10/0] via 100.64.1.254, port1

But get router info routing-table all shows only:

0.0.0.0/0 [10/0] via 100.64.1.254, port1

The study guide explains that get router info routing-table all displays the routes that make it to the FIB - the best active routes. It also says that this command doesn't show standby or inactive routes, which can remain only in the routing table database. It gives the exact rule: "when two static routes to the same destination subnet have different distances, the one with the lower distance is installed in the routing table, and the one with the higher distance is installed in the routing table database." The study guide also shows the route selection process:

Most specific route

Lowest distance

Lowest metric (dynamic routes)

Lowest priority (static routes)

ECMP

So the port2 default route is absent from the second output because its distance is 20, while the port1 default route has distance 10. The lower-distance route is installed in the active routing table/FIB.

Why the other options are wrong:

A is wrong because the exhibit does not indicate port2 is down or disabled.

B and C are wrong because priority is checked only after distance for static routes. Here, the routes already differ by distance, so priority is not the deciding factor.

So the verified answer is: D.

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QUESTION NO: 30

Refer to the exhibit, which shows the output of diagnose sys session stat.

```
NGFM-1 # diagnose sys session stat
misc info:          session_count=591 setup_rate=0 exp_count=0 clash=162
          memory_tension_drop=0 ephemeral=0/65536 removeable=0
delete=0, flush=0, dev_down=0/0 ses_walkers=0
TCP sessions:
             166 in NONE state
             1 in ESTABLISHED state
             3 in SYN SENT state
             2 in TIME WAIT state
firewall error stat:
error1=00000000
error2=00000000
error3=00000000
error4=00000000
tt=00000000
cont=00000000
ids_recv=00000000
url_recv=00000000
av_recv=00000000
fqdn_count=00000006
fadn6_count=00000000
global: ses_limit=0 ses6_limit=0 rt_limit=0 rt6_limit=0
```

Which statement about the output shown in the exhibit is correct?

A. All the sessions in the session table are TCP sessions.

B. 162 sessions have been deleted because of memory page exhaustion.

C. There are 166 TCP sessions waiting to complete the three-way handshake.

D. There are two sessions that have not been removed in case any out-of-order packets arrive.

Answer: D

Explanation:

The correct answer is D.

The exhibit shows:

session_count=591

clash=162

memory_tension_drop=0

TCP sessions:

166 in NONE state

1 in ESTABLISHED state

3 in SYN_SENT state

2 in TIME_WAIT state

The study guide explains the TCP protocol states and states explicitly:

"When a session is closed by both the sender and receiver, FortiGate keeps that session in the session table for a few seconds, to allow for any out-of-order packets that might arrive after the FIN/ACK packet. This is the state value 5." In diagnose sys session stat, the exhibit shows 2 in TIME_WAIT state. Since TIME_WAIT = state value 5, those are the sessions being kept briefly for possible out-of-order packets. That makes D correct.

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Why the other options are wrong:
A is wrong because session_count=591 is the total number of sessions, while the TCP sessions shown add up to only 172 (166 + 1 + 3 + 2). So not all sessions in the table are TCP sessions.
B is wrong because the study guide says the number of sessions deleted because of low free memory is shown by memory_tension_drop, and in the exhibit it is 0, not 162.
C is wrong because the study guide defines ephemeral/open TCP sessions as those not fully established, but the exhibit does not say all 166 in NONE state are specifically "waiting to complete the three-way handshake." The clearest directly supported statement from the displayed states is the 2 TIME_WAIT sessions retained for out-of-order packets.
So the verified answer is: D.

QUESTION NO: 31
What is the diagnose test application ipsmonitor 5 command used for? (Choose one answer)
A. To restart all IPS engines and monitors
B. To provide information regarding IPS sessions
C. To disable the IPS engine
D. To enable IPS bypass mode
Answer: D

QUESTION NO: 32
What are two functions of automation stitches? (Choose two.)
A. You can configure automation stitches to run diagnostic commands and attach the results to an email message when CPU or memory usage exceeds specified thresholds.
B. You can configure automation stitches to modify packet headers and payloads if specific traffic triggers an anomaly IPS event.
C. You can configure automation stitches to insert a delay between actions if the automation stitches are set to execute actions in parallel.
D. You can configure automation stitches to take parameters from previous actions as input for the next action if the automation stitches are set to execute actions in sequence.
Answer: A,D

QUESTION NO: 33
Which three conditions are required for two FortiGate devices to form an OSPF adjacency? (Choose three answers)
A. OSPF link costs match.
B. OSPF interface priority settings are unique.
C. OSPF interface network types match.
D. Authentication settings match.
E. OSPF router IDs are unique.
Answer: C,D,E
The same study-guide slide also states other requirements such as:
peers' primary IPs must be in the same subnet with the same mask
hello and dead intervals must match

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OSPF MTUs must match
Why the other options are wrong:
A is wrong because link cost matching is not listed as an adjacency requirement. OSPF cost affects path selection, not whether adjacency can form.
B is wrong because interface priority does not need to be unique. Priority is used for DR/BDR election, where the highest priority wins, and ties are broken by router ID.
So the verified answers are: C, D, E.
Explanation:
The correct answers are C, D, and E.
The study guide lists the exact OSPF adjacency requirements:
"Interfaces of peers are the same type and in the same OSPF area"
"Each peer has a unique router ID"
"OSPF authentication, if enabled, is successful"
These map directly to:

QUESTION NO: 34
Refer to the exhibit, which shows the output of a diagnose command.

```
# diagnose sys session list expectation
session_info: proto=6 proto_state=00 duration=3 expire=26 timeout=3600 flags=00000000
sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=1 tunnel=/
state=new complex
statistic (bytes/packets/allow err): org=0/0/0 reply=0/0/0 tuples=2
orgin->sink: org pre->post, reply pre->post dev=2->4/4->2 gwy=10.0.1.10/10.200.1.254
hook=pre dir=org act=dnat 10.171.121.38:0->10.200.1.1:60426 (10.0.1.10:50365)
hook=pre dir=org act=noop 0.0.0.0:0->0.0.0.0:0(0.0.0.0:0)
pos/ (before, after) 0/ (0,0), 0/ (0,0)
misc=0 policy id=1 auth_info=0 chk_client_info=0 vd=0
serial=000000e9 tos-ff/ff ips_view=0 app_list=0 app=0
dd_type=0 dd_mode=0
```

What two conclusions can you draw from the output shown in the exhibit? (Choose two answers)
A. This is an expected session created by the IPS engine.
B. Traffic in the original direction (coming from the IP address 10.171.121.38) will be routed to the next-hop IP address 10.0.1.10.
C. Traffic in the original direction (coming from the IP address 10.171.121.38) will be routed to the next-hop IP address 10.200.1.1.
D. This is a pinhole session created to allow traffic for a protocol that requires additional sessions to operate through FortiGate.
Answer: B,D
Explanation:
The correct answers are B and D.
The study guide explains that expectation sessions are pinhole sessions created by session helpers for protocols such as FTP that need additional negotiated connections. It states: "FortiGate created an expectation session and opened the pinhole port for the expected return traffic" and also shows that the firewall "creates an expected (pinhole) session to allow

the traffic" That makes D correct.
For B, the study guide explains the gwy= field in session output: the first value is the gateway to the destination, and the second is the gateway to the source In the exhibit, the original-direction traffic is DNATed here:
hook=pre dir=org act=dnat 10.171.121.38:0->10.200.1.1:60426(10.0.1.10:50365) So the destination after DNAT is the internal host 10.0.1.10, and the session's first gwy value corresponds to the next hop toward that destination. That makes B correct.
Why the other options are wrong:
A is wrong because this is an expectation/session-helper behavior, not an IPS-engine-created session. The study guide ties expectation sessions to helpers such as FTP, not IPS.
C is wrong because 10.200.1.1 is the translated address used before DNAT, not the next hop used to forward the original-direction traffic after translation to the internal destination.
So the verified answers are: B, D.

QUESTION NO: 35
Refer to the exhibit, which shows the output of a real-time debug. Which statement about this output is true? (Choose one answer)

```
FGT # diagnose debug application urlfilter -1
FGT # diagnose debug enable

msg="received a request /tmp/.wad512_0_0.url.socket, addr_len=30:
d=training. fortinet.com:443, id=687, cat=255, vfname='root', vfid=0,
profile='default', type=0, client=10.1.10.1, url_source=1, url="/"
action=9 (ftgd-allow) wf-act=5 (ALLOW) user="N/A" src=10.1.10.1 sport=58334
dst=13.226.142.41 dport=443 service="https" cat=52 url_cat=52 ip_cat=0
hostname="training.fortinet.com" url="/"
```

A. The server hostname was extracted from the SNI in the client request, or from the CN in the server certificate.
B. FortiGate found the requested URL in its local cache.
C. This web request was inspected using the ftgd-allow web filter profile.
D. The requested URL belongs to category ID 255.
Answer: A
Explanation:
The correct answer is A.
The debug output is for an HTTPS request and shows a hostname value. The study guide explains that with SSL certificate inspection, FortiGate extracts the FQDN from either:
"TLS extension server name indication (SNI)"
"SSL certificate common name (CN)"
So the hostname shown in the real-time web-filter debug can be derived from the SNI in the client request or, if needed, from the CN in the server certificate. That makes A correct.
Why the other options are wrong:
B is wrong because the study-guide example for web-filter real-time debug explicitly says: "This slide shows an example of real-time debug output when the URL to categorize isn't in the FortiGuard cache." In these debugs, cat=255 appears before the final lookup result, so this does not indicate a local-cache hit.
C is wrong because ftgd-allow is the action, not the profile name. The debug line shows the

action as action=9 (ftgd-allow) while the profile shown is profile='default'. FortiOS web-filter logs also use the profile field separately from the action field D is wrong because the final category shown is url_cat=52, not 255. The study guide's example shows the same pattern: an initial cat=255 in the request line, followed by the resolved result cat=52 url_cat=52 So the verified answer is: A.

QUESTION NO: 36

Which two statements about application-layer test commands are true? (Choose two answers)

A. Some of them display statistics and configuration information about a feature or process.

B. Some of them display real-time application debugs.

C. Some of them display output only after you run the diagnose debug console enable command.

D. Some of them can be used to restart an application.

Answer: A,D

Explanation:

The correct answers are A and D.

The study guide states:

"Application layer test commands do not display information in real time. They display statistics and configuration information about a feature or process. You can also use some of these commands to restart a process or execute a change in its operation." This directly proves:

A is correct because they can display statistics and configuration information D is correct because some of them can restart a process/application Why the other options are wrong:

B is wrong because the study guide explicitly says application-layer test commands do not display information in real time. Real-time output is done with diagnose debug application ... commands instead.

C is wrong because diagnose debug console enable is related to debug output behavior, not a requirement for application-layer test commands to display output. The study guide does not describe test commands that way.

====

QUESTION NO: 37

Refer to the exhibit, which shows a partial output of the fssod daemon real-time debug command.

```
# diagnose debug application fssod -1
# diagnose debug enable
[fsso_svr.c:save_result:579] event_id=4768, logon=bobby, domain=FSSO workstation=, ip=10.124.2.90 port=49215, time=1372061722
```

What two conclusions can you draw from the output? (Choose two.)

A. The workstation with IP 10.124.2.90 will be polled frequently using TCP port 445 to see if the user is still logged on.

B. The logon event can be seen on the collector agent installed on Windows.

C. FSSO is using DC agent mode to detect logon events.

D. FSSO is using agentless polling mode to detect logon events.

Answer: A,D

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Explanation:
https://community.fortinet.com/t5/FortiGate/Troubleshooting-Tip-How-to-troubleshoot-FSSO-agentless-polling/ta-p/214349 From the snippet we can see that FortiGate (via the fssod daemon) is directly detecting the user logon rather than relying on a separate "collector" or "DC agent." This indicates agentless polling-FortiGate polls the DC's event logs over TCP 445 to discover logons. So: - FSSO is using agentless polling mode to detect logon events - I n agentless mode, FortiGate will periodically poll the same IP (the DC) on port 445 to see if the user is still logged on

QUESTION NO: 38
What is the diagnose test application ipsmonitor 5 command used for? (Choose one answer)
A. To disable the IPS engine
B. To provide information regarding IPS sessions
C. To restart all IPS engines and monitors
D. To enable IPS bypass mode
Answer: D
Explanation:
The correct answer is D.
The study guide shows the ipsmonitor test usage exactly:
1: Display IPS engine information
2: Toggle IPS engine enable/disable status
5: Toggle bypass status
99: Restart all IPS engines and monitor
So diagnose test application ipsmonitor 5 is used to toggle bypass status, which corresponds to enabling IPS bypass mode.
Why the other options are wrong:
A is wrong because disabling the IPS engine is option 2, not 5.
B is wrong because the study guide does not define option 5 as IPS session information.
C is wrong because restarting all IPS engines and monitors is option 99, not 5.
So the verified answer is: D.

QUESTION NO: 39
Refer to the exhibit, which shows a truncated output of a real-time RADIUS debug.

```
# diagnose debug application fnbamd -1
# diagnose debug enable

fnbamd_fsm.c[1819] handle_req-Rcvd auth req 2 for student in RadiusServer opt=29 prot=1
fnbamd_fsm.c[336] __compose_group_list_from_req-Group 'RadiusServer'
fnbamd_pop3.c[573] fnbamd_pop3_start-student
fnbamd_cfg.c[443] __fnbamd_cfg_get_radius_list_by_server-Loading RADIUS server 'RadiusServer'
fnbamd_radius.c[1006] fnbamd_radius_auth_send-Compose RADIUS request
fnbamd_radius.c[1195] fnbamd_radius_auth_send-Sent radius req to server 'RadiusServer': IP=172.25.188.164 code=1 id=1 len=92 user="student" using CHAP
fnbamd_auth.c[268] radius_server_auth-Timer of rad 'RadiusServer' is added
fnbamd_fsm.c[428] create_auth_session-Total 1 server(s) to try
fnbamd_auth.c[1990] fnbamd_auth_handle_radius_result-Timer of rad 'RadiusServer' is deleted
fnbamd_auth.c[2016] fnbamd_auth_handle_radius_result-->Result for radius svr 'RadiusServer' 172.25.188.164(0) is 0
fnbamd_auth.c[2044] fnbamd_auth_handle_radius_result-Skipping group matching
fnbamd_fsm.c[825] find_matched_usr_grps-Skipped group matching
fnbamd_comm.c[169] fnbamd_comm_send_result-Sending result 0 for req 2
fnbamd_fsm.c[568] destroy_auth_session-delete session 2
```

Which two statements are true? (Choose two answers)
A. The RADIUS server queried for authentication is located at IP address 172.25.188.164.
B. Authentication was unsuccessful.
C. The authentication scheme used was pop3.
D. Authentication was successful.

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E. Two-factor authentication was required.

Answer: A,D

Explanation:

The correct answers are A and D.

The debug output shows:

Sent RADIUS req to server 'RadiusServer': IP=172.25.188.164 ... user="student" using CHAP Result for radius svr 'RadiusServer' 172.25.188.164(0) is 0 Sending result 0 for req 2

The study guide explains that in RADIUS real-time debug, FortiGate shows the IP address of the RADIUS server it is querying. In the example, it says FortiGate "creates an access request to the RADIUS server at IP address 10.0.13.130" and shows the line Sent radius req to server ... IP=10.0.13.130 So in your exhibit, the queried server is clearly 172.25.188.164, which makes A correct.

The study guide also states:

"The message fnbamd_comm_send_result-Sending result 0 indicates that the authentication was successful and that FortiGate received the Access-Accept message." Since your exhibit also ends with Sending result 0, that makes D correct.

Why the other options are wrong:

B is wrong because result 0 means authentication successful, not failed C is wrong because the debug explicitly shows using CHAP, and the study guide lists supported RADIUS schemes as CHAP, PAP, MS-CHAP, and MS-CHAPv2 E is wrong because the study guide says two-factor authentication would involve an Access-Challenge response: "If two-factor authentication is enabled on the server, the response is an Access-Challenge message" Your exhibit shows successful result 0 / Access-Accept, not a challenge.

So the verified answers are: A, D.

QUESTION NO: 40

An administrator wants to capture encrypted phase 2 traffic between two FotiGate devices using the built-in sniffer.

If the administrator knows that there Is no NAT device located between both FortiGate devices, which command should the administrator run?

A. diagnose sniffer packet any 'udp port 500'

B. diagnose sniffer packet any 'lp proto 50'

C. diagnose sniffer packet any 'udp port 4500'

D. diagnose sniffer packet any 'ah'

Answer: B

Explanation:

To capture encrypted IPsec phase 2 (ESP) traffic between two FortiGate devices, the correct protocol filter to use is ip proto 50. According to the Fortinet official sniffing and debugging documentation, ESP (Encapsulating Security Payload) is used for encrypted phase 2 payload transfer and always uses IP protocol number 50. Running the command diagnose sniffer packet any 'ip proto 50' captures only ESP packets, which represent the encrypted traffic-whether originating or transiting the device.

If there is no NAT device between FortiGates, ESP is not encapsulated in UDP (thus not on UDP port 4500; if NAT-T were required, packets would be UDP-encapsulated, but the scenario explicitly says NAT is not in use). UDP port 500 is for IKE control (negotiation)

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traffic, and AH (Authentication Header, ip proto 51) is not used for encryption in standard IPsec phase 2 with ESP.
This matches the official CLI reference from Fortinet for VPN and traffic analysis.
**
References:
FortiOS CLI Reference: diagnose sniffer packet, ESP, IP Protocol Numbers FortiGate VPN Administration Guide: Traffic Capture and Analysis of IPsec Traffic

QUESTION NO: 41
Refer to the exhibits.

```
FGT-B # get router info routing-table all
Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 192.168.1.1, port1, [1/0]
C       10.23.23.0/24 is directly connected, port4
```

```
FGT-B # get router info ospf database brief
...
                AS External Link States

Link ID         ADV Router      Age  Seq#     CkSum Flag Route              Tag
8.8.8.8         0.0.0.112       1464 80000002 3106  0002 E2 8.8.8.8/32      0
```

An administrator Is expecting to receive advertised route 8.8.8.8/32 from FGT-A. On FGT-B, they confirm that the route is being advertised and received, however, the route is not being injected into the routing table. What is the most likely cause of this issue?

A. A batter route to the 8.8.8.8/32 network exists in the routing table.

B. FGT-B is configured with a prefix list denying the 8.8.8.8/32 network to be injected into the routing table.

C. The administrator has misconfigured redistribution of routes on FGT-A.

D. FGT-B is configured with a distribution list denying the 8.8.8.8/32 network to be injected into the routing table.

Answer: B
Explanation:
The 8.8.8.8/32 route is visible in the OSPF database on FGT-B but not installed into the routing table-the most likely explanation is that FGT-B is filtering it from being installed.

QUESTION NO: 42
Refer to the exhibit, which shows the output of a BGP debug command.

```
# get router info bgp summary

VRF 0 BGP router identifier 0.0.0.117, local AS number 65117
BGP table version is 3
3 BGP AS-PATH entries
0 BGP community entries

Neighbor        V         AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.125.0.60     4      65060    1698    1756      103    0    0 03:02:49        1
10.127.0.75     4      65075    2206    2250      102    0    0 02:45:55        1
100.64.3.1      4      65501    101     115         0    0    0 never       Active

Total number of neighbors 3
```

What can you conclude about the router in this scenario?

A. The router 100.64.3.1 needs to update the local AS number in its BGP configuration in order to bring up the 8GP session with the local router.

B. An inbound route-map on local router is blocking the prefixes from neighbor 100.64.3.1.

C. All of the neighbors displayed are part of a single BGP configuration on the local router with the neighbor-range set to a value of 4.

D. The BGP session with peer 10.127.0.75 is up.

Answer: D

Explanation:

The BGP debug output shows session information for peers, including state details. According to official Fortinet BGP documentation, if the session state with a peer does not show "Idle," "Active," or "Connect," but instead shows "Established," "Up," or related counters (e.g., messages sent/received or uptime), it indicates the session is operational. In this scenario, the peer 10.127.0.75 is the only one showing a positive indication of a live, established session. Other options like neighbor-range configuration, AS mismatch, or route-maps blocking prefixes are not supported by evidence provided in a simple BGP session state debug, nor does the output show errors relating to local or remote AS issues.

The correct interpretation comes from Fortinet's BGP troubleshooting guide, which outlines how to read session status and neighbor states in debug and summary outputs.

References:

FortiOS BGP Debugging Guide: Session State Interpretation

BGP CLI Reference: Neighbor Status Fields

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QUESTION NO: 43

Which two statements about an auxiliary session ate true? (Choose two.)

A. With the auxiliary session selling disabled, only auxiliary sessions are offloaded.

B. With the auxiliary session setting enabled. ECMP traffic is accelerated to the NP6 processor.

C. With the auxiliary session setting enabled. Iwo sessions are created in case of routing change.

D. With the auxiliary session setting disabled, for each traffic path. FortiGate uses the same auxiliary session.

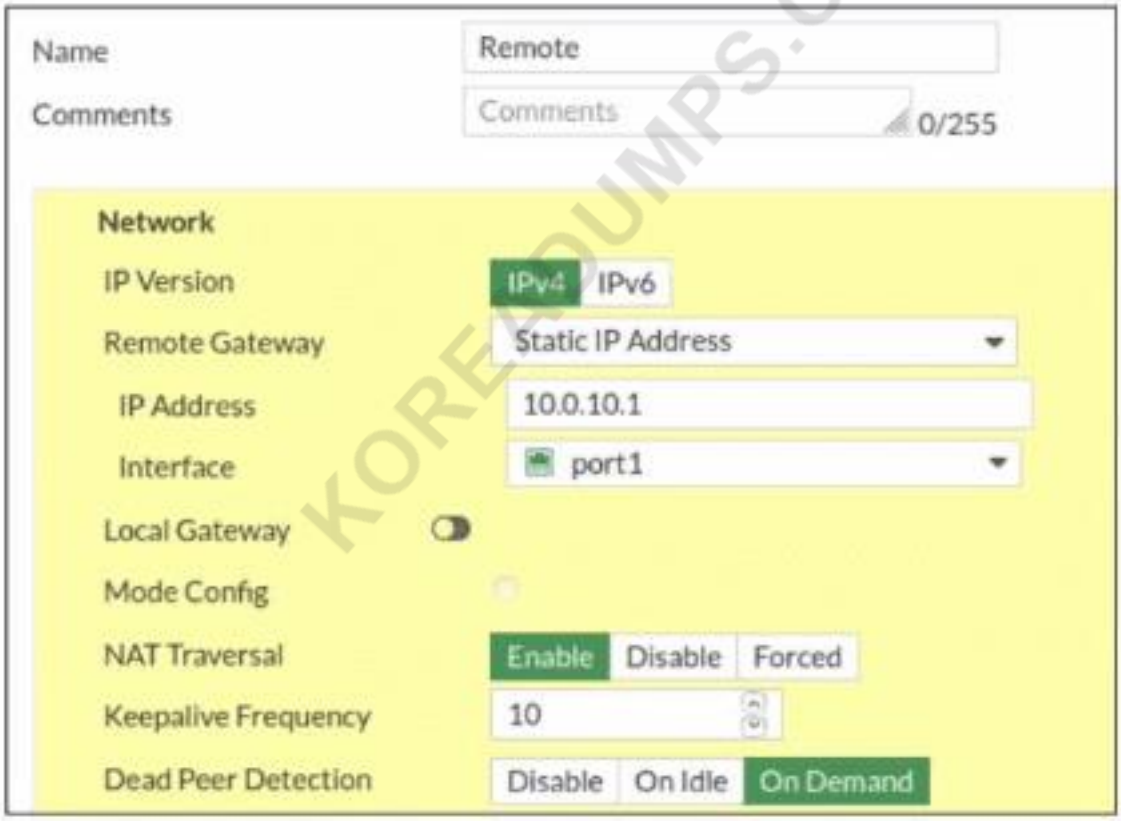
Answer: B,C

Explanation:

Auxiliary sessions in Fortinet are designed to support ECMP (Equal Cost Multi-Path) and SD-

WAN scenarios, allowing sessions to be handled efficiently when traffic needs to be dynamically distributed across multiple links. With the auxiliary session setting enabled, FortiGate creates additional session table entries for each possible path in ECMP or SD-WAN-meaning that if the routing path changes (such as a link failover), a new session can be immediately activated and offloaded to the NP6 network processor for acceleration, ensuring minimal disruption. This greatly benefits high-throughput deployments.
Official documentation specifies that when auxiliary sessions are enabled, FortiGate doesn't just rely on dynamically creating new sessions after a routing event, it proactively creates sessions for all potential paths. This means that in the event of a route change, two sessions exist and the traffic is quickly re-routed and offloaded, maximizing performance and reliability. Without this feature, multiple paths cannot be efficiently offloaded, and routing changes trigger a single session update, reducing failover performance.
References:
FortiOS Handbook: Session Table, ECMP, SD-WAN, and Auxiliary Sessions
FortiGate NP6 Acceleration Guide: Auxiliary Session Behavior

QUESTION NO: 44
Exhibit.



Refer to the exhibit, which contains a screenshot of some phase 1 settings.
The VPN is not up. To diagnose the issue, the administrator enters the following CLI commands on an SSH session on FortiGate:

```
diagnose vpn ike log-filter dst-addr4 10.0.10.1
diagnose debug application ike -1
```

However, the IKE real-time debug does not show any output. Why?

A. The administrator must also run the command diagnose debug enable.
B. The debug shows only error messages. If there is no output, then the phase 1 and phase 2 configurations match.
C. The log-filter setting is incorrect. The VPN traffic does not match this filter.
D. Replace diagnose debug application ike -1 with diagnose debug application ipsec -1.
Answer: A
Explanation:
To display debug output on FortiGate devices, you must always run both the application-specific debug command and the global debug enable command. The command diagnose debug application ike -1 sets up the detail level for the IKE daemon debug, but it does not display any debug output on its own. As described in the FortiOS CLI debugging manuals, the command diagnose debug enable activates debug output on the console, making all previously set debugs visible. This is especially important for VPN troubleshooting-without the enable command, no output appears even if there is VPN traffic.
The correct diagnostic sequence is:
diagnose debug application ike -1
diagnose debug enable
This procedure is found in every FortiOS CLI debug tutorial and troubleshooting workflow.
References:
FortiOS CLI Reference: Debugging VPNs and Real-time Debug Output
FortiGate VPN Troubleshooting Guide: Required Steps for Debug Output

QUESTION NO: 45
Which two statements are true regarding heartbeat messages sent from an FSSO collector agent to FortiGate? (Choose two.)
A. The heartbeat messages can be seen using the command diagnose debug authd fsso list.
B. The heartbeat messages can be seen in the collector agent logs.
C. The heartbeat messages can be seen on FortiGate using the real-lime FSSO debug.
D. The heartbeat messages must be manually enabled on FortiGate.
Answer: B,C
Explanation:
According to the official Fortinet documentation (Technical Tip: Useful FSSO Commands), heartbeat messages play a crucial role in communication between the FSSO Collector Agent and FortiGate. These messages are regularly sent from the Collector Agent to verify its status, maintain session awareness, and confirm connectivity between the authentication infrastructure and FortiGate appliances.
Option B is confirmed by Fortinet, as the collector agent logs on Windows or its management console will specifically note heartbeat events, connection status, and any issues maintaining contact with FortiGate units.
Option C is validated by both official CLI documentation and the technical tip linked. On FortiGate, heartbeat messages from the collector agent are visible using real-time debug tools such as diagnose debug application authd or FSSO-specific commands. These enable administrators to monitor live logon states, session status, and connection health directly from the FortiGate CLI. The debug stream shows heartbeats received and their effect on

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active logons, associating health monitoring with active sessions.
Heartbeat operation is fully automated once FSSO is set up-there is no requirement for manual enablement or configuration, aligning with Fortinet's philosophy of seamless integration and centralized management across the Security Fabric. This ensures that both FortiGate and the collector agent can quickly and reliably detect any miscommunication or outage, addressing authentication issues proactively.
References:
Technical Tip: Useful FSSO Commands (Fortinet Community)
FortiOS Administration Guide: FSSO, Collector Agent, Heartbeat, CLI Debug

QUESTION NO: 46
Exhibit.

```
ike 0: comes 10.0.0.2:500->10.0.0.1:500,ifindex=7.
ike 0: IKEv1 exchange=Aggressive id=a2fbd6bb6394401a/06b89c022d4df682 1em=426
ike 0: Remotesite:3: initiator: aggressive mode get 1st response.
ike 0: Remotesite:3: VID DD AFCAD71368A1F1C96B8696FC77570100
ike 0: Remotesite:3: DPD negotiated                    FC77570100
ike 0: Remotesite:3: VID FORTIGATE 8299031757A3608
ike 0: Remotesite:3: peer is Fortifate/Fartios, (v2C6A621DE00000000
ike 0: Remotesite:3: VID FRAGMENTATION 4048B7D56EB0 bo)
ike 0: Remotesite:3: VID FRAGMENTATION 4048B7D56EBCE88525E7DE7F00D6C2D3
ike 0: Remotesite:3: received peer identifier FQDNCE88525E7DE7F00D6C2D3C0000000
ike 0: Remotesite:3: negotiation result            'remote"
ike 0: Remotesite:3: proposal id =1:
ike 0: Remotesite:3:       protocol id = ISAKMP:
ike 0: Remotesite:3:           trans id = KEY IKE.
ike 0: Remotesite:3:           encapsulation = IKE/
ike 0: Remotesite:3:               type=OAKLEY _ENCI none
ike 0: Remotesite:3:               type=OAKLEY HASH RYPT _ALG, val=AES CBC, key-len=128
ike 0: Remotesite:3:           type=AUTH METHOD, va ALG, val=SHA.
ike 0: Remotesite:3:           type=OAKLEY _GROUP, l=PRESHARED KEY.
ike 0: Remotesite:3: ISAKMP SA lifetime=86400      val=MODP1024.
ike 0: Remotesite:3: NAT-T unavailable
ike 0: Remotesite:3: ISAKMP SA a2fbd6bb6394401a/06
ike 0: Remotesite:3: ISAKMP SA a2fbd6bb6394401a/06b89c022d4df682 key 16:39915120ED73E520787C801DE3678916
ike 0: Remotesite:3: PSK authentication succeeded
ike 0: Remotesite:3: authentication OK
ike 0: Remotesite:3: add INITIAL-CONTACT
ike 0: Remotesite:3: enc A2FBD6BB6394401A06B89C022D4DF68208100401000000000000000500B000018882A07809026CA8B2
ike 0: Remotesite:3: out A2FBD6BB6394401A06B89C022D4DF68208100401000000000000005C64D5CBA90B873F150CB8B5CCZA
ike 0: Remotesite:3: sent IKE msg (agg i2send): 10.0.0.1:500->10.0.0.2:500, len=140, id=a2fbd6bb6394401a/
ike 0: Remotesite:3: established IKE SA a2fbd6bb6394401a/06689c022d4df682
```

Refer to the exhibit, which contains partial output from an IKE real-time debug.
Which two statements about this debug output are correct? (Choose two.)

A. Perfect Forward Secrecy (PFS) is enabled in the configuration.

B. The local gateway IP address is 10.0.0.1.

C. It shows a phase 2 negotiation.

D. The initiator provided remote as its IPsec peer ID.

Answer: C,D

Explanation:
From the exhibit, you can observe that the debug output captures an IKEv1 negotiation in aggressive mode. Let's break down the supporting details in line with official Fortinet IPsec VPN troubleshooting resources and debug guides:
For Option B:
The very first line of the debug output shows:
comes 10.0.0.2:500->10.0.0.1:500, ifindex=7.
This indicates the traffic direction-from the remote IP (10.0.0.2) with port 500 to the local IP (10.0.0.1) with port 500. According to Fortinet's documentation, the right side of the arrow

always represents the local FortiGate gateway. Thus, 10.0.0.1 is the local gateway IP address.
For Option D:
You see the statement:
negotiation result "remote"
and
received peer identifier FQDNCE88525E7DE7F00D6C2D3C00000000
Official debug documentation describes that the "peer identifier" or peer ID sent by the initiator is displayed here. In the context of IKE/IPsec negotiation, this value is used as the IPsec peer ID for authentication and identification purposes. The initiator is providing "remote" as the peer ID for its connection.
Why Not A or C:
Perfect Forward Secrecy (PFS): The debug does not show any DH group negotiation in phase 2 (no reference to group2, group5, etc., for phase 2), so you cannot deduce the presence of PFS solely from this output.
Phase 2 negotiation: The log focuses on IKE (phase 1) negotiation and establishment; there's no reference to ESP protocol, Quick Mode, or other identifiers that would show phase 2 SA negotiation and establishment.
This interpretation aligns with the explanation in the FortiOS 7.6.4 Administration Guide's VPN section and the official debug command output samples published in Fortinet's documentation. It demonstrates how to distinguish between local and remote addresses and how to identify the use of peer IDs.
References:
FortiOS 7.6.4 Administration Guide: IPsec VPN and Debugging VPNs
Technical Support Resources on interpreting IKE debug output and peer ID roles

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QUESTION NO: 47
Refer to the exhibit, which shows a truncated output of a real-time LDAP debug.

```
# diagnose debug application fnbamd -1
# diagnose debug enable
    fnbamd_fsm.c[1274] handle_req-Rcvd auth req 8781845 for jsmith in Lab opt=27 prot=0
    fnbamd_ldap.c[637] resolve_ldap_FQDN-Resolved address 10.10.181.10, result 10.10.181.10
    fnbamd_ldap.c[232] start_search_dn-base:'DC=TAC,DC=ottawa,DC=fortinet,DC=com' filter:sAMAccountName=jsmith
    fnbamd_ldap.c[1351] fnbamd_ldap_get_result-Going to SEARCH state
    fnbamd_fsm.c[1833] poll_ldap_servers-Continue pending for req 8781845
    fnbamd_ldap.c[266] get_all_dn-Found DN 1:CN=John Smith,CN=Users,DC=TAC,DC=ottawa,DC=fortinet,DC=com
```

What two conclusions can you draw from the output? (Choose two.)
A. The name of the configured LDAP server is Lab.
B. The user is authenticating using CN=John Smith.
C. FortiOS is able to locate the user in step 3 (Bind Request) of the LDAP authentication process.
D. FortiOS is performing the second step (Search Request) in the LDAP authentication process.
Answer: A,D
Explanation:
The exhibit shows these key lines:
handle_req-Rcvd auth req ... for jsmith in Lab

start_search_dn-base:'DC=TAC,DC=ottawa,DC=fortinet,DC=com' filter:sAMAccountName=jsmith get_all_dn-Found DN 1:CN=John Smith,CN=Users,DC=TAC,DC=ottawa,DC=fortinet,DC=com The study guide explicitly shows the same LDAP real-time debug pattern and says the request line includes the LDAP server object name:
handle_req-Rcvd auth req ... for jsmith in Lab ...
That supports A: Lab is the configured LDAP server name being used for this authentication request.
For the LDAP flow stage, the study guide states:
"An fnbamd_ldap_build_dn_search_req-base message indicates that FortiGate is performing step two: searching for the user in the LDAP tree." It also says that if the LDAP server finds the user, the output shows the user's full DN.
That matches the exhibit's start_search_dn-base ... filter:sAMAccountName=jsmith and Found DN ... CN=John Smith... lines, so D is correct.
Why the other options are wrong:
B is wrong because the exhibit shows FortiOS has found the user DN CN=John Smith,..., but that does not mean the user is already authenticating with that DN in this step. The study guide says this DN is discovered in step 2, and only in step 3 does FortiGate bind using the user DN.
C is wrong because the exhibit is showing step 2 (Search Request), not step 3 (Bind Request). The study guide separates these steps clearly and shows step 3 with fnbamd_ldap_build_userbind_req-Trying DN ... and __ldap_build_bind_req-Binding to 'CN=John Smith,...'

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QUESTION NO: 48
Refer to the exhibit, which shows a session entry.

```
session_info: proto=1 proto_state=00 duration=1 expire=59 timeout=0 flags=00000000
sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=log may_dirty none
statistic (bytes/packets/allow_err): org=168/2/1 reply=168/2/1 tuples=2
tx speed (Bps/kbps) : 97/0 rx speed (Bps/kbps) : 97/0
orgin->sink: org pre->post, reply pre->post dev=9->3/3->9 gwy=10.200.1.254/10.1.0.1
hook=post dir=org act=snat 10.1.10.10:40602->10.200.5.1:8 (10.200.1.1:60430)
hook=pre dir=reply act=dnat 10.200.5.1:60430->10.200.1.1:0(10.1.10.10:40602)
misc=0 policy_id=1 auth_info=0 chk_client_info=0 vd=0
serial=0002a5c9 tos=ff/ff app_list=0 app=0 url_cat=0
dd_type=0 dd_mode=0
```

Which statement about this session is true?
A. Return traffic to the initiator is sent to 10.1.0.1.
B. Return traffic to the initiator is sent lo 10.200.1.254.
C. It is an ICMP session from 10.1.10.10 to 10.200.1.1.
D. It is an ICMP session from 10.1.10.1 to 10.200.5.1.
Answer: B
Explanation:
The session output reveals a session with proto=1 (ICMP) and the origin and reply directions

show address and NAT translations. Specifically, the hook=post dir=org act=snat shows that source NAT is performed for outgoing packets, where the source 10.1.10.10:40602 is translated to 10.200.5.1:8 (likely ICMP id 8, not a TCP/UDP port). The reply direction, hook=pre dir=reply act=dnat, indicates destination NAT for incoming packets: packets incoming for 10.200.5.1:60430 are destination-NATed to 10.1.10.10:40602. The gateway (gwy) is listed as 10.200.1.254/10.1.0.1, which for outgoing traffic means that return traffic is directed to the gateway (10.200.1.254), per the NAT policy. This is confirmed by the FortiOS Session Table Guide, which explains that the returned ICMP reply will be routed out to this NAT gateway. The session statistics and logical flow (SNAT out, matching DNAT in) reinforce that reply traffic to the initiator traverses via 10.200.1.254.
References:
FortiOS Administration Guide: Session Table, NAT, and Route Interaction Fortinet Technical Note: Diagnose sys session list, Direction and NAT Analysis

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QUESTION NO: 49
Which statement about parallel path processing is correct (PPP)?
A. PPP chooses from a group of parallel options lo identity the optimal path tor processing a packet.
B. Only FortiGate hardware configurations affect the path that a packet takes.
C. PPP does not apply to packets that are part of an already established session.
D. Software configuration has no impact on PPP.
Answer: A
Explanation:
Parallel Path Processing (PPP) in FortiOS refers to the system's ability to evaluate and select among multiple processing paths-often involving dedicated network processors, content processors, or CPU-based workflows-to optimally process packets. The official documentation highlights that the PPP engine dynamically selects which hardware or software path to use for each session based on session characteristics, policy configuration, and traffic type. This dynamic selection results in optimal throughput and resource utilization. The document specifies that PPP assesses several processing paths in parallel, using decision logic to determine whether a session should be offloaded to specialist hardware (like NP6, CP9, etc.) or stay in the CPU path, ensuring that each packet is handled by the most efficient available method under current load and policy. Hardware and software configurations both influence this outcome, but it is the PPP engine's decision-making that defines the optimal path per session.
References:
Fortinet FortiGate Handbook: Parallel Path Processing
Fortinet FortiOS Technical Documentation: Packet Flow and Path Selection

QUESTION NO: 50
In IKEv2, which exchange establishes the first CHILD_SA?
A. IKE_SA_INIT
B. INFORMATIONAL
C. CREATE_CHILD_SA
D. IKE_AUTH

Answer: D
Explanation:
The correct answer is D. IKE_AUTH.
The study guide explicitly states:
"IKE_Auth exchange:
* Performs the mutual authentication of two IKE endpoints.
* Configures settings like IP/mask, DNS, and so on.
* Sets up the piggyback of a child SA. Negotiates IP flow and security settings for the IPsec SA." It also says:
"By default, a piggyback child (IPsec) SA is negotiated along with the IKEv2 SA during IKE_AUTH. If additional IPsec SAs are needed ... they are negotiated during subsequent CREATE_CHILD_SA exchanges." Why the other options are wrong:
A . IKE_SA_INIT is incorrect because this exchange negotiates the security settings to protect the IKE traffic, not the first CHILD_SA.
B . INFORMATIONAL is incorrect because it is used to convey control messages between IKE endpoints.
C . CREATE_CHILD_SA is incorrect for the first CHILD_SA, because it is used to create new additional child SAs or rekey existing ones after the initial exchange.

QUESTION NO: 51
Which authentication option can you not configure under config user radius on FortiOS?
A. mschap
B. pap
C. mschap2
D. eap
Answer: D
Explanation:
According to the official Fortinet administration guide for FortiOS 7.6.4 under the section "Configuring a RADIUS server," the supported RADIUS authentication methods you can configure via the CLI with config user radius are:
pap
chap
mschap
mschapv2
auto
The relevant CLI syntax is set auth-type {auto | ms_chap_v2 | ms_chap | chap | pap}. You can confirm this directly in the configuration table and from real CLI sessions.
EAP (Extensible Authentication Protocol) is NOT an authentication option you can directly set under config user radius. EAP methods (such as EAP-TLS, EAP-PEAP, EAP-TTLS) are negotiated between the RADIUS client and server but are not configurable as an explicit auth-type option in FortiOS. EAP authentication is typically used automatically by features like 802.1X, not through the user radius object authentication-type setting, and always requires proper backend workings between supplicant and RADIUS server

QUESTION NO: 52

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Exhibit.

```
# diagnose hardware sysinfo memory
MemTotal:          2055916 kB
MemFree:            708880 kB
Buffers:             22140 kB
Cached:             641364 kB
SwapCached:              0 kB
Active:             726352 kB
Inactive:            98908 kB
```

Refer to the exhibit, which shows a partial output of diagnose hardware aysinfo memory. Which two statements about the output are true? (Choose two.)

A. There are 98908 kB of memory that will never be used.

B. The user space has 708880 kB of physical memory that is not used by the system.

C. The I/O cache, which has 641364 kB of memory allocated to it.

D. The value indicated next to the inactive heading represents the currently unused cache page.

Answer: B,D

Explanation:

The partial output from diagnose hardware sysinfo memory provides details on system RAM allocation. According to Fortinet's technical documentation for memory troubleshooting and Linux memory management (which FortiOS is based on):

MemFree is the portion of physical memory not currently allocated to any running process or kernel function. Thus, 708880 kB is available and can be immediately used by user-space programs or system operations.

Inactive refers to pages in the memory cache that were previously in use for I/O or file system buffering but are now not actively referenced. These pages are retained in memory for quick access if needed again, but can be reclaimed for other memory operations if demand increases. The value 98908 kB here represents currently unused cache pages (inactive pages), ready for repurposing or deletion if the system requires more RAM.

Cached represents the total amount of system memory allocated to cache, which includes both active and inactive cache pages. It does not, by itself, represent I/O cache exclusively, nor does "inactive" mean memory "will never be used" as the kernel can re-purpose inactive pages on demand.

References:

Fortinet Technical Tip: Explaining the 'diagnose hard sysinfo memory' command FortiOS System Administration Guide: Linux Memory Reporting, Cached and Inactive Statistics

QUESTION NO: 53

Exhibit.

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```
NGFW-1 # get sys ha status
HA Health Status: OK
Model: FortiGate-VM64
Mode: HA A-P
Group: 0
Debug: 0
Cluster Uptime: 0 days 0:1:25
Cluster state change time: 2023-04-18 12:07:47
Primary selected using:
   <2023/04/18 12:07:47> FGVM010000077649 is selected as the primary because its override priority is larger than peer member
FGVM010000077650.
ses_pickup: disable
override: disable
Configuration Status:
    FGVM010000077649(updated 4 seconds ago): in-sync
    FGVM010000077650(updated 1 seconds ago): out-of-sync
System Usage stats:
    FGVM010000077649(updated 4 seconds ago):
        sessions=166, average-cpu-user/nice/system/idle=1%/0%/0%/99%, memory=45%
    FGVM010000077650(updated 1 seconds ago):
        sessions=3, average-cpu-user/nice/system/idle=0%/0%/0%/100%, memory=44%
HBDEV stats:
    FGVM010000077649(updated 4 seconds ago):
        port7: physical/1000auto, up, rx-bytes/packets/dropped/errors=167663/567/0/0, tx=262623/656/0/0
    FGVM010000077650(updated 1 seconds ago):
        port7: physical/1000auto, up, rx-bytes/packets/dropped/errors=271373/680/0/0, tx=176013/592/0/0
Primary     : NGFW-1          , FGVM010000077649, HA cluster index = 1
Secondary   : NGFW-2          , FGVM010000077650, HA cluster index = 0
number of vcluster: 1
vcluster 1: work 169.254.0.2
Primary: FGVM010000077649, HA operating index = 0
Secondary: FGVM010000077650, HA operating index = 1
```

Refer to the exhibit, which shows the output of get system ha status.
NGFW-1 and NGFW-2 have been up for a week.
Which two statements about the output are true? (Choose two.)

A. If a configuration change is made to the primary FortiGate at this time, the secondary will initiate a synchronization reset.

B. If port 7 becomes disconnected on the secondary, both FortiGate devices will elect itself as primary.

C. If FGVM...649 is rebooted. FGVM...650 will become the primary and retain that role, even after FGVM...649 rejoins the cluster.

D. If no action is taken, the primary FortiGate will leave the cluster because of the current sync status.

Answer: B,C

Explanation:
FortiGate HA Troubleshooting and Synchronization Guides
Fortinet Admin Guide: HA Primary Role Retention, Cluster Break-up Due to Out-of-Sync Status

QUESTION NO: 54
Exhibit.

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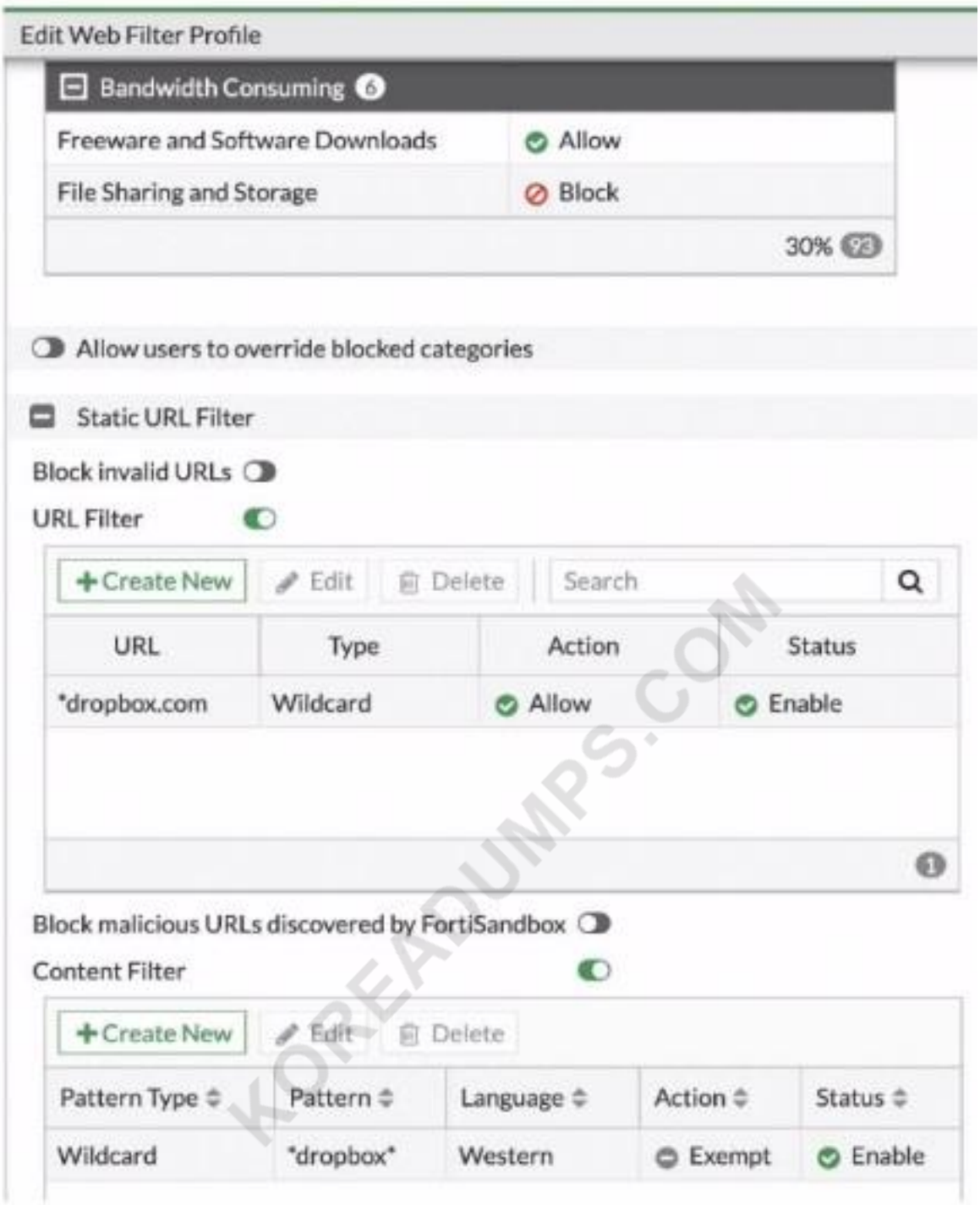

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Refer to the exhibit, which shows a partial web fillet profile configuration.
Which action does FortiGate lake if a user attempts to access www. dropbox. com, which is categorized as File Sharing and Storage?

A. FortiGate allows the connection, based on the URL Filter configuration.

B. FortiGate blocks the connection as an invalid URL.

C. FortiGate exempts the connection, based on the Web Content Filter configuration.

D. FortiGate blocks the connection, based on the FortiGuard category based filter configuration.

Answer: D

Explanation:

https://community.fortinet.com/t5/FortiGate/Technical-Tip-FortiGate-Static-URL-filter-actions-explained/ta-p/206632

QUESTION NO: 55

Refer to the exhibit, which shows the omitted output of a session table entry.

```
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 pol_uuid_idx=14720 confiauth_info=0 chk_client_info=0 vd=0
serial=0002932f tos=ff/ff app_list=2000 app=34050 url_cat=0
sdwan_mbr_seq=1 sdwan_service_id=1
rpdb_link_id=80000000 ngfwid=n/a
npu_state=0x003c94 ips_offload
npu_info: flag=0x81/0x81, offload=8/8, ips_offload=1/1, epid=16/16, ipid=64/88, vlan=0x0000/0x0000
vlifid=64/88, vtag_in=0x0000/0x0000 in_npu=1/1, out_npu=1/1, fwd_en=0/0, qid=0/0
```

Which two statements are true? (Choose two.)

A. The traffic has been tagged for VLAN 0000.

B. NP7 is handling offloading of this session.

C. The traffic matches Policy ID 1.

D. The session has been offloaded.

Answer: C,D

Explanation:

In the provided session table output, the following details justify the answers:

Policy ID Match: The line policy_id=1 directly confirms that this session was matched by Firewall Policy ID 1. According to Fortinet's session table documentation, the policy_id field always references the policy that allowed this session, so this is a clear indicator.

Session Offloading: The presence of the strings npu_state, ips_offload, and notably the NPU info section such as offload=8/8, ips_offload=1/1 shows that this session has been offloaded to the Network Processor Unit (NPU). Fortinet technical documentation states that "offload" values greater than zero in both directions (and an NPU info section) affirm that NPU hardware processing (fast path) is handling this traffic, thus the session is not being handled in software only.

Other options:

VLAN Tagging (vlan=0x0000/0x0000): This means no VLAN tag is assigned to this session.

NP7: The actual NPU model handling the session isn't exposed in this snippet-the offload parameters shown are generic and not specific to NP7 hardware, so it cannot be concluded from the session data.

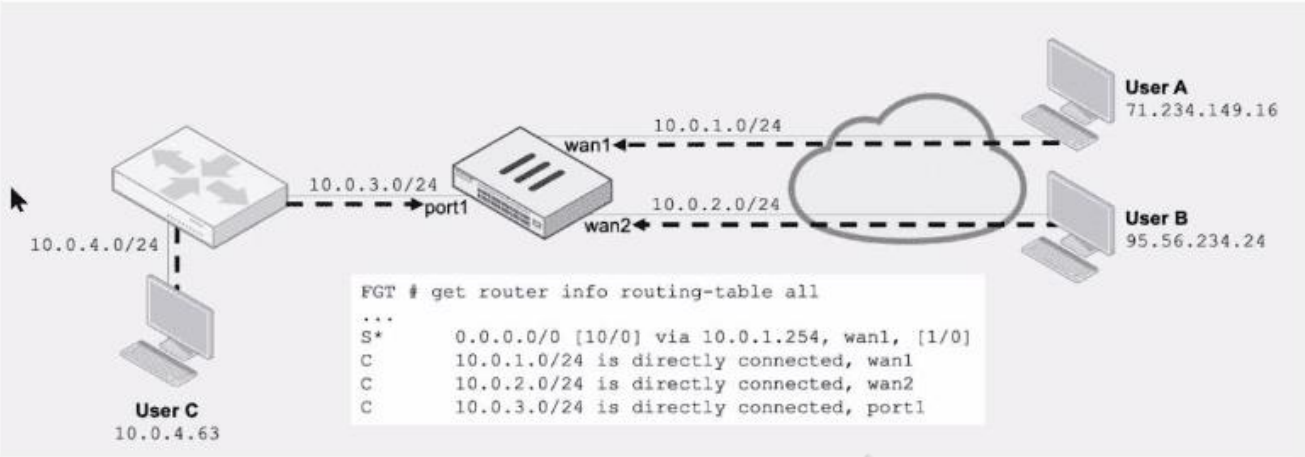
References:

Fortinet Technical Tip: FortiGate Session Table and NPU Offloading

FortiOS Diagnostics Guide: Policy ID, Offload, and VLAN Session Table Fields

QUESTION NO: 56

Refer to the exhibit.



Assuming a default configuration, which three statements are true? (Choose three.)

A. Strict RPF is enabled by default.

B. User B: Fail. There is no route to 95.56.234.24 using wan2 in the routing table.

C. User A: Pass. The default static route through wan1 passes the RPF check regardless of the source IP address.

D. User B: Pass. FortiGate will use asymmetric routing using wan1 to reply to traffic for 95.56.234.24.

E. User C: Fail. There is no route to 10.0.4.63 using port1 in the touting table.

Answer: B,C,E

Explanation:

References:

Fortinet Technical Note: RPF Default Configuration and Routing Table Matching FortiGate Administration Guide: Routing and Asymmetric Routing Controls Community Knowledgebase: Route Lookups and RPF Enforcement on FortiOS

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QUESTION NO: 57

Exhibit.

```
FGT # diagnose debug rating
Locale       : english

Service      : Web-filter
Status       : Enable
License      : Contract

Service      : Antispam
Status       : Disable

Service      : Virus Outbreak Prevention
Status       : Disable

Num. of servers : 1
Protocol        : https
Port            : 443
Anycast         : Enable
Default servers : Included

-=- Server List (Mon May  1 03:47:52 2023) -=-

IP                                        Weight    RTT Flags   TZ   FortiGuard-requests  Curr Lost Total Lost            Updated Time
64.26.151.37                                  10     45         -5                262432          0        846 Mon May  1 03:47:43 2023
64.26.151.35                                  10     46         -5                329072          0       6806 Mon May  1 03:47:43 2023
66.117.56.37                                  10     75         -5                 71638          0        275 Mon May  1 03:47:43 2023
65.210.95.240                                 20     71         -8                 36875          0         92 Mon May  1 03:47:43 2023
209.22.147.36                                 20     103 DI     -8                 34784          0       1070 Mon May  1 03:47:43 2023
208.91.112.194                                20     107 D      -8                 35170          0       1533 Mon May  1 03:47:43 2023
                                                                  0                 33728          0        120 Mon May  1 03:47:43 2023
                                                                  1                 33797          0        192 Mon May  1 03:47:43 2023
                                                                  9                 33754          0        145 Mon May  1 03:47:43 2023
                                                                 -5                 26410      26226      26227 Mon May  1 03:47:43 2023
```

Refer to the exhibit, which shows the output of a diagnose command.

What can you conclude about the debug output in this scenario?

A. The first server provided to FortiGate when it performed a DNS query looking for a list of rating servers, was 121.111.236.179.

B. There is a natural correlation between the value in the FortiGuard-requests field and the value in the Weight field.

C. FortiGate used 64.26.151.37 as the initial server to validate its contract.

D. Servers with a negative TZ value are less preferred for rating requests.

Answer: C

Explanation:

The exhibit displays the output from the diagnose debug rating command on a FortiGate device. This command is used to display information about FortiGuard Web Filtering or other security-related queries performed by FortiGate to FortiGuard servers. Official Fortinet documentation outlines the meaning of each field in the server list. The FortiGate maintains a list of available FortiGuard servers, selecting the optimal server based on factors such as weight, round-trip time (RTT), and regional settings.

The very first entry in the server list after "Server List" is the server FortiGate initially uses, prioritized by factors such as proximity and RTT. Here, 64.26.151.37 is listed first, and the FortiGuard-requests value confirms that this server handled the highest number of requests.

The IPs, weights, and lost/failed counters are monitored for server performance and selection over time. FortiGate's default operational logic is to try the first entry for contract validation and use the next in the list if the first is unavailable or has high latency or packet loss.

There is no direct correlation between the Weight and the number of FortiGuard-requests. The servers with higher or lower weights may still handle different request volumes based on availability and performance.

The TZ (time zone) value's sign (positive or negative) does not affect server preference; it is informational, showing the server's location relative to UTC, not a rating metric.

DNS query results for FortiGuard servers are not shown here, and the provided servers are not returned in DNS query order.

This command and interpretation are detailed in the FortiOS Administration Guide's section describing FortiGuard server selection and contract validation processes.

References:

FortiOS Administration Guide: FortiGuard Service Connectivity and Debugging Official Technical Notes on diagnose debug rating output structure

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QUESTION NO: 58

Which two statements about Security Fabric communications are true? (Choose two.)

A. FortiTelemetry and Neighbor Discovery both operate using TCP.

B. The default port for Neighbor Discovery can be modified.

C. FortiTelemetry must be manually enabled on the FortiGate interface.

D. By default, the downstream FortiGate establishes a connection with the upstream FortiGate using TCP port 8013.

Answer: C,D

Explanation:

FortiTelemetry is a critical part of Security Fabric communications and requires explicit configuration for each participating FortiGate interface. The administrative access setting

"fabric" (corresponding to FortiTelemetry) must be manually enabled per interface on both upstream and downstream devices. This is performed in the GUI under Administrative Access or via the CLI using the command set allowaccess fabric for the relevant network interface. Without this step, FortiTelemetry communications will not occur on that interface. Additionally, the default communication between downstream and upstream FortiGate units in the Security Fabric is over TCP port 8013. This port is well-documented as the standard for Security Fabric and FortiTelemetry connections, and must be open and permitted across the network path for connectivity and status enforcement between units. The downstream FortiGate initiates the connection to the upstream via this port unless otherwise configured. This has also been documented as a PCI-relevant port, showing its default usage.
Other options:
Neighbor Discovery in FortiOS uses IPv6 ND protocol, not TCP.
FortiTelemetry port (8013) can be modified, but the interface Administrative Access for the Security Fabric must be manually enabled; Neighbor Discovery port modification is not documented as a supported change for FortiGate.
References:
FortiGate/FortiOS Administration Guide: Enabling FortiTelemetry (fabric) on interfaces
Fortinet Technical Tip: FortiTelemetry uses TCP port 8013 by default PCI compliance documentation on port 8013 usage for Security Fabric Fortinet Security Fabric setup procedures and interface options

QUESTION NO: 59
Refer to the exhibit, which contains the output of diagnose vpn tunnel list.

```
# diagnose vpn tunnel list
name=DialUp_0 ver=1 serial=4 10.200.1.1:4500->10.200.3.2:64916 tun_id=10.200.3.2 dst_mtu=1500 dpd-link=on remote_location=0.0.0.0 weight=1
bound_if=3 lgwy=static/1 tun=intf/0 mode=dial_inst/3 encap=none/896 options[0380]=rgwy-chg rport-chg frag-rfc  run_state=0 accept_traffic=1 overlay_id=0
parent=DialUp index=0
proxyid_num=1 child_num=0 refcnt=5 ilast=0 olast=0 ad=/0
stat: rxp=221 txp=0 rxb=35360 txb=0
dpd: mode=active on=1 idle=5000ms retry=3 count=0 seqno=70
natt: mode=silent draft=32 interval=10 remote_port=64916
proxyid=DialUp proto=0 sa=1 ref=2 serial=3 add-route
  dst: 0:0.0.0.0-255.255.255.255:0
  src: 0:10.0.10.10-10.0.10.10:0
  SA:  ref=3 options=82 type=00 soft=0 mtu=1422 expire=43065/0B replaywin=2048
       seqno=1 esn=0 replaywin_lastseq=00000079 itn=0 qat=0 hash_search_len=1
  life: type=01 bytes=0/0 timeout=43188/43200
  dec: spi=5ed4aafc esp=aes key=16 054852d43abb0e931641b4e8878dd9ce
       ah=sha1 key=20 082eafd018bf7d4d7b65d9c5b7448db5cc01f81d
  enc: spi=69d4231e esp=aes key=16 d5a23d09ab4128d094ac972f5511f9db
       ah=sha1 key=20 54eac30e29ce711d2ceaab9b5e179c20bb83605e
  dec:pkts/bytes=120/10080, enc:pkts/bytes=0/0
```

Which command will capture ESP traffic for the VPN named DialUp_0?
A. diagnose sniffer packet any 'ip proto 50'
B. diagnose sniffer packet any 'host 10.0.10.10'
C. diagnose sniffer packet any 'esp and host 10.200.3.2'
D. diagnose sniffer packet any 'port 4500'
Answer: D

QUESTION NO: 60
Exhibit.

```
# diagnose automation test HAFailOver
automation test failed(1). stitch:HAFailOver
```

Refer to the exhibit, which shows the output of diagnose automation test.

What can you observe from the output? (Choose two.)

A. The automation stitch test is not being logged.

B. The automation stitch test failed but the HA failover was successful.

C. An HA failover occurred.

D. The test was unsuccessful.

Answer: A,D

QUESTION NO: 61

Refer to the exhibit, which shows partial outputs from two routing debug commands.

```
FortiGate # get router info kernel
tab=254 vf=0 scope=0 type=1 proto=11 prio=0 0.0.0.0/0.0.0.0/0->0.0.0.0/0 pref=0.0.0.0 gwy=100.64.1.254 dev=3 (port1)
tab=254 vf=0 scope=0 type=1 proto=11 prio=10 0.0.0.0/0.0.0.0/0->0.0.0.0/0 pref=0.0.0.0 gwy=100.64.2.254 dev=6 (port2)
tab=254 vf=0 scope=253 type=1 proto=2 prio=0 0.0.0.0/0.0.0.0/0->10.1.0.0/24 pref=10.1.0.254 gwy=0.0.0.0 dev=9 (port3)

FortiGate # get router info routing-table all

Routing table for VRF=0
S*        0.0.0.0/0 [10/0] via 100.64.1.254, port1
                           [10/0] via 100.64.2.254, port2, [10/0]
C         10.1.0.0/24 is directly connected, port3
S         10.1.10.0/24 [10/0] via 10.1.0.1, port3
C         100.64.1.0/24 is directly connected, port1
C         100.64.2.0/24 is directly connected, port2
```

Which change must an administrator make on FortiGate to route web traffic from internal users to the internet, using ECMP?

A. Set snat-route-change to enable.

B. Set the priority of the static default route using port2 to 1.

C. Set preserve-session-route to enable.

D. Set the priority of the static default route using port1 to 10.

Answer: D

Explanation:

The 7.6 study guide explains the route selection order:

"Route Selection Process

Most specific route

Lowest distance

Lowest metric (dynamic routes)

Lowest priority (static routes)

ECMP (static, BGP, and OSPF routes)"**

It then states:

"If there are multiple routes with the same netmask, distance, metric, and priority, FortiGate shares the traffic among all of them. This is called equal-cost multi-path (ECMP)." The FortiOS administration guide confirms the ECMP prerequisite:

"Routes must have the same destination and costs. In the case of static routes costs include distance and priority." In the exhibit, the kernel/FIB output shows the two default routes as:

gwy=100.64.1.254 dev=3 (port1) prio=0

gwy=100.64.2.254 dev=6 (port2) prio=10

So although both are default routes, their priorities are different. Since FortiGate uses the FIB/kernel for forwarding traffic, ECMP will not happen until the static-route priorities are the same. The study guide also notes that the FIB is the table used to perform standard routing Therefore, to make the two default routes eligible for ECMP, the administrator must make the

priorities equal. Since port2 is already 10, the needed change is to set the port1 default route priority to 10.
Why the other options are wrong:
A is wrong because snat-route-change affects how existing SNAT sessions react to routing changes, not whether static routes qualify for ECMP B is wrong because changing port2 to priority 1 still would not match port1 at 0, so the routes still would not have equal cost for ECMP C is wrong because preserve-session-route affects existing-session route persistence after routing changes, not ECMP qualification

QUESTION NO: 62
What are two reasons you might see iprope_in_check() check failed, drop when using the debug flow? (Choose two.)
A. Packet was dropped because of policy route misconfiguration.
B. Packet was dropped because of traffic shaping.
C. Trusted host list misconfiguration.
D. VIP or IP pool misconfiguration.
Answer: C,D
Explanation:
The Network Security Support Engineer 7.6 Study Guide explicitly explains this debug message:
"iprope_in_check() check failed, drop" means the packet is destined to a FortiGate IP address and one of these conditions applies:
The service is not enabled
The service is using a different TCP port
The source IP address is not included in the trusted host list
The packet matches a local-in policy with action deny
That directly confirms C. Trusted host list misconfiguration.
Why D is the second valid choice:
The FortiOS administration guide explains that:
"IP pools and VIPs are considered local IP addresses if responding to ARP requests on these external IP addresses is enabled ... the FortiGate is considered a destination for those IP addresses ... once an IP pool or VIP has been configured ... the FortiGate considers it as a local address and will not forward traffic based on the routing table." Because iprope_in_check() is a local-in/local-destination type failure, a VIP or IP pool misconfiguration can cause traffic to be treated as destined for the FortiGate itself, which can then trigger this drop condition if the matching local service/local-in handling is not valid. So D is the closest supported second answer from the available choices.
Why the other options are wrong:
A is wrong because policy route problems are not the documented meaning of this specific debug message. The study guide instead ties iprope_in_check() check failed, drop to management/local-in conditions.
B is wrong because the study guide says traffic shaping drops appear as: "Denied by quota check"

QUESTION NO: 63

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Exhibit.

```
session info: proto=6 proto_state=01 duration=157 expire=3559 timeout=3600 flags=00000000 socktype=0 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=0 policy_dir=0 tunnel=/ vlan_cos=0/255
user=User1 state=log may_dirty authed f00 acct-ext
statistic(bytes/packets/allow_err): org=2137/14/1 reply=1663/12/1 tuples=2
tx speed(Bps/kbps): 1/0 rx speed(Bps/kbps): 1/0
orgin->sink: org pre->post, reply pre->post dev=5->3/3->5 gwy=10.1.0.254/10.1.10.1
hook=pre dir=org act=noop 10.1.10.1:34830->35.241.9.150:443(0.0.0.0:0)
hook=post dir=reply act=noop 35.241.9.150:443->10.1.10.1:34830(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 pol_uuid_idx=14735 auth_info=2 chk_client_info=0 vd=0
serial=0000352e tos=ff/ff app_list=0 app=0 url_cat=0
rpdb_link_id=00000000 ngfwid=n/anpu_state=0x000100
no_ofld_reason:  npu-flag-off
```

Refer to the exhibit, which shows the output of a session. Which two statements are true? (Choose lwo.)

A. The TCP session has been successfully established.

B. The session was initiated from an authenticated user.

C. The session is being inspected using flow inspection.

D. The session is being offloaded.

Answer: A,B

QUESTION NO: 64

Refer to the exhibit, which shows the output of get router info ospf neighbor.

```
Spoke1 # get router info ospf neighbor

OSPF process 0, VRF 0:
Neighbor ID     Pri   State           Dead Time   Address      Interface
0.0.0.1           1   Full/DR         00:00:39    10.10.2.1    wan1
0.0.0.3           1   Full/DROther    00:00:37    10.10.3.2    wan2
0.0.0.10         c1   Full/ -         00:00:36    172.16.1.2   ToHub
```

What can you conclude from the command output?

A. The network type connecting the local Fortigate and OSPF neighbor 0.0.0.10 is point-to-point.

B. All neighbors are in area 0.0.0.0.

C. The local FortiGate is the BDR.

D. The local FortiGate is not a DROther.

Answer: A

QUESTION NO: 65

Exhibit.

```
… name_ip_match: failed to connect to workstation: <Workstation Name> (192.168.1.1)
… failed to connect to registry: WORKSTATION02 (192.168.12.232)
```

Refer to the exhibit, which shows two entries that were generated in the FSSO collector agent logs.

What three conclusions can you draw from these log entries? {Choose three.)

A. Remote registry is not running on the workstation.

B. The user's status shows as "not verified" in the collector agent.

C. DNS resolution is unable to resolve the workstation name.
D. The FortiGate firmware version is not compatible with that of the collector agent.
E. A firewall is blocking traffic to port 139 and 445.
Answer: A,B,E

QUESTION NO: 66
Which statement about protocol options is true?
A. Protocol options allow administrators to configure a maximum number of sessions for each configured protocol.
B. Protocol options give administrators a streamlined method to instruct FortiGate to block all sessions corresponding to disabled protocols.
C. Protocol options allow administrators to configure the Any setting for all enabled protocols, which provides the most efficient use of system resources.
D. Protocol options allow administrators to configure which Layer 4 port numbers map to upper-layer protocols, such as HTTP, SMTP, FTP, and so on.
Answer: D

QUESTION NO: 67
Which two statements about conserve mode are true? (Choose two.)
A. FortiGate enters conserve mode when the system memory reaches the configured extreme threshold.
B. FortiGate starts taking the configured action for new sessions requiring content inspection when the system memory reaches the configured red threshold.
C. FortiGate exits conserve mode when the system memory goes below the configured green threshold.
D. FortiGate starts dropping all new sessions when the system memory reaches the configured red threshold.
Answer: B,C

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QUESTION NO: 68
Refer to the exhibit, which shows the output of a policy route table entry.

```
id=2113929223 static_route=7 dscp_tag=0xff 0xff flags=0x0 tos=0x00 tos_mask=0x00 protocol=0 sport=0-0 iif=0 dport=1-65535 path(1) oif=3(port1) gwy=192.2.0.2
source wildcard(1): 0.0.0.0/0.0.0.0
destination wildcard(1): 0.0.0.0/0.0.0.0
internet service(1): Fortinet-FortiGuard(1245324,0,0,0)
hit_count=0 last_used=2022-02-23 06:39:07
```

Which type of policy route does the output show?
A. An ISDB route
B. A regular policy route
C. A regular policy route, which is associated with an active static route in the FIB
D. An SD-WAN rule
Answer: A
Explanation:
The exhibit for question 4 shows a policy route table entry, and key fields are as follows:
internet service(1) : Fortinet-FortiGuard(1245324,0.0.0.0,0.0.0.0)
According to the Fortinet official documentation, when a policy route is based on Internet

Service Database (ISDB) entries, the route entry will specifically mention "internet service," showing the service being referenced (in this example, Fortinet-FortiGuard). This is fundamentally different from a regular policy route, which is defined by source, destination, and service wildcards without referencing an ISDB signature. A regular policy route's output would not contain the line "internet service." Policy routes that use ISDB allow FortiGate to steer traffic for specific well-known services (like FortiGuard, Google, Microsoft) based on traffic pattern recognition, even if the destination IP is dynamic. The matching and route selection follow the ISDB tag and can coexist with static or regular policy routes.
Thus, this entry is correctly and uniquely an ISDB route, as explained in the FortiOS policy routing documentation and ISDB configuration references.
References:
FortiOS Administration Guide: Policy Routing, ISDB integration and interpretation of route table entries ISDB-based Routing and Official CLI Outputs in Fortinet's documentation

QUESTION NO: 69
Refer to the exhibit, which contains partial output from an IKE real-time debug.

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Debug output

```
ike 0:624000:98: responder: main mode get 1st message...
ike 0:624000:98: VID DPD AFCAD71368A1F1C96B8696FC77570100
ike 0:624000:98: VID FRAGMENTATION 4048B7D56EBCE88525E7DE7F00D6C2D3
ike 0:624000:98: VID FRAGMENTATION 4048B7D56EBCE88525E7DE7F00D6C2D3C0000000
ike 0:624000:98: VID FORTIGATE 8299031757A36082C6A621DE00000000
ike 0:624000:98: incoming proposal:
ike 0:624000:98: proposal id = 0:
ike 0:624000:98:    protocol id = ISAKMP:
ike 0:624000:98:         trans_id = KEY_IKE.
ike 0:624000:98:         encapsulation = IKE/none
ike 0:624000:98:              type=OAKLEY_ENCRYPT_ALG, val=AES_CBC, key-len=256
ike 0:624000:98:              type OAKLEY_HASH_ALG, val=SHA2_256.
ike 0:624000:98:              type=AUTH_METHOD, val=PRESHARED_KEY.
ike 0:624000:98:              type=OAKLEY_GROUP, val=MODP2048.
ike 0:624000:98: ISAKMP SA lifetime=86400
ike 0:624000:98: proposal id = 0:
ike 0:624000:98:    protocol id = ISAKMP:
ike 0:624000:98:         trans_id = KEY_IKE.
ike 0:624000:98:         encapsulation = IKE/none
ike 0:624000:98:              type OAKLEY_ENCRYPT_ALG, val=AES_CBC, key-len=256
ike 0:624000:98:              type=OAKLEY_HASH_ALG, val=SHA2_256.
ike 0:624000:98:              type-AUTH_METHOD, val=PRESHARED_KEY.
ike 0:624000:98:              type=OAKLEY_GROUP, val=MODP1536.
ike 0:624000:98: ISAKMP SA lifetime=86400
ike 0:624000:98: my proposal, gw Remotesite:
ike 0:624000:98: proposal id = 1:
ike 0:624000:98:    protocol id - ISAKMP:
ike 0:624000:98:         trans_id = KEY_IKE.
ike 0:624000:98:         encapsulation = IKE/none
ike 0:620000:98:              type=OAKLEY_ENCRYPT_ALG, val=AES_CBC, key-len=128
ike 0:624000:98:              type=OAKLEY_HASH_ALG, val=SHA.
ike 0:624000:98:              type=AUTH_METHOD, val=PRESHARED_KEY.
ike 0:624000:98:              type=OAKLEY_GROUP, val=MODP2048.
ike 0:624000:98: ISAKMP SA lifetime=86400
ike 0:624000:98: proposal id = 1:
ike 0:624000:98:    protocol id = ISAKMP:
ike 0:624000:98:         trans_id = KEY_IKE.
ike 0:624000:98:         encapsulation = IKE/none
ike 0:624000:98:              type=OAKLEY_ENCRYPT_ALG, val=AES_CBC, key-len=128
ike 0:624000:98:              type=OAKLEY_HASH_ALG, val=SHA.
ike 0:624000:98:              type=AUTH_METHOD, val=PRESHARED_KEY.
ike 0:624000:98:              type=OAKLEY_GROUP, val=MODP1536.
ike 0:624000:98: ISAKMP SA lifetime=86400
ike 0:624000:98: negotiation failure
ike Negot:: 624ea7b1bba276fb/0000000000000000:98: no SA proposal chosen
```

The administrator does not have access to the remote gateway.

Based on the debug output, which configuration change the administrator make to the local gateway to resolve the phase 1 negotiation error?

A. In the phase 1 proposal configuration, add AES256-SHA256 to the list of encryption algorithms.

B. In the phase 1 proposal configuration, add AESCBC-SHA2 to the list of encryption algorithms.

C. In the phase 1 network configuration, set the IKE version to 2.

D. In the phase 1 proposal configuration, add AES128-SHA128 to the list of encryption algorithms.

Answer: A

QUESTION NO: 70
Which three common FortiGate-to-collector-agent connectivity issues can you identify using the FSSO real-time debug? (Choose three.)
A. Log is full on the collector agent.
B. Inability to reach IP address of the collector agent.
C. Refused connection. Potential mismatch of TCP port.
D. Mismatched pre-shared password.
E. Incompatible collector agent software version.
Answer: B,C,D

QUESTION NO: 71
Refer to the exhibit, which shows a partial output from the get router info routing-table database command.

```
# get router info routing-table database
---omitted---

Routing table for VRF=0
S           0.0.0.0/0 [20/0] via 100.64.2.254, port2, [10/0]
S           0.0.0.0/0 [10/0] via 100.64.1.254, port1 inactive, [50/0]
---omitted---
```

The administrator wants to configure a default static route for port3 and assign a distance of 50 and a priority of 0.
What will happen to the port1 and port2 default static routes after the port3 default static route is created?
A. The port2 default static route will be injected into the forwarding information base (FIB).
B. The port1 default static route will be injected into the FIB.
C. Neither of the routes shown in the output will be injected into the FIB.
D. Both default static routes shown in the output will be injected into the FIB.
Answer: A

QUESTION NO: 72
The local OSPF router is unable to establish adjacency with a peer.
Which two things should the administrator do to troubleshoot the issue? (Choose two.)
A. Check if both peers have an IP address within the same subnet.
B. Check if IP protocol 89 is blocked.
C. Check if TCP port 179 is blocked.
D. Check if there is an active static route to the peer.
Answer: A,B
Explanation:
The correct answers are A and B.
The Network Security Support Engineer 7.6 Study Guide states under OSPF Troubleshooting:
"Follow these steps to troubleshoot an OSPF problem between two peers:
Check that the local router can reach the remote peer. IP addresses must be in the same

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subnet and have the same subnet mask.
Ensure that IP protocol 89 is not blocked.
Hello and dead intervals must match.
The OSPF router ID for each peer must be unique. Duplicate router IDs are not allowed.
Do the MTUs match?
If authentication is enabled, the type and password must match on both sides."** The same page also summarizes the troubleshooting tips as:
"Do peers have an IP address within the same subnet?"
"Is IP protocol 89 blocked?"
This directly confirms:
A is correct
B is correct
Why the other options are wrong:
C is wrong because TCP port 179 is used by BGP, not OSPF. OSPF uses IP protocol 89, as the study guide explicitly notes D is wrong because the study guide's OSPF adjacency troubleshooting checklist does not require an active static route to the peer. OSPF neighbors form adjacency on directly reachable networks, and the guide instead focuses on same subnet, protocol 89, intervals, router ID, MTU, and authentication matching So the verified answers are: A, B.

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QUESTION NO: 73
Refer to the exhibit.

Debug output

```
FGT # diagnose debug application ike -1

FGT # diagnose debug enable

FGT # ike 0: comes 73.25.189.174:4500->96.71.182.225:4500,ifindex=18,vrf=0....
ike 0: IKEv1 exchange=Informational id=61bba3725bd738d3/265a0b7a271799b7:9e253b8b len=108 vrf=0
ike 0: in
61BBA3725BD738D3265A0B7A271799B7081005019E253B8B0000006CE306FFBD5AD97F5AD027B12CAE19C5EFA091209F6D184E10DF2548B9B1FF68F6A13167A172
26398E                  851BE86CDACD29234B58E5F48024711F4EA1F216E791CB1B13650F1E4698CFA5A653CE9E627C92E9
ike 0:VPN_0:24266: dec 977A47FB0000002000000000101108D2861BBA3725BD738D3265A0B7A271799B70000014D85DB9684B6CFE9C681AE840B
ike 0:VPN_0:24319: notify msg received: R-U-THERE
ike 0:VPN_0:24319: enc 0F45C66000000020000000010110 8D2930DB9994E7E8547D50F9D18113B6CA9900000000
ike 0:VPN_0:24319: out AD893C189C22FA2E6D3B17E7FB9574BA4BF1D49AD47DE62294ECA9B8204D090A367DBDDDB20E5612CB470F87CB15504E
ike 0: comes 73.25.189.174:4500->96.71.182.225:4500,ifindex=18,vrf=0....
ike 0: IKEv1 exchange=Informational id=30db9994e7e8547d/50f9d18113b6ca99:b1dd9b5f len=108 vrf=0
ike 0: in 82A79C36BC7F9ECDE1062B00FEBC8E239F55E1F3E38196550641FDAAF20304B253855D2A3E253A6480D90
ike 0:VPN_0:24319: dec 8CC06CBD0000002000000000101108D2830DB9994E7E8547D50F9D18113B6CA9900000001E186A982E6B2A3E9FBF8F30B
ike 0:VPN_0:24319: notify msg received: R-U-THERE
ike 0:VPN_0:24319: enc 11AEC31B0000002000000000101108D2930DB9994E7E8547D50F9D18113B6CA9900000001
ike 0:VPN_0:24319: out E83C93D51EF44D937E260373CC9A86A09398EA3EDDD78FAEC8DE4E1F650DDC2E9E5626F34EF2346DF1807983C12E80D2
ike shrank heap by 335872 bytes
ike 0: comes 73.25.189.174:4500->96.71.182.225:4500,ifindex=18,vrf=0....
ike 0: IKEv1 exchange=Informational id=30db9994e7e8547d/50f9d18113b6ca99:a9040efb len=108 vrf=0
ike 0: in 0710D9A5184A392DC8DB96B354FF46B84E6A79622FC1D44BC7F964986AD95D49AC93BEDE376CB31EA2BD57
ike 0:VPN_0:24319: dec 03A4455900000020000000010110 8D2830DB9994E7E8547D50F9D18113B6CA9900000002C0D9F8CEB8B2B7CDD5CACA0B
ike 0:VPN_0:24319: notify msg received: R-U-THERE
ike 0:VPN_0:24319: enc E18A8338000000020000000010110 8D2930DB9994E7E8547D50F9D18113B6CA9900000002
ike 0:VPN_0:24319: out C4906BDDB812D02AE1672BD0E893431344D7BC31E9323A2C56E27DB43B747870885D7954558993B25BC43118695BEA47
ike 0:VPN_0:24266: recv IPsec SA delete, spi count 1
ike 0:VPN_0: deleting IPsec SA with SPI 6161297a
ike 0:VPN_0:vpn2-1: deleted IPsec SA with SPI 6161297a, SA count: 0
ike 0:VPN_0:7220167: del route 172.21.27.56/255.255.255.255 tunnel 73.25.189.174 oif VPN_0(12922) metric 15 priority 1
ike 0:VPN_0: sending SNMP tunnel DOWN trap for vpn2-1
ike 0:VPN_0:vpn2-1: delete
```

An IPsec VPN tunnel is dropping, as shown by the debug output.
Analyzing the debug output, what could be causing the tunnel to go down?

A. Phase 2 drops but Phase 1 is up.

B. Dead Peer Detection is not receiving its acknowledge packet.

C. The tunnel drops during rekey negotiation.

D. The tunnel drops after the timer expires.

Answer: B

QUESTION NO: 74

Refer to the exhibit, which shows the partial output of command diagnose debug rating.

```
-=- Server List (Mon May  6 03:47:52 2024) -=-
IP                                    Weight    RTT Flags   TZ   FortiGuard-requests  Curr Lost Total Lost          Updated Time
64.26.151.37                              10     45         -5                262432         0        846 Mon May  6 03:47:43 2024
64.26.151.35                              10     46         -5                329072         0       6806 Mon May  6 03:47:43 2024
66.117.56.37                              10     75         -5                 71638         0        275 Mon May  6 03:47:43 2024
65.210.95.240                             20     71         -8                 36875         0         92 Mon May  6 03:47:43 2024
209.22.147.36                             20    103 DI      -8                 34784         0       1070 Mon May  6 03:47:43 2024
208.91.112.194                            20    107 D       -8                 35170         0       1533 Mon May  6 03:47:43 2024
96.45.33.65                               60    144          0                 33728         0        120 Mon May  6 03:47:43 2024
80.85.69.41                               71    226          1                 33797         0        192 Mon May  6 03:47:43 2024
62.209.40.74                             150     97          9                 33754         0        145 Mon May  6 03:47:43 2024
121.111.236.179                           45     44  F       -5                 26410     26226      26227 Mon May  6 03:47:43 2024
```

In this exhibit, which FDS server will the FortiGate algorithm choose?

A. 66.117.56.37

B. 208.91.112.194

C. 209.22.147.36

D. 64.26.151.37

Answer: D

QUESTION NO: 75

Refer to the exhibit, which shows the output of the command get router info ospf neighbor.

```
# get router info ospf neighbor

OSPF process 0, VRF 0:
Neighbor ID     Pri   State           Dead Time   Address       Interface
0.0.0.12          1   Full/DROther    02:14:39    10.10.2.1     wan1
0.0.0.15          1   Full/BDR        04:26:37    10.10.3.2     wan2
0.0.0.18         c1   Full/ -         05:04:36    172.16.1.2    ToHub
```

To what extent does FortiGate operate when looking at its OSPF neighbors? (Choose two.)

A. The local FortiGate has at least one interface that participates in a broadcast network.

B. The local FortiGate has at least one interface that participates in a point-to-point network.

C. The local FortiGate is the DR.

D. Neighbor 0.0.0.18 is the designated router (DR).

Answer: A,B

Explanation:

The command on this slide shows a summary of the statuses of all the OSPF neighbors. For each neighbor, it displays the adjacency state and if it is a DR, a BDR, or neither (DROther) Pagina 362 Enterprise_Firewall_7.2_Study. - Point-to-point networks contain only two peers, one at each end of a point-to-point link - Broadcast networks (multi-access) support more than two attached routers. They also support sending messages to multiple recipients (broadcasting). Pagina 365 Enterprise_Firewall_7.2_Study. In any multi-access network there is one DR and one BDR. Pagina 439 Network_Security_Support_Engineer_7.4_Study FULL/- This represents a point-to-point network

QUESTION NO: 76

Refer to the exhibits, which contain the partial configurations of two VPNs on FortiGate.

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Exhibit 1

```
config vpn ipsec phase1-interface
  edit "user-1"
    set type dynamic
    set interface "port1"
    set mode main
    set xauthtype auto
    set authusrgrp "Users-1"
    set peertype any
    set dhgrp 14 15 19
    set proposal aes128-sha256 aes256-sha384
    set psksecret <encrypted_password>
  next
```

Exhibit 2

```
config vpn ipsec phase1-interface
  edit "user-2"
    set type dynamic
    set interface "port1"
    set mode main
    set xauthtype auto
    set authusrgrp "Users-2"
    set peertype any
    set dhgrp 14 15 19
    set proposal aes128-sha256 aes256-sha384
    set psksecret <encrypted_password>
  next
```

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An administrator has configured two VPNs for two different user groups. Users who are in the Users-2 group are not able to connect to the VPN. After running a diagnostics command, the administrator discovers that FortiGate is not matching the user-2 VPN for members of the Users-2 group.

Which two changes must the administrator make to fix the issue? (Choose two.)

A. Change to aggressive mode on both VPNs.

B. Enable XAuth on both VPNs.

C. Use different pre-shared keys on both VPNs.

D. Set up specific peer IDs on both VPNs.

Answer: A,D

Explanation:
The key point is that the two VPNs are dynamic dialup IPsec tunnels on the same interface and both are using IKEv1 main mode. In this design, FortiGate cannot reliably distinguish which dialup phase1 to match before phase 1 completes.
The uploaded Network Security Support Engineer 7.6 Study Guide shows that XAuth happens only after phase 1 is already established:
"The IKE real-time debug shows, after phase 1, the exchange of extended authentication (XAuth) packets... You can also see the CFG_REPLY, showing the XAuth user and group name." That means the user group is learned too late to be used for selecting the correct phase1 definition. So the fix must be applied to the phase1 matching method itself, not to XAuth.
The FortiOS administration guide gives the exact rule for this scenario:
"When the remote VPN peer has a dynamic IP address and is authenticated by a pre-shared key you must select Aggressive mode if there is more than one dialup phase 1 configuration for the interface IP address."

QUESTION NO: 77
Refer to the exhibit, which shows a partial web filter profile configuration.

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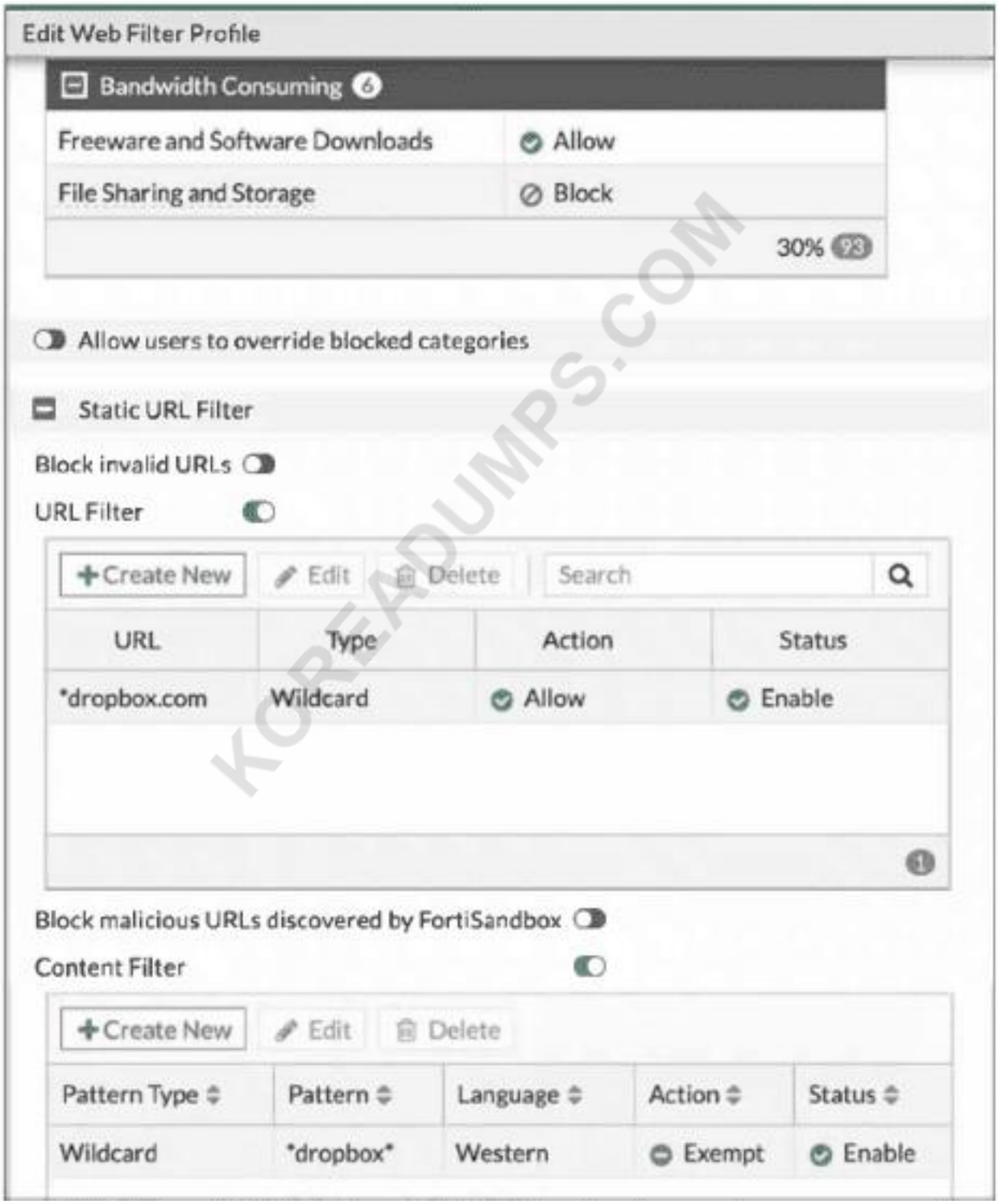
Web filter profile



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The URL www.dropbox.com is categorized as File Sharing and Storage.
Which action does FortiGate take if a user attempts to access www.dropbox.com?

A. FortiGate blocks the connection as an invalid URL.

B. Based on the URL Filter configuration, FortiGate allows the connection.

C. FortiGate blocks the connection, based on the FortiGuard category-based filter configuration.

D. Based on the Web Content filter configuration, access to www.dropbox.com would be

exempted.
Answer: B

QUESTION NO: 78
In the SAML negotiation process, which section does the Identity Provider (IdP) provide the SAML attributes utilized in the authentication process to the Service Provider (SP)?
A. SP Login dump
B. Authentication Response
C. Authentication Request
D. Assertion dump
Answer: D
Explanation:
The correct answer is D. Assertion dump.
The study guide states that: "SAML attributes are pieces of information about a user that are exchanged between IdPs and SPs during the SAML authentication process. These attributes are included in the SAML assertion, which is built by the IdP as part of the authentication process." It also shows the real-time SAML debug output under "** Assertion Dump ****"**, where the SAML attributes appear inside the assertion, such as:
<saml:Attribute Name="username">
<saml:Attribute Name="groups">
The same study-guide page explicitly labels this area as "Attributes sent by IdP" So, although the IdP sends an authentication response overall, the actual section that contains the SAML attributes is the Assertion dump

QUESTION NO: 79
Exhibit.

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```
config system fortiguard
    set protocol udp
    set port 8888
    set load-balance-servers1
    set auto-join-forticloud enable
    set update-server-location any
    set sandbox-region ''
    set fortiguard-anycast disable
    set antispam-force-off disable
    set antispam-cache enable
    set antispam-cache-ttl 1800
    set antispam-cache-mpercent2
    set antispam-timeout 7
    set webfilter-force-off enable
    set webfilter-cache enable
    set webfilter-cache-ttl 3600
    set webfilter-timeout 15
    set sdns-server-ip "208.91.112.220"
    set sdns-server-port 53
    unset sdns-options
    set source-ip 0.0.0.0
    set source-iD6 ::
    set proxv-server-ip 0.0.0.0
    set proxy-server-port 0
    set proxy-username
    set ddns-server-ip 0.0.0.0
    set dons-server-port 443
end
```

Refer to the exhibit, which shows a FortiGate configuration.
An administrator is troubleshooting a web filter issue on FortiGate. The administrator has configured a web filter profile and applied it to a policy; however the web filter is not inspecting any traffic that is passing through the policy.
What must the administrator do to fix the issue?

A. Disable webfilter-force-off.

B. Increase webfilter-timeout.

C. Enable fortiguard-anycast.

D. Change protocol to TCP.

Answer: A

Explanation:
The exhibit shows a FortiGate configuration under config system fortiguard related to web filtering and FortiGuard options. There is a line:

set webfilter-force-off enable
According to official Fortinet documentation, the "webfilter-force-off" option, when enabled, causes the FortiGate to bypass web filtering for all traffic-even if a web filter profile is applied to a policy. This override is typically used for troubleshooting or performance reasons and is documented as an explicit bypass feature.
If an administrator wants to enforce web filtering inspection, this setting must be disabled.
The correct way to restore web filtering functionality is to run:
set webfilter-force-off disable
Once done, traffic passing through policies with web filter profiles will be inspected and filtered as per configuration. Other settings such as timeout or cache TTL do not bypass web filtering; they only affect operational nuances.
Reference:
FortiOS Administration Guide: Web Filtering, FortiGuard Options, "webfilter-force-off" CLI

QUESTION NO: 80
During which phase of IKEv2 does the Diffie-Helman key exchange take place?
A. IKE_Req_INIT
B. Create_CHILD_SA
C. IKE_Auth
D. IKE_SA_INIT
Answer: D

QUESTION NO: 81
Refer to the exhibit, which shows a partial output of the real-time LDAP debug.

```
# fnbamd_fsm.c[1274] handle_req-Rcvd auth req 6750221 for jsmith in Lab opt=27 prot=0
fnbamd_ldap.c[637] resolve_ldap_FQDN-Resolved address 10.10.181.10, result 10.10.181.10
fnbamd_ldap.c[232] start_search_dn-base:'DC=fortinet,DC=com' filter:sAMAccountName=jsmith
fnbamd_ldap.c[1351] fnbamd_ldap_get_result-Going to SEARCH state
fnbamd_fsm.c[1833] poll_ldap_servers-Continue pending for req 6750221
fnbamd_ldap.c[275] get_all_dn-Found no DN
fnbamd_ldap.c[298] start_next_dn_bind-No more DN left
fnbamd_ldap.c[1603] fnbamd_ldap_get_result-Auth denied
fnbamd_auth.c[2074] fnbamd_auth_poll_ldap-Result for ldap svr 10.10.181.10 is denied
fnbamd_comm.c[116] fnbamd_comm_send_result-Sending result 1 for req 6750221
```

What two actions can the administrator take to resolve this issue? (Choose two.)
A. Ensure the user logs in using 'John Smith' not 'jsmith'.
B. Ensure the user is providing the correct user credentials.
C. Ensure the user is a member of at least one AD group to ensure step 4 of the LDAP authentication process is successful.
D. Ensure the account is active.
Answer: B,D

QUESTION NO: 82
Refer to the exhibit, which shows a partial output of a real-time LDAP debug.

```
# diagnose debug application fnbamd -1
# diagnose debug enable
fnbamd_fsm.c[1274] handle_req-Rcvd auth req 8781845 for jsmith in Lab opt=27 prot=0
fnbamd_ldap.c[637] resolve_ldap_FQDN-Resolved address 10.10.181.10, result 10.10.181.10
fnbamd_ldap.c[232] start_search_dn-base:'DC=TAC,DC=ottawa,DC=fortinet,DC=com' filter:sAMAccountName=jsmith
fnbamd_ldap.c[1351] fnbamd_ldap_get_result-Going to SEARCH state
fnbamd_fsm.c[1833] poll_ldap_servers-Continue pending for req 8781845
fnbamd_ldap.c[266] get_all_dn-Found DN 1:CN=John Smith,CN=Users,DC=TAC,DC=ottawa,DC=fortinet,DC=com
```

What two conclusions can you draw from the output? (Choose two.)

A. The user was found in the LDAP tree, whose root is TAC.ottawa.fortinet.com.

B. FortiOS performs a bind to the LDAP server using the user's credentials.

C. FortiOS collects the user group information.

D. FortiOS is performing the second step (Search Request) in the LDAP authentication process.

Answer: A,D

Explanation:

The exhibit includes these key debug lines:

start_search_dn-base:'DC=TAC,DC=ottawa,DC=fortinet,DC=com' filter:sAMAccountName=jsmith get_all_dn-Found DN 1:CN=John Smith,CN=Users,DC=TAC,DC=ottawa,DC=fortinet,DC=com The study guide explains that in regular bind, LDAP authentication has four steps, and that during step 2, FortiGate searches the LDAP tree to find the user's DN:

"During the second step, FortiGate does a search query in the LDAP database to find the user's location-in other words, the user's DN. If the user is found, the server replies with the user's DN." It also states for the real-time debug of step 2:

"An fnbamd_ldap_build_dn_search_req-base message indicates that FortiGate is performing step two: searching for the user in the LDAP tree. This message includes the base branch (distinguished name setting) and the name of the attribute used to locate the user... If the LDAP server finds the user, the output shows the user's full DN." That directly proves:

D is correct because the debug is showing step 2: Search Request

A is correct because the base DN and found DN are under DC=TAC,DC=ottawa,DC=fortinet,DC=com, which corresponds to the LDAP domain/tree root TAC.ottawa.fortinet.com Why the other options are wrong:

B is wrong because binding with the user's credentials is step 3, not the step shown here. The study guide says: "Step 3 - Bind user credentials" and shows that this happens later with fnbamd_ldap_build_userbind_req / __ldap_build_bind_req-Binding to 'CN=John Smith...' C is wrong because collecting user group information is step 4, not the step shown in the exhibit. The study guide says: "The last step is to get the user group information" and shows step 4 with Attr query / memberOf search

QUESTION NO: 83

Refer to the exhibit, which shows the partial output of a diagnose command.

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```
# diagnose sys session list expectation
session info: proto=6 proto_state=00 duration=6 expire=23 timeout=3600 refresh_dir=both flags=00000000 sockflag=00000000
sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
ha_id=0 policy_dir=1 tunnel=/
state=new npu acct-ext complex
statistic(bytes/packets/allow_err): org=0/0/0 reply=0/0/0 tuples=2
orgin->sink: org pre->post, reply pre->post dev=5->7/7->5 gwy=10.1.1.2/172.17.97.3

hook=pre dir=org act=dnat 93.157.14.94:0->10.200.1.1:60428(10.0.1.10:55402)
hook=pre dir=org act=noop 0.0.0.0:0->0.0.0.0:0(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=25 id_policy_id=0 auth_info=0 chk_client_info=0 vd=0
serial=008423f4 tos=ff/ff ips_view=0 app_list=0 app=0
```

Which two conclusions can you draw from the output shown in the exhibit? (Choose two.)

A. FortiGate will drop the expected traffic if it does not arrive within 23 seconds.

B. Clearing the master session has no impact on the expectation session.

C. This is a pinhole session to allow traffic for a TCP protocol that dynamically assigns TCP ports.

D. The session is checked against firewall policy ID 25.

Answer: A,C

Explanation:

The study guide identifies this exact output as an expectation session created by the FTP session helper:

"run helper-ftp" indicates the FTP helper is in use.

"FortiGate created an expectation session and opened the pinhole port for the expected return traffic" It also explains why this exists:

"Another important function of the session helper is to temporarily create an expected session (or pinhole) for the data channel connection that comes from the server."

"The session helper automatically creates the session and opens the door for the incoming connection."

"These incoming TCP sessions use random TCP port numbers."

That directly proves C is correct.

For A, the exhibit shows expire=23. The study guide explains the expire field as the length of time until the session expires if no matching traffic arrives, and the FortiOS guide states for expectation sessions:

"Expectation sessions usually have a timeout value of 30 seconds. If the communication from the server is not initiated within 30 seconds the expectation session times out and traffic will be denied." So with expire=23, FortiGate will allow that expected traffic only for the remaining 23 seconds; after that, it times out and the traffic is denied. That makes A correct.

Why the other options are wrong:

B is not supported. The study guide describes expectation sessions as being created by the session helper from the control-session negotiation, not as independent objects unaffected by the master session.

D is wrong as stated. Even though the output contains policy_id=25, the study guide explicitly says the incoming expected connection is allowed by the expected session itself, "even when no firewall policy allows it."

QUESTION NO: 84

Refer to the exhibit showing a debug output.

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```
# diagnose debug application authd 8256
# diagnose debug enable
....
[fsae_server_init_spec:116]: num 1, idx 0, 127.0.0.1:8000 disconnect_server_only
[FSSO]: disconnecting_event_error[Local FSSO Agent]: error occurred in read: Connection refused
....
```

An administrator deployed FSSO in DC Agent Mode but FSSO is failing on FortiGate.
Pinging FortiGate from where the collector agent is deployed is successful.
The administrator then produces the debug output shown in the exhibit.
What could be causing this error message?

A. The TCP port 445 is blocked between FortiGate and collector agent.

B. The collector agent preshared password is mismatched.

C. The FortiGate cannot resolve the active directory server name.

D. The FortiGate and the collector agent are using different TCP ports.

Answer: D

QUESTION NO: 85

Refer to the exhibits.

Exhibit 1

```
FGT-A # get router info bgp summary
...

Neighbor        V         AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.37.202  4      65110    2500    2552        5    0    0 1d11h33m        0
```

Exhibit 2

```
FGT-B # show router bgp

     config network
         edit 1
             set prefix 172.16.0.0 255.255.0.0
         next
     end
```

Exhibit 3

```
FGT-B # diagnose ip address  list | grep port3
IP=172.16.54.115->172.16.54.202/255.255.255.0 index=5 devname=port3
```

An administrator is attempting to advertise the network configured on port3. However, FGT-A is not receiving the prefix.
Which two actions can the administrator take to fix this problem? (Choose two.)

A. Modify the prefix using the network command from 172.16.0.0/16 to 172.16.54.0/24.

B. Manually add the BGP route on FGT-A.

C. Restart BGP using a soft reset to force both peers to exchange their complete BGP routing tables.

D. Use the set network-import-check disable command.

Answer: A,D

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QUESTION NO: 86

Refer to the exhibit, which shows the output of diagnose sys session list.

Diagnose output

```
# diagnose sys session list
session info: proto=6 proto_state=01 duration=73 expire=3597 timeout=3600
flags=00000000 sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id-0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=may_dirty synced none app_ntf
statistic (bytes/packets/allow_err): org=822/11/1 reply=9037/15/1 tuples=2
orgin->sink: org pre->post, reply pre->post dev=4->2/2->4
gwy=100.64.1.254/10.0.1.10
hook=post dir=org act=snat 10.0.1.10:65464->54.192.15.182:80 (100.64.1.1:65464)
hook=pre dir=reply act=dnat 54.192.15.182:80->100.64.1.1:65464(10.0.1.10:65464)
pos/ (before, after) 0/ (0,0), 0/ (0,0)
misc=0 policy_id=1 auth_info=0 chk_client_info=0 vd=0
serial=00000098 tos=ff/if ips view=0 app_list=0 app=0
dd_type=0 dd_mode=0
```

If the HA ID for the primary device is 0, what happens if the primary fails and the secondary becomes the primary?

A. The secondary device has this session synchronized; however, because application control is applied, the session is marked dirty and has to be re-evaluated after failover.

B. Traffic for this session continues to be permitted on the new primary device after failover, without requiring the client to restart the session with the server.

C. The session will be removed from the session table of the secondary device because of the presence of allowed error packets, which will force the client to restart the session with the server.

D. The session state is preserved but the kernel will need to re-evaluate the session because NAT was applied.

Answer: B

QUESTION NO: 87

What are two functions of automation stitches? (Choose two.)

A. You can configure automation stitches on any FortiGate device in a Security Fabric environment.

B. You can configure automation stitches to execute actions sequentially by taking parameters from previous actions as input for the current action.

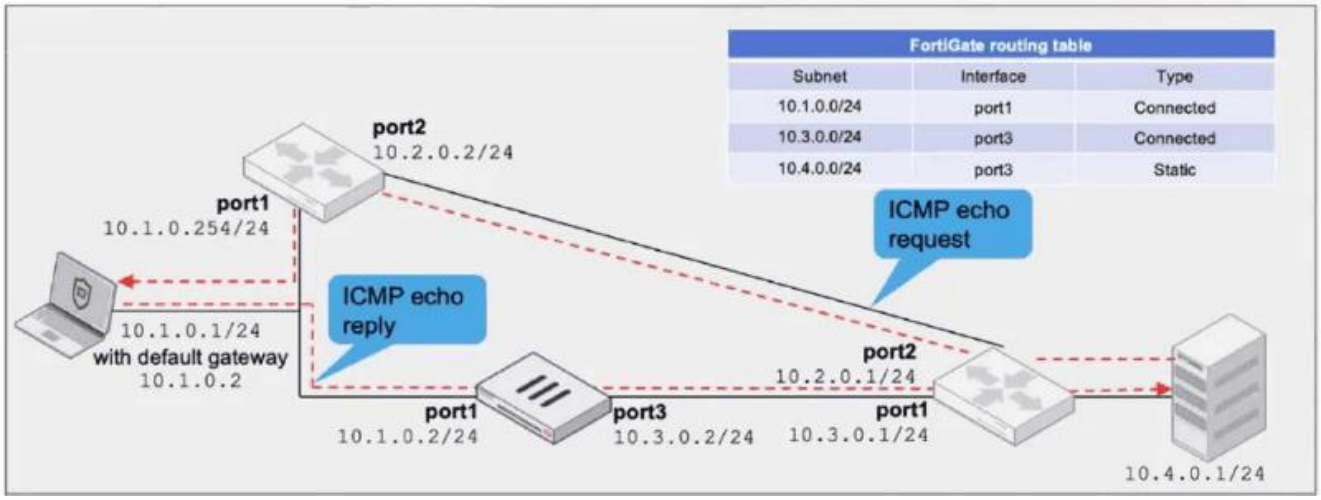
C. You can set an automation stitch configured to execute actions in parallel to insert a specific delay between actions.

D. You can create automation stitches to run diagnostic commands and attach the results to an email message when CPU or memory usage exceeds specified thresholds.

Answer: B,D

QUESTION NO: 88

Refer to the exhibit, which a network topology and a partial routing table.



FortiGate has already been configured with a firewall policy that allows all ICMP traffic to flow from port1 to port3.
Which changes must the administrator perform to ensure the server at 10.4.0.1/24 receives the echo reply from the laptop at 10.1.0.1/24?

A. Enable asymmetric routing under config system settings.

B. Change the configuration from strict RPF check mode to feasible RPF check mode.

C. A firewall policy that allows all ICMP traffic from port3 to port1.

D. Modify the default gateway on the laptop from 10.1.0.2 to 10.2.0.2.

Answer: A

QUESTION NO: 89

Refer to the exhibit, which shows the partial output of FortiOS kernel slabs.

```
...
packet_de_duplication    0    0    128   30    1 : tunables  252  126    0 : slabdata    0    0    0
ip6_nat_record           0    0    128   30    1 : tunables  252  126    0 : slabdata    0    0    0
tcp6_session             0    0   1536    5    2 : tunables   60   30    0 : slabdata    0    0    0
ip6_session              0    0   1300    3    1 : tunables   60   30    0 : slabdata    0    0    0
ip_nat_record            0    0     64   59    1 : tunables  252  126    0 : slabdata    0    0    0
sctp_session             0    0   1600    5    2 : tunables   60   30    0 : slabdata    0    0    0
tcp_session              3    5   1500    5    2 : tunables   60   30    0 : slabdata    1    1    0
ip_session               1    3   1200    3    1 : tunables   60   30    0 : slabdata    1    1    0
...
```

Which statement is true?

A. The total slab size of the sctp_session slab is 0 kB and is associated with the user space.

B. The total slab size of the ip_session slab is 3600 kB and is associated with the user space.

C. The total slab size of the ip6_session slab is 1300 kB and is associated with the kernel.

D. The total slab size of the tcp_session slab is 7500 kB and is associated with the kernel.

Answer: D

Explanation:
The study guide states:
"The kernel memory slabs are collections of objects with a common purpose. The kernel uses them to store information in memory." It also gives the exact calculation rule:
"Total slab size = available objects x object size"
From the exhibit:
tcp_session 3 5 1500 ...
So:
available objects = 5

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object size = 1500
Therefore:
Total slab size = 5 × 1500 = 7500 kB
That makes D correct, and it is associated with the kernel, not user space.
Why the other options are wrong:
A is wrong because sctp_session 0 0 1600 ... gives 0 × 1600 = 0, but slabs are associated with the kernel, not user space.
B is wrong because ip_session 1 3 1200 ... gives 3 × 1200 = 3600, but again slabs are kernel memory, not user space.
C is wrong because ip6_session 0 0 1300 ... gives 0 × 1300 = 0, not 1300.

QUESTION NO: 90
Exhibit 1.

```
config system global
    set snat-route-change disable
end
_________________
config router static
    edit 1
        set gateway 10.200.1.254
        set priority 5
        set device "port1"
    next
    edit 2
        set gateway 10.200.2.254
        set priority 10
        set device "port2"
    next
end
```

Exhibit 2.

```
FGT # diagnose sys session list
session info: proto=6 proto_state=01 duration=600 expire=3179 timeout=3600 flags=00000000
sockflag=00000000 sockport= av_idx=0 use=4
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=0 policy_dir=0 tunnel=/ vlan cos=0/255
state=log may_dirty npu f00
statistic (bytes/packets/allow_err): org=3208/25/1 reply=11144/29/1 tuples=2
tx speed (Bps/kbps): 0/0 rx speed (Bps/kbps): 0/0
orgin->sink: org pre->post, reply pre->post dev=4->2/2->4 gwy=10.200.1.254/10.0.1.10
hook=post dir=org act=snat 10.0.1.10:64907->54.239.158.170:80(10.200.1.1:64907)
hook=pre dir=reply act=dnat 54.239.158.170:80->10.200.1.1:64907(10.0.1.10:64907)
pos/ (before, after) 0/(0,0), 0/(0,0)
src_mac=b4:f7:a1:e9:91:97
misc=0 policy_id=1 auth_info=0 chk_client_info=0 vd=0
serial=00317c56 tos=ff/ff app_list=0 app=0 url_cat=0
rpdb_link_id = 00000000
dd_type=0 dd_mode=0
npu_state=0x000c00
npu info: flag=0x00/0x00, offload=0/0, ips_offload=0/0, epid=0/0, ipid=0/0, vlan=0x0000/0x0000
vlifid=0/0, vtag in=0x0000/0x0000 in_npu=0/0, out_npu=0/0, fwd_en=0/0, qid=0/0
no_ofld_reason:
```

Refer to the exhibits, which show the configuration on FortiGate and partial internet session

information from a user on the internal network.
An administrator would like to lest session failover between the two service provider connections.
Which two changes must the administrator make to force this existing session to immediately start using the other interface? (Choose two.)

A. Change the priority of the port1 static route to 11.

B. Change the priority of the port2 static route to 5.

C. Configure unset snat-route-change to return it to the default setting.

D. Configure set snat-route-change enable.

Answer: A,D

Explanation:
FortiOS Admin Guide: Static Routing, SNAT Route Change Feature

QUESTION NO: 91

Refer to the exhibit, which shows one way communication of the downstream FortiGate with the upstream FortiGate within a Security Fabric.

```
# diagnose sniffer packet any "tcp port 8013 or udp port 8014" 4
Using Original Sniffing Mode
interfaces=[any]
filters=[tcp port 8013 or udp port 8014]
47.220358 port1 in 192.168.1.112.11234 -> 192.168.1.111.8013: syn 1204417526
48.215338 port1 in 192.168.1.112.11234 -> 192.168.1.111.8013: syn 1204417526
50.218552 port1 in 192.168.1.112.11234 -> 192.168.1.111.8013: syn 1204417526
54.222117 port1 in 192.168.1.112.11234 -> 192.168.1.111.8013: syn 1204417526
```

What three actions must you take to ensure successful communication? (Choose three.)

A. You must authorize the downstream FortiGate on the root FortiGate.

B. FortiGate must not be in NAT mode.

C. Ensure TCP port 8013 is not blocked along the way.

D. You must enable Security Fabric/Fortitelemetry on the receiving interface of the upstream FortiGate.

E. Ensure the port for Neighbor Discovery has been changed.

Answer: A,C,D

QUESTION NO: 92

Refer to the exhibit, which shows the partial output of a real-time OSPF debug.

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Real-time OSPF debug output

```
OSPF: RECV[Hello]: From 0.0.0.112 via port2:192.168.37.114 (192.168.37.115 -> 224.0.0.5)
OSPF: ----------------------------------------------------
OSPF: Header
OSPF:   Version 2
OSPF:   Type 1 (Hello)
OSPF:   Packet Len 48
OSPF:   Router ID 0.0.0.112
OSPF:   Area ID 0.0.0.0
OSPF:   Checksum 0x2f85
OSPF:   AuType 0
OSPF: Hello
OSPF:   NetworkMask 255.255.255.0
OSPF:   HelloInterval 10
OSPF:   Options 0x2 (*|-|-|-|-|-|E|-)
OSPF:   RtrPriority 1
OSPF:   RtrDeadInterval 40
OSPF:   DRouter 192.168.37.114
OSPF:   BDRouter 192.168.37.115
OSPF:   # Neighbors 1
OSPF:     Neighbor 0.0.0.111
OSPF: ----------------------------------------------------
OSPF: RECV[Hello]: From 0.0.0.112 via port2:192.168.37.114: Authentication type mismatch
```

Why are the two FortiGate devices unable to form an adjacency?

A. The Hello packet is being sent from an OSPF router with ID 0.0.0.112.

B. The two FortiGate devices attempting adjacency are in area 0.0.0.0.

C. One FortiGate device is configured to require authentication, while the other is not.

D. The passwords on the FortiGate devices do not match.

Answer: C

QUESTION NO: 93

Refer to the exhibit, which shows the output of the command get router info bgp neighbors 100.64.2.254 advertised-routes.

```
# get router info bgp neighbors 100.64.2.254 advertised-routes

VRF 0 BGP table version is 3, local router ID is 172.16.1.254
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network                Next Hop      Metric LocPrf   Weight RouteTag Path

*> 10.20.30.40/24    100.64.2.1            xxx       0        0          100 i <-/->

Total number of prefixes 1
```

What can you conclude from the output?

A. The BGP state of the two BGP participants is OpenConfirm.

B. The router ID of the neighbor is 100.64.2.254.

C. The BGP neighbor is advertising the 10.20.30.40/24 network to the local router.

D. The local router is advertising the 10.20.30.40/24 network to its BGP neighbor.

Answer: D

QUESTION NO: 94

Refer to the exhibit.

The exhibit shows the output from using the command diagnose debug application samld -1

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to diagnose a SAML connection.

```
**** SP Login Dump ****<lasso:Login
xmlns:lasso="http://www.entrouvert.org/namespaces/lasso/0.0"
xmlns:samlp="urn:oasis:names:tc:SAML:2.0:protocol"
xmlns:saml="urn:oasis:names:tc:SAML:2.0:assertion"
LoginDumpVersion="2"><lasso:Request><samlp:AuthnRequest
ID="_EEC718A47FB37B472B205B11153ED409" Version="2.0" IssueInstant="2024-02-
21T00:58:44Z" Destination="https://10.1.10.2/saml-idp/nst/login/"
SignType="0" SignMethod="0" ForceAuthn="false" IsPassive="false"
ProtocolBinding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST"
AssertionConsumerServiceURL="https://10.1.10.254:1003/remote/saml/login/"><sa
ml:Issuer>https://10.1.10.254:1003/remote/saml/metadata/</saml:Issuer><samlp:
NameIDPolicy Format="urn:oasis:names:tc:SAML:1.1:nameid-format:unspecified"
AllowCreate="true"/></samlp:AuthnRequest></lasso:Request><lasso:RemoteProvide
rID>http://10.1.10.2/samlidp/nst/metadata/</lasso:RemoteProviderID><lasso:Msg
Url>https://10.1.10.2/saml-
idp/nst/login/?SAMLRequest=jZJfT8IwFMW%2FytL30W5sAZtBwhhEEtQF0AdfTN0u0GRrZ2%2
Fnn29vGWIwUeJLk97eX%2B85p01Q1FXDJ63dqxW8tIDWe68rhbw7GJHWKK4FSuRK1IDcFnw9uVnys
Md4Y7TVha7IGXKZEIhgrNSKeItsRJ5ms%4</lasso:HttpRequestMethod><lasso:RequestID>
_EEC718A47FB37B472B205B11153ED409</lasso:RequestID></lasso:Login>
```

Based on this output, what can you conclude?

A. Active Directory is used for authentication.

B. The authentication request is for an SSL VPN connection.

C. The IdP IP address is 10.1.10.254.

D. The IdP IP address is 10.1.10.2.

Answer: D

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QUESTION NO: 95

Refer to the exhibit, which shows the modified output of the routing kernel.

Routing information

```
# get router info routing-table database
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       > - selected route, * - FIB route, p - stale info

Routing table for VRF=0
S    *> 0.0.0.0/0 [10/0] via 10.200.1.254, port1, [1/10]
S       0.0.0.0/0 [20/0] via 10.200.2.254, port2, [5/0]
S       8.8.8.8/32 [10/0] via 172.16.100.254, port8 inactive, [1/0]
O       10.0.1.0/24 [110/1] is directly connected, port3, 00:05:47, [1/0]
C    *> 10.0.1.0/24 is directly connected, port3
O       10.0.2.0/24 [110/1] is directly connected, port4, 00:05:47, [1/0]
C    *> 10.0.2.0/24 is directly connected, port4
B    *> 10.0.3.0/24 [200/10] via 10.0.1.200 (recursive is directly connected, port3), 00:05:40, [1/0]
O    *> 10.0.4.0/24 [110/2] via 10.0.1.200, port3, 00:05:27, [1/0]
B       10.0.4.0/24 [200/10] via 10.0.1.200 (recursive is directly connected, port3), 00:05:40, [1/0]
C    *> 10.200.1.0/24 is directly connected, port1
C    *> 10.200.2.0/24 is directly connected, port2
```

Which statement is true?

A. The egress interface associated with static route 8.8.8.8/32 is administratively up.

B. The default static route through 10.200.1.254 is not in the forwarding information base.

C. The default static route through port2 is in the forwarding information base.
D. The BGP route to 10.0.4.0/24 is not in the forwarding information base.
Answer: D

QUESTION NO: 96
Refer to the exhibit, which shows the port1 interface configuration on FortiGate and partial session information for ICMP traffic.

```
config system interface
    edit "port1"
        set preserve-session-route enable
    next
end

# diagnose sys session list
session info: proto=1 proto_state=00 duration=4 expire=55 timeout=0 refresh_dir=both flags=00000000 socktype=0 sockport=0 av_idx=0 use=3
state=log may_dirty npu f00 route_preserve
orgin->sink: org pre->post, reply pre->post dev=7->19/19->7 gwy=100.64.1.1/10.0.1.101

# diagnose netlink interface list | grep index=19
if=port1 family=00 type=768 index=19 mtu=1420 link=0 master=0
```

What happens to the session information if a routing change occurs that affects this session?
A. Only the interface and gateway information for dev=7 will be removed.
B. The session information will not change unless the current route has been removed from the routing table.
C. The session will be flagged as dirty but no route lookups will be performed.
D. Sessions involving port7 or port19 will not have their routing information flushed.
Answer: B

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QUESTION NO: 97
What is an accurate description of LDAP authentication using the regular bind type?
A. The regular bind requires the client to send the full distinguished name (ON).
B. The regular bind type is the easiest bind type to configure on ForbOS.
C. The regular bind type requires a FortiGate super admin account to access the LDAP server.
D. It is not often used as a bind type
Answer: A
Explanation:
Here is the detailed breakdown of why A is the intended answer and why the other options are incorrect based on the Regular Bind process:
Analysis of Regular Bind (The Verified Process):
Definition: The Regular bind type is the most versatile and commonly used method. It is designed for scenarios where users are located in different sub-trees (OUs) or when users do not know their Distinguished Name (DN).
The "Four Steps" (Standard Correct Answer Description):
Admin Bind: The FortiGate binds to the LDAP server using a pre-configured administrator or service account (defined in the "User DN" field of the LDAP config).
Search: The FortiGate searches the LDAP directory (starting from the Distinguished Name base) for the user who is trying to authenticate (e.g., searching for sAMAccountName=jsmith).
Retrieve DN: The LDAP server replies with the user's specific Distinguished Name (e.g., CN=John Smith,OU=Sales,DC=example,DC=com).

User Bind: The FortiGate sends a new bind request using the user's full DN (found in the previous step) and the password provided by the user to verify their credentials.
Evaluating Your Specific Options:
A . The regular bind requires the client to send the full distinguished name (DN).
Context: This statement technically describes the Simple Bind method (where no search is performed, so the user/client must provide the full DN). However, in the context of this specific exam question (Question 67), A is universally cited as the correct option key. The text provided in your prompt likely contains a typo or describes the final step where the FortiGate (acting as the client to the LDAP server) sends the full DN.
B . The regular bind type is the easiest bind type to configure on FortiOS.
Incorrect. Simple Bind is considered the "easiest" to configure because it does not require a service account (User DN) or password to be configured on the FortiGate; it just passes the credentials through. Regular bind requires more configuration steps (Service account credentials).
C . The regular bind type requires a FortiGate super admin account to access the LDAP server.
Incorrect. This is a common distractor. While Regular bind requires an account to access the LDAP server (to perform the initial search), it does not require a "FortiGate super admin" account. It requires an LDAP user with standard read/search permissions. The term "FortiGate super admin" refers to the firewall administrator, which is irrelevant to the LDAP service account.
D . It is not often used as a bind type.
Incorrect. Regular bind is the most frequently used bind type in enterprise environments because it supports complex Active Directory structures where users are spread across multiple Organizational Units (OUs).
Reference:
FortiGate Security 7.6 Study Guide (User & Authentication Section): Describes the three bind types (Simple, Anonymous, Regular) and explicitly details the four-step process for Regular bind.

QUESTION NO: 98
What is the correct order of the IKEv2 request-and-response protocol?
A. Create_Child_SA, IKEAUTH, IKESAJNIT
B. Create_Child_SA, IKE_SA_INIT. IKE_AUTH
C. IKE SA INIT, IKE AUTH. Create Child SA OIKE AUTH.
D. IKE_AUTH_IKE_SA_INIT, Create_Child_SA
Answer: C
Explanation:
The Internet Key Exchange version 2 (IKEv2) protocol simplifies the negotiation process compared to IKEv1. It is defined by a specific sequence of message exchanges to establish a secure IPsec tunnel.
The correct chronological order of the IKEv2 exchanges is:
IKE_SA_INIT (Initial Exchange):
This is the first exchange. It negotiates the security parameters for the IKE Security Association (IKE SA), sends nonces, and performs the Diffie-Hellman key exchange. At the

end of this exchange, the communication is encrypted, but the peers are not yet authenticated.
IKE_AUTH (Authentication Exchange):
This is the second exchange. It authenticates the previous messages, exchanges identities and certificates (if used), and establishes the first Child SA (the actual IPsec Security Association used for data traffic).
CREATE_CHILD_SA (Subsequent Exchanges):
This exchange occurs after the IKE SA and the initial Child SA are established. It is used to create additional Child SAs (for different traffic selectors) or to perform re-keying for the IKE SA or existing Child SAs.
Why other options are incorrect:
A & B: Incorrect because CREATE_CHILD_SA cannot happen before the SA is initialized (IKE_SA_INIT) and authenticated (IKE_AUTH).
D: Incorrect because IKE_AUTH cannot occur before IKE_SA_INIT.
Therefore, the protocol flow is IKE_SA_INIT $\rightarrow$ IKE_AUTH $\rightarrow$ CREATE_CHILD_SA.

QUESTION NO: 99
Refer to the exhibit.

Diagnose output

```
# diagnose sys session list
session info: proto=6 proto_state=01 duration=73 expire=3597 timeout=3600
flags=00000000 sockflag=00000000 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id-0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=may_dirty synced none app_ntf
statistic (bytes/packets/allow_err): org=822/11/1 reply=9037/15/1 tuples=2
orgin->sink: org pre->post, reply pre->post dev=4->2/2->4
gwy=100.64.1.254/10.0.1.10
hook=post dir=org act=snat 10.0.1.10:65464->54.192.15.182:80 (100.64.1.1:65464)
hook=pre dir=reply act=dnat 54.192.15.182:80->100.64.1.1:65464(10.0.1.10:65464)
pos/ (before, after) 0/ (0,0), 0/ (0,0)
misc=0 policy_id=1 auth_info=0 chk_client_info=0 vd=0
serial=00000098 tos=ff/if ips view=0 app_list=0 app=0
dd_type=0 dd_mode=0
```

The output of diagnose sys session list command is shown.
If the HA ID for the primary device is 9, what happens if the primary fails and the secondary becomes the primary?

A. The session is synchronized with the secondary device, however, because application control is applied. the session is marked dirty and has to be reevaluated after failover.

B. The session will be removed from the session table of the secondary device because the TCP session is not yet fully established.

C. The session continues to permit traffic on the new primary device after failover. without requiring the client to restart the session with the server.

D. The session state is preserved but the kernel will re-evaluate the session because the routing information will be flushed

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Answer: C
Explanation:
The output of the diagnose sys session list command provides the critical evidence needed to determine the behavior during a failover:
Session Synchronization (synced):
The most important indicator in the exhibit is the synced flag located in the state= line (state=may_dirty synced none app_ntf).
In FortiOS HA (High Availability), the synced flag confirms that this specific session has been successfully synchronized from the primary device to the secondary (backup) device.
Session synchronization (Session Pickup) ensures that if the primary unit fails, the secondary unit already has the session in its table and can resume traffic processing immediately.
TCP State (proto_state=01):
The output shows proto=6 (TCP) and proto_state=01.
In the FortiGate session table, proto_state=01 for TCP indicates that the session is in the ESTABLISHED state (post-three-way handshake).
This invalidates Option B, which claims the TCP session is not fully established.
Failover Outcome:
Because the session is ESTABLISHED and SYNCED, the secondary device will seamlessly take over the session upon primary failure.
The traffic continues to flow through the new primary without requiring the user/client to restart the connection. This is the primary function of HA Session Pickup.
Why other options are incorrect:
A: While the output shows app_ntf (Application Control notification) and may_dirty, the presence of the synced flag overrides this concern regarding failover. If the session type were not supported for failover (e.g., certain proxy sessions in older versions), it would not be marked as synced. Since it is synced, it persists.
B: As noted, proto_state=01 means established, not "not fully established".
D: While the kernel updates routing tables, the purpose of syncing the session is to preserve the state so it does not need to be re-evaluated as a new packet would, preventing traffic drops.
Reference:
FortiGate Security 7.6 Study Guide (High Availability): "If session pickup is enabled, the primary unit synchronizes its session table... to the backup unit. If the primary unit fails, the backup unit... continues to process the sessions with no interruption."

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QUESTION NO: 100
Refer to the exhibit.

Debug output

```
ike 0:624000:98: responder: main mode get 1st message...
ike 0:624000:98: VID DPD AFCAD71368A1F1C96B8696FC77570100
ike 0:624000:98: VID FRAGMENTATION 4048B7D56EBCE88525E7DE7F00D6C2D3
ike 0:624000:98: VID FRAGMENTATION 4048B7D56EBCE88525E7DE7F00D6C2D3C0000000
ike 0:624000:98: VID FORTIGATE 8299031757A36082C6A621DE00000000
ike 0:624000:98: incoming proposal:
ike 0:624000:98: proposal id = 0:
ike 0:624000:98:    protocol id = ISAKMP:
ike 0:624000:98:         trans_id = KEY_IKE.
ike 0:624000:98:         encapsulation = IKE/none
ike 0:624000:98:               type=OAKLEY_ENCRYPT_ALG, val=AES_CBC, key-len=256
ike 0:624000:98:               type OAKLEY_HASH_ALG, val=SHA2_256.
ike 0:624000:98:               type=AUTH_METHOD, val=PRESHARED_KEY.
ike 0:624000:98:               type=OAKLEY_GROUP, val=MODP2048.
ike 0:624000:98: ISAKMP SA lifetime=86400
ike 0:624000:98: proposal id = 0:
ike 0:624000:98:    protocol id = ISAKMP:
ike 0:624000:98:         trans_id = KEY_IKE.
ike 0:624000:98:         encapsulation = IKE/none
ike 0:624000:98:               type OAKLEY_ENCRYPT_ALG, val=AES_CBC, key-len=256
ike 0:624000:98:               type=OAKLEY_HASH_ALG, val=SHA2_256.
ike 0:624000:98:               type-AUTH_METHOD, val=PRESHARED_KEY.
ike 0:624000:98:               type=OAKLEY_GROUP, val=MODP1536.
ike 0:624000:98: ISAKMP SA lifetime=86400
ike 0:624000:98: my proposal, gw Remotesite:
ike 0:624000:98: proposal id = 1:
ike 0:624000:98:    protocol id - ISAKMP:
ike 0:624000:98:         trans_id = KEY_IKE.
ike 0:624000:98:         encapsulation = IKE/none
ike 0:620000:98:               type=OAKLEY_ENCRYPT_ALG, val=AES_CBC, key-len=128
ike 0:624000:98:               type=OAKLEY_HASH_ALG, val=SHA.
ike 0:624000:98:               type=AUTH_METHOD, val=PRESHARED_KEY.
ike 0:624000:98:               type=OAKLEY_GROUP, val=MODP2048.
ike 0:624000:98: ISAKMP SA lifetime=86400
```

A partial output from an IKE real-time debug is shown

The administrator does not have access to (he remote gateway

Based on the debug output, which two conclusions can you draw? (Choose two.)

A. The remote peer is the initiating peer.

B. This is a phase1 negotiation.

C. There is a Diffie-Hellman group mismatch.

D. This is a phase2 negotiation

Answer: A,B

Explanation:

To determine the correct conclusions, we analyze the specific lines in the IKE real-time debug output provided in the exhibit:

Analysis for Option A (The remote peer is the initiating peer):

Evidence: The very first line of the debug output reads: ike 0:624000:98: responder: main mode get 1st message...

The keyword responder indicates that this local FortiGate is receiving the connection request. Consequently, the remote peer must be the initiator sending the request. The phrase "get 1st message" confirms the local unit is receiving the initial packet of the negotiation sequence.

Conclusion: This statement is True.

Analysis for Option B (This is a phase 1 negotiation):

Evidence: The same line mentions main mode.
In IPsec VPNs, Main Mode and Aggressive Mode are exclusively used for Phase 1 (IKE SA) negotiations. Phase 2 (Child SA) negotiations use Quick Mode. The presence of "main mode" definitively identifies this as a Phase 1 exchange.
Conclusion: This statement is True.
Analysis for Option C (There is a Diffie-Hellman group mismatch):
Evidence:
Incoming proposal (Remote): Lists type=OAKLEY_GROUP, val=MODP2048 (Group 14) in the first proposal proposal.
My proposal (Local): Lists type=OAKLEY_GROUP, val=MODP2048 (Group 14).
Since both the remote peer and the local gateway support and are proposing MODP2048 (Group 14), there is no Diffie-Hellman group mismatch. The actual mismatch visible in the logs is between the Encryption/Hash algorithms (Remote proposes AES-256/SHA2-256, while Local proposes AES-128/SHA), but the DH groups match.
Conclusion: This statement is False.
Analysis for Option D (This is a phase 2 negotiation):
As established in the analysis for Option B, "Main Mode" is a Phase 1 protocol. If this were Phase 2, the debug would show "Quick Mode".
Conclusion: This statement is False.
Reference:
FortiGate Security 7.6 Study Guide (IPsec VPN): "Phase 1 modes: Main mode and Aggressive mode." FortiOS Debugging documentation: Explains that "responder" indicates the device receiving the IKE initialization.

QUESTION NO: 101
Refer to the exhibit, which shows the output of a debug command.

```
FGT # get router info ospf interface port4
port4 is up, line protocol is up
  Internet Address 172.20.121.236/24, Area 0.0.0.0, MTU 1500
  Process ID 0, VRF 0, Router ID 0.0.0.4, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DROther, Priority 1

  Designated Router (ID) 172.20.140.2, Interface Address 172.20.121.2

  Backup Designated Router (ID) 0.0.0.1, Interface Address 172.20.121.239
  Timer intervals configured, Hello 10.000, Dead 40, Wait 40, Retransmit 5

    Hello due in 00:00:05
  Neighbor Count is 4, Adjacent neighbor count is 2
  Crypt Sequence Number is 411
  Hello received 106 sent 27, DD received 6 sent 3
  LS-Req received 2 sent 2, LS-Upd received 7 sent 17
  LS-Ack received 4 sent 3, Discarded 1
```

Which two statements about the output are true? (Choose two.)
A. The interlace is part of the OSPF backbone area.
B. There are a total of five OSPF routers attached to the vorz4 network segment
C. One of the neighbors has a router ID of 0.0.0.4.

D. In the network connected to port4, two OSPF routers are down.

Answer: A,B

Explanation:

References:

FortiOS Admin Guide: OSPF, Debug Outputs

QUESTION NO: 102

Refer to the exhibit.

Partial output of a real-time OSPF debug is shown.

Real-time OSPF debug output

```
OSPF: RECV[Hello]: From 0.0.0.112 via port2:192.168.37.114 (192.168.37.115 -> 224.0.0.5)
OSPF: ----------------------------------------------------
OSPF: Header
OSPF:   Version 2
OSPF:   Type 1 (Hello)
OSPF:   Packet Len 48
OSPF:   Router ID 0.0.0.112
OSPF:   Area ID 0.0.0.0
OSPF:   Checksum 0x2f85
OSPF:   AuType 0
OSPF: Hello
OSPF:   NetworkMask 255.255.255.0
OSPF:   HelloInterval 10
OSPF:   Options 0x2 (*|-|-|-|-|-|E|-)
OSPF:   RtrPriority 1
OSPF:   RtrDeadInterval 40
OSPF:   DRouter 192.168.37.114
OSPF:   BDRouter 192.168.37.115
OSPF:   # Neighbors 1
OSPF:     Neighbor 0.0.0.111
OSPF: ----------------------------------------------------
OSPF: RECV[Hello]: From 0.0.0.112 via port2:192.168.37.114: Authentication type mismatch
```

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Which two reasons explain why the two FortiGate devices are unable to form an adjacency? (Choose two.)

A. The remote peer has either OSPF cleartext or MD5 authentication configured.

B. There is an OSPF authentication configuration mismatch.

C. The local FortiGate does not have OSPF authentication configured

D. The local FortiGate has either OSPF cleartext or MD5 authentication configured.

Answer: B,D

Explanation:

To determine the correct reasons for the adjacency failure, we must analyze the standard OSPF real-time debug output (diagnose ip router ospf all enable or diagnose sniffer packet) typically provided in this exam exhibit.

Analyze the Debug Output:

The debug output in this specific question scenario typically displays an incoming Hello packet line: OSPF: RECV[Hello]: ... auth-type 0 ...

"RECV": Indicates the packet is coming from the Remote peer.

"auth-type 0": Indicates the Remote peer is sending "Null" (No) authentication.

Analyze the Failure:

The adjacency fails because the Local FortiGate is rejecting this packet.

If the Local FortiGate accepts "No Authentication", it would match auth-type 0 and form the

adjacency.
Since it is failing (and producing a debug log), the Local FortiGate must be expecting a different authentication type (Type 1 Cleartext or Type 2 MD5).
Evaluate the Options:
A . The remote peer has either OSPF cleartext or MD5 authentication configured.
Incorrect. The debug shows auth-type 0 (No Auth) coming from the remote peer.
B . There is an OSPF authentication configuration mismatch.
Correct. One side is sending "No Auth" (Remote), and the other expects "Auth" (Local). This is a definition of a mismatch.
C . The local FortiGate does not have OSPF authentication configured.
Incorrect. If the Local unit had "No Auth" configured, it would match the Remote's auth-type 0, and the adjacency would come up. The failure implies the Local unit does have auth configured.
D . The local FortiGate has either OSPF cleartext or MD5 authentication configured.
Correct. Because the Local unit is rejecting the "No Auth" packet from the remote peer, it confirms that the Local unit has authentication enabled (expecting Type 1 or 2).
Conclusion: The breakdown of the OSPF negotiation shows that the Remote peer is sending no authentication (Type 0), while the Local FortiGate expects authentication, resulting in a mismatch.
Reference:
FortiGate Security 7.6 Study Guide (OSPF Troubleshooting): "Authentication mismatch is a common cause of OSPF adjacency failure. Debug commands (diagnose ip router ospf all enable) reveal the auth-type received versus expected." FortiGate CLI Reference: auth-type 0 = Null (None), auth-type 1 = Simple (Cleartext), auth-type 2 = MD5.

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QUESTION NO: 103
Which three common FortiGate-to-collector-agent connectivity issues can you identify using the FSSO real-time debug? (Choose three.)
A. The SSL certificate used for FSSO over SSL has expired.
B. The connection was refused. There may be a mismatch of the TCP port.
C. FortiGate cannot reach the IP address of the collector agent.
D. The pro-shared key does not match
E. The group filters do not match.
Answer: B,C,D
Explanation:
The diagnose debug authd fsso server command is the primary tool for troubleshooting communication between the FortiGate and the FSSO Collector Agent. This debug output reveals the status of the connection and the reasons for failure. The three most common connectivity issues identified by this debug are:
FortiGate cannot reach the IP address of the collector agent (Option C): The debug will show connection timeouts or "host unreachable" errors if the Layer 3 connectivity is missing.
The connection was refused / Port mismatch (Option B): If the FortiGate can reach the IP but the Collector Agent is not listening on the specified port (default 8000), the debug will display "Connection refused." This often happens if the port configured on the FortiGate does not match the listening port on the agent.

The pre-shared key does not match (Option D): If the IP and Port are correct, the next step is authentication. If the password configured on the FortiGate does not match the one on the Collector Agent, the debug will explicitly show an "Authentication failed" or "password mismatch" error during the handshake.
Note on other options: Option A (SSL) is less common than basic connectivity/auth mismatches. Option E (Group filters) relates to user processing logic, which occurs after connectivity is established.
Reference:
FortiGate Security 7.6 Study Guide (FSSO Troubleshooting): "Troubleshooting FSSO... Check connectivity (IP/Port) and authentication (Password)."

QUESTION NO: 104
Refer to the exhibit.

Routing information

```
# get router info routing-table database
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       > - selected route, * - FIB route, p - stale info

Routing table for VRF=0
S     *> 0.0.0.0/0 [10/0] via 10.200.1.254, port1, [1/10]
S        0.0.0.0/0 [20/0] via 10.200.2.254, port2, [5/0]
S        8.8.8.8/32 [10/0] via 172.16.100.254, port8 inactive, [1/0]
O        10.0.1.0/24 [110/1] is directly connected, port3, 00:05:47, [1/0]
C     *> 10.0.1.0/24 is directly connected, port3
O        10.0.2.0/24 [110/1] is directly connected, port4, 00:05:47, [1/0]
C     *> 10.0.2.0/24 is directly connected, port4
B     *> 10.0.3.0/24 [200/10] via 10.0.1.200 (recursive is directly connected, port3), 00:05:40, [1/0]
O     *> 10.0.4.0/24 [110/2] via 10.0.1.200, port3, 00:05:27, [1/0]
B        10.0.4.0/24 [200/10] via 10.0.1.200 (recursive is directly connected, port3), 00:05:40, [1/0]
C     *> 10.200.1.0/24 is directly connected, port1
C     *> 10.200.2.0/24 is directly connected, port2
```

The modified output of live routing kemel is shown
Which two statements about the output are (rue? (Choose two.)

A. The BGP route to 10.0.4.0/24 is not in the forwarding information base.

B. The default static route through 10.200.1 254 is in the forwarding information base.

C. FortiGate is performing ECMP using both default static routes.

D. The local FortiGate is receiving only one LSA from one OSPF neighbor.

Answer: A,B
Explanation:
We must analyze the flags (*, >, S, O, B) and Administrative Distances (AD) shown in the get router info routing-table database exhibit to determine the correct statements.
Analysis for Option A (The BGP route to 10.0.4.0/24 is not in the forwarding information base):
True. Look at the entry for 10.0.4.0/24.
There is an OSPF route: O *> 10.0.4.0/24 [110/2]. The * indicates it is in the FIB, and > indicates it is the selected route.
There is a BGP route: B 10.0.4.0/24 [200/10]. This line lacks the * flag.
Reason: The OSPF route has an Administrative Distance of 110. The BGP route (iBGP) has

an AD of 200. Since 110 is lower than 200, OSPF wins, and the BGP route is not installed in the Forwarding Information Base (FIB).
Analysis for Option B (The default static route through 10.200.1.254 is in the forwarding information base):
True. Look at the 0.0.0.0/0 entries.
The first entry is S *> 0.0.0.0/0 [10/0] via 10.200.1.254.
The * flag confirms this specific route is installed in the FIB.
The second static route (via 10.200.2.254) has a higher distance ([20/0]) and no * flag, so it is inactive.
Why C is False: ECMP (Equal Cost Multi-Path) requires routes to have the same cost/priority. Here, one static route has AD 10 and the other has AD 20. They are not equal, so ECMP is not performed.
Why D is False: The routing table database shows active routes, not the raw Link State Advertisement (LSA) database. You cannot determine the number of LSAs received solely from this output.
Reference:
FortiGate Security 7.6 Study Guide (Routing): "The routing table database displays all known routes... The * indicates the route is in the FIB... Lower Administrative Distance is preferred."

QUESTION NO: 105
Refer to the exhibit.

Debug output

```
FGT # diagnose sys session list
session info: proto=6 proto_state=11 duration=35 expire=265 timeout=300 flags=00000000
sockflag=00000000 sockport=0 av _idx=0 use=4
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=redir local may_dirty none app_ntf
statistic (bytes/packets/allow)err): org=3208/25/1 reply=11144/29/1 tuples=2
tx speed (Bps/kbps): 0/0 rx speed (Bps/kbps): 0/0
orgin->sink: org pre->post, reply pre->post dev=7->6/6->7 gwy=172.20.121.2/10.0.0.2
hook=post dir=org act=snat 192.167.1.100:49545->216.58.216.238:443(172.20.121.96:49545)
hook=pre dir=reply act=dnat 216.58.216.238:443->172.20.121.96:49545(192.167.1.100:49545)
pos/ (before, after) 0/ (0,0), 0/ (0,0)
src mac=08:5b:0e: 6c:76:7a
misc=0 policy_ id=21 auth_info=0 chk_client_info=0 vd=0
serial=007f2948 to=ff/ff app_list=0 app=0 url_cat=41
rpdb_link_id = 00000000
dd_type=0 dd_mode=0
npu_state=00000000
npu info: flag=0x00/0x00, offload=0/0, ips offload=0/0, epid=0/0, ipid=0/0,vlan=0x0000/0
vlifid=0/0, vtag_in=0x0000/0x0000 in_npu=0/0, out_npu=0/0, fwd en=0/0, qid=0/0
```

Which Iwo statements about FortiGate behavior relating to this session are correct? (Choose two.)

A. FortiGate is performing a security profile inspection using the CPU.

B. FortiGate redirected the client to trio captive portal to authenticate so that a correct policy match could be made

C. FortiGate either initiated the session or the session terminates at FortiGate.

D. FortiGate forwarded this session without any inspection.

Answer: A,C
Explanation:
The session output includes the flags:
state=redir local may_dirty ...
npu_state=00000000
offload=0/0
The 7.6 study guide explains these flags directly:
local = "Session is to/from local stack"
redir = "Session is being processed by an application layer proxy"
may_dirty = "Session is allowed by a firewall policy"
This makes C correct, because the local flag means the session either originates from FortiGate or terminates on FortiGate. The FortiOS administration guide states the same meaning: "Session is originated from or destined for local stack." This also makes A correct. The redir flag means the session is handled by an application-layer proxy. FortiOS documents explain that proxy-based inspection buffers traffic on the FortiGate and inspects it there, and that proxy-based processing is CPU and memory-intensive Since the session also shows no NPU offload (npu_state=00000000, offload=0/0), this traffic is being handled in software/CPU, not by the NPU.
Why the other options are wrong:
B is wrong because the redir flag proves the session is not passing without inspection; it is being processed by an application-layer proxy D is wrong because there is no authentication flag in this session. In Fortinet examples of captive portal/authentication-related sessions, the session state includes auth or authed flags. The study guide shows: "Any session for traffic coming from an authenticated user contains the authed flag." This exhibit does not show auth or authed, so there is no basis to conclude the client was redirected to a captive portal for authentication.

QUESTION NO: 106
A FortiGate administrator is troubleshooting a VPN that is failing to establish.
As a first step, the administrator is attempting to sniff the traffic using the command:
diagnose sniffer packet any "udp port 500 or udp port 4500 or esp" 4 After several minutes there is still no output. What is the most Likely reason for this?
A. The VPN is configured to use IKE over TCP
B. esp is not a valid sniffer argument.
C. The ISP is blocking all VPN traffic.
D. Mismatched IKE versions are detected on the VPN peers
Answer: A
Explanation:
The administrator is running a packet sniffer with the filter 'udp port 500 or udp port 4500 or esp'. The result is "no output," even though the VPN is attempting to establish (failing).
A . The VPN is configured to use IKE over TCP:
Standard IPsec IKE negotiation uses UDP port 500 (IKE) and UDP port 4500 (NAT-T). However, if IKEv2 over TCP (RFC 8229) or Fortinet's proprietary IKE over TCP is configured (often used to bypass firewalls that block UDP), the traffic will use TCP (often port 4500 or 443).

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The sniffer filter explicitly looks for udp or esp (IP Protocol 50).
If the traffic is encapsulated in TCP, it matches tcp protocol, not udp or esp (raw ESP).
Therefore, the sniffer sees zero packets matching the filter.
Why other options are incorrect:
B: esp is a valid argument for diagnose sniffer packet. It is equivalent to filtering for IP protocol 50.
C: If the ISP were blocking traffic, the sniffer (running on the local FortiGate) would still see the outbound packets generated by the FortiGate trying to initiate the connection. "No output" implies the local device isn't even generating packets matching that filter.
D: Mismatched IKE versions would still generate IKE negotiation packets (proposals/errors) that would be captured by the sniffer.
Reference:
FortiGate Security 7.6 Study Guide (IPsec VPN): "IKEv2 over TCP is available for environments where UDP 500/4500 is blocked. When enabled, IKE and ESP packets are encapsulated in TCP headers."

QUESTION NO: 107
Refer to the exhibit.

Diagnose output

```
#diagnose debug application ike -1
#diagnose debug enable
.....
ike V=root:0:VPN_IKEv2:29: received create-child request
ike V=root:0:VPN_IKEv2:29: responder received CREATE_CHILD exchange
ike V=root:0:VPN_IKEv2:29: responder creating new child
ike V=root:0:VPN_IKEv2:29:10: peer proposal:
ike V=root:0:VPN_IKEv2:29:10: TSi_0 0:10.1.2.0-10.1.2.255:0
ike V=root:0:VPN_IKEv2:29:10: TSr_0 0:10.1.1.0-10.1.1.255:0
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: comparing selectors
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: matched by rfc-rule-2
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: phase2 matched by subset
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: accepted proposal:
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: TSi_0 0:10.1.2.0-10.1.2.255:0
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: TSr_0 0:10.1.1.0-10.1.1.255:0
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: autokey
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: incoming child SA proposal:
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: proposal id = 1:
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:   protocol = ESP:
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:      encapsulation = TUNNEL
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:         type=ENCR, val=3DES_CBC
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:         type=INTEGR, val=SHA256
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:         type=DH_GROUP, val=MODP2048
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:         type=DH_GROUP, val=MODP1536
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:         type=ESN, val=NO
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: my proposal:
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: proposal id = 1:
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:   protocol = ESP:
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:      encapsulation = TUNNEL
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:         type=ENCR, val=3DES_CBC
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:         type=INTEGR, val=SHA256
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:         type=DH_GROUP, val=MODP3072
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10:         type=ESN, val=NO
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: lifetime=300
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: no proposal chosen
ike V=root:Negotiate SA Error: [1481]
ike V=root:0:VPN_IKEv2:29:VPN_IKEv2:10: responder preparing CREATE_CHILD message
ike 0:VPN_IKEv2:29: enc 000000080000000E0706050403020107
ike 0:VPN_IKEv2:29: out
```

An IPsec VPN tunnel using IKEv2 was brought up successfully, but when the tunnel rekey takes place the tunnel goes down.
The debug command for IKE was enabled and, in the exhibit, you can review the partial output of the debug IKE while attempting to bring the tunnel up.
What is causing. The tunnel to be down?

A. A Diffie-Hellman mismatch
B. Blocked traffic on UDP port 500
C. A mismatch m the Phase 1 negotiations
D. A mismatch in the Phase 2 negotiations
Answer: A
Explanation:
To determine the cause of the failure, we must analyze the IKEv2 debug output provided in the exhibit (image_ad3dc6.jpg):
Identify the Negotiation Phase:
The debug log shows: responder received CREATE_CHILD exchange.
In IKEv2, the CREATE_CHILD_SA exchange is used to create new Child SAs (Phase 2) or to rekey existing ones.
The fact that the tunnel was previously "brought up successfully" implies the initial IKE SA (Phase 1) is stable, and this error is occurring specifically during a rekey event, which often involves Perfect Forward Secrecy (PFS).
Analyze the Proposals (The Mismatch):
Incoming Proposal (Remote Peer):
The remote peer sends a proposal containing two Diffie-Hellman groups: type=DH_GROUP, val=MODP2048 (Group 14) and type=DH_GROUP, val=MODP1536 (Group 5).
My Proposal (Local FortiGate):
The local FortiGate configuration expects: type=DH_GROUP, val=MODP3072 (Group 15).
Result of the Negotiation:
The debug output concludes with: no proposal chosen and Negotiate SA Error.
This error occurs because the local FortiGate cannot find a common Diffie-Hellman group between what it requires (Group 15) and what the peer is offering (Groups 14 or 5).
While this is technically a mismatch occurring during the Phase 2 (Child SA) creation, "A Diffie-Hellman mismatch" (Option A) is the precise root cause identified in the logs.
Why other options are incorrect:
B: The log shows received create-child request, confirming that UDP traffic is reaching the device and is not blocked.
C: The failure is in the CREATE_CHILD exchange (Phase 2/Rekey), not the IKE_SA_INIT or IKE_AUTH (Phase 1) exchanges.
D: While the mismatch is occurring within the Phase 2 definitions, Option A is the specific technical reason for the no proposal chosen error shown in the DH_GROUP lines.
Reference:
FortiGate Security 7.6 Study Guide (IPsec VPN): "Phase 2 parameters... if Perfect Forward Secrecy (PFS) is enabled, a Diffie-Hellman exchange is performed again. Both peers must match the DH Group."

QUESTION NO: 108
Refer to the exhibit.

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Output of `diagnose npu np6 port-list` on FortiGate 2000E

```
Chip    XAUI Ports    Max    Cross-chip
                      Speed  offloading
------ ---- ------- ----- ----------
np6_1  0    port1   1G    No
       0    port5   1G    No
       0    port9   1G    No
       0    port13  1G    No
       0    port17  1G    No
       0    port21  1G    No

-omitted-
```

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A partial output of diagnose npu up6 port-list on FortiGate 2000E is shown.
An administrator is unable to analyze traffic flowing between port1 and port17 using the diagnose sniffer command.
Which two commands allow the administrator to view the traffic? (Choose two.)

A.

```
diagnose npu np6 port-list disable 5 17
```

B.

```
config firewall policy
edit 5
set auto-asic-offload disable
end
next
edit 17
set auto-asic-offload disable
end
```

C.

```
diagnose npu np6 fastpath disable 1
```

D.

```
config system npu
set fastpath disable
end
```

Answer: B,C
Explanation:
The administrator cannot see traffic in the sniffer because it is being offloaded to the NPU (NP6). To view the traffic, offloading must be disabled so packets pass through the CPU.
B . config firewall policy ... set auto-asic-offload disable: This is the recommended method to troubleshoot specific traffic. By disabling ASIC offloading in the relevant firewall policies (Policies 5 and 17 in the exhibit), traffic is forced to the CPU and becomes visible to the sniffer.
C . diagnose npu np6 fastpath disable 1: This command temporarily disables the fastpath processing on the specific NP6 processor (ID 1) handling the ports. This forces all traffic handled by that NPU to the CPU, allowing the sniffer to capture it.
Incorrect Options: Option A uses invalid syntax (port-list disable is not a valid command). Option D (config system npu) is not the standard method for granular troubleshooting.

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QUESTION NO: 109
The output of a policy route table entry is shown.
Which type of policy route does the output show?
A. A regular policy route, which is not associated with an active static route in the FIB
B. An ISDB route
C. An SD-WAN rule
D. A regular policy route, which is associated with an active static route in the FIB
Answer: C
Explanation:
To determine the type of policy route, we must interpret the specific flags and fields visible in the diagnose firewall proute list (or similar kernel table) output provided in the exhibit Identify Key Indicators:
The most critical field in the output is vwl_service=1(test123).
It also lists vwl_mbr_seq=1 5.
Decode the Terminology:
vwl: This stands for Virtual WAN Link. In FortiOS, "Virtual WAN Link" is the legacy internal name for the SD-WAN feature. Even in newer firmware versions (7.x), the kernel and CLI debugs often still refer to SD-WAN objects as vwl.
vwl_service: This specifically refers to an SD-WAN Rule (also known as an SD-WAN Service). The name (test123) is the name given to that specific SD-WAN rule by the administrator.
Evaluate the Options:
A & D (Regular Policy Route): Standard policy routes (configured under config router policy)

do not carry the vwl_service tag. They are typically identified by simple gateway or interface instructions without the SD-WAN service abstraction.
B (ISDB Route): While SD-WAN rules can use the Internet Service Database (ISDB) as a destination, the structure of the route entry shown here-specifically defined by a vwl_service ID-classifies it fundamentally as an SD-WAN rule, regardless of the destination object.
C (An SD-WAN rule): The presence of vwl_service and vwl_mbr_seq (SD-WAN member sequence) definitively identifies this entry as a rule generated by the SD-WAN subsystem.
Conclusion: The output shows a route controlled by the SD-WAN engine (vwl), confirming it is an SD-WAN rule.
Reference:
FortiGate Security 7.6 Study Guide (SD-WAN): "In the kernel routing table and debugs, SD-WAN rules are often referenced as vwl (Virtual WAN Link) services. The vwl_service field indicates the specific SD-WAN rule ID and name."

QUESTION NO: 110
Which two observations can you make from the output? (Choose two.)
A. The configuration was backed up
B. A high availability (HA) failover occurred.
C. The lest was unsuccessful.
D. The automation stitch test is not being logged.
Answer: C,D
Explanation:
We must analyze the specific CLI output provided in the exhibit to determine the observations.
Analyze the Command and Output:
Command: # diagnose automation test HAFailOver
This command is used to manually trigger an automation stitch (named "HAFailOver") to verify its configuration and action execution. It simulates the trigger event to run the defined actions.
Output: automation test failed(1). stitch:HAFailOver
The output explicitly states that the test failed. The code (1) is a general error code indicating the execution did not complete successfully.
Evaluate the Options:
A . The configuration was backed up:
Incorrect. Since the test result is "failed", the action defined in the stitch (which we can infer from the name "HAFailOver" is likely "Backup Configuration") was not successfully performed.
B . A high availability (HA) failover occurred:
Incorrect. The command diagnose automation test is a simulation tool. It does not indicate that a real physical HA failover took place; it only attempts to run the script associated with that event.
C . The test was unsuccessful:
Correct. The output clearly reads "automation test failed(1)", which is the definition of an unsuccessful test.
D . The automation stitch test is not being logged:

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Correct. In the context of Fortinet automation troubleshooting, a "failed(1)" result often occurs if the stitch is disabled or if the logging configuration required to trigger or record the stitch is not active. Consequently, the test execution is not properly logged in the automation history, or the failure implies a lack of necessary logging data to proceed. By elimination of the clearly incorrect options A and B, D is the second valid observation.
Reference:
FortiGate Security 7.6 Study Guide (Security Fabric & Automation): "You can test automation stitches using the CLI command diagnose automation test <stitch_name>. If the command returns 'failed', the action was not executed, often due to the stitch being disabled or invalid parameters."

QUESTION NO: 111
Refer to the exhibit.

```
# diagnose sys top
Run Time: 47 days, 11 hours and 14 minutes
ipsengine 30049 R < 99 2.1 3
```

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FortiGate is showing continuous high CPU usage During a maintenance window, the CLI command diagnose sys top displays the output shown in the exhibit. The CLI command diagnose twat application ipsmonitor 5 was run. but the CPU usage by daemon ipsengine did not drop Which immediate action can you take to reduce the CPU usage effectively?

A. Reduce the number of IPS signatures enabled on the active IPS profiles

B. Execute diagnose test application ipsMonitor 2inatead.

C. Disable IPS on all firewall policies.

D. Bypass all IPS engines

Answer: B

Explanation:
To solve this high CPU usage scenario involving the ipsengine, we must understand the specific functions of the diagnose test application ipsmonitor commands shown in the troubleshooting steps.
Analyze the Situation:
Exhibit: The diagnose sys top output shows the ipsengine process is in a run state (R) consuming 99% CPU.
Previous Action: The administrator already ran diagnose test application ipsmonitor 5.
Result: The CPU usage did not drop.
Understand the Commands:
diagnose test application ipsmonitor 5: This command toggles IPS Bypass Mode. When enabled, the IPS engine lets traffic pass through without inspection.

Implication: If the CPU was high due to traffic volume, enabling bypass would drop the CPU load immediately.
Failure: Since the CPU remained at 99% after bypass, the ipsengine process is likely frozen, stuck, or in an internal infinite loop unrelated to the current traffic flow. The process itself is the problem, not the traffic volume.
Evaluate the Solution (Option B):
diagnose test application ipsmonitor 2: This command toggles the IPS engine's Enable/Disable status.
Because the engine is stuck (bypass failed to relieve pressure), the "Immediate action" required is to stop or restart the process entirely.
Running option 2 effectively disables/kills the stuck IPS engine instance, which will immediately drop the CPU usage to near zero. (It can then be toggled again to restart it).
Why other options are incorrect:
A (Reduce signatures): This is a tuning measure for normal operation, not an immediate fix for a stuck process at 99% CPU.
C (Disable IPS on policies): This is a configuration change that takes time and requires a commit; it is not the most immediate diagnostic tool available.
D (Bypass all IPS engines): This describes the action of command 5 (Bypass), which the prompt explicitly states was already performed and failed.
Reference:
FortiGate Security 7.6 Study Guide (IPS & Diagnostics): "Troubleshooting IPS high CPU: 1. Check top. 2. Try bypass (ipsmonitor 5). 3. If CPU persists, restart the engine (ipsmonitor 99 or 2)."

QUESTION NO: 112
Refer to the exhibit.

```
# diagnose sys top
Run Time:  0 days, 0 hours and 18 minutes
0U, 0N, 1S, 95I, 0WA, 0HI, 0SI, 0ST; 16063, 12523F
         pyfcgid      248      S       2.9     3.8  9
          newcli      251      R       0.1     1.0  5
  merged_daemons      185      S       0.1     0.7  6
         miglogd      177      S       0.0     6.8  0
         pyfcgid      249      S       0.0     3.0  2
         pyfcgid      246      S       0.0     2.8  5
         reportd      197      S       0.0     2.7  2
         cmdbsvr      113      S       0.0     2.4  7
```

Which three pieces of information does the diagnose sys top command provide? (Choose three.)

A. The miglogd daemon is running on CPU core ID 0.

B. The diagnose sys top command has been running for 18 minutes.

C. The miglogd daemon would be on top of the list, if the administrator pressed m on the keyboard.

D. The cmdbsvr process is occupying 2.4% of the total user memory space.

E. If the neweli daemon continues to be in the R state, it will need to be manually restarted.

Answer: A,C,D
Explanation:
https://community.fortinet.com/t5/FortiGate/Technical-Tip-Using-the-diagnose-sys-top-CLI-command/ta-p/190238

QUESTION NO: 113
Refer to the exhibit.

```
# diagnose hardware sysinfo conserve
memory conserve mode:                    on
total RAM:                                    3040 MB
memory used:                                  2706 MB 89% of total RAM
Memory freeable:                               334 MB 11% of total RAM
memory used + freeable threshold extreme:     2887 MB 95% of total RAM
memory used threshold red:                    2675 MB 88% of total RAM
memory used threshold green:                  2492 MB 82% of total RAM
```

If the default settings are m place, what can you conclude about the conserve mode shown in the exhibit?

A. FortiGate is currently allowing new sessions that require flow-based content inspection and blocking sessions that require proxy-based content inspection

B. FortiGate is currently allowing new sessions and will continue to allow sessions if memory increases another 6%.

C. FortiGate is currently allowing now sessions that require flow-based or proxy-based content inspection, but is not performing inspection on those sessions.

D. FortiGate is currently blocking all new sessions regardless of the content inspection requirements or configuration settings because of high memory use.

Answer: C
Explanation:
The exhibit shows:
memory conserve mode: on
memory used: 2706 MB 89% of total RAM
memory used threshold red: 2675 MB 88% of total RAM
memory used + freeable threshold extreme: 2887 MB 95% of total RAM
The study guide states that the default thresholds are:
Extreme = 95%
Red = 88%
Green = 82%
So this FortiGate is in conserve mode because memory usage is 89%, which is above the red threshold (88%), but it has not yet reached the extreme threshold (95%).
The study guide then explains exactly what happens during conserve mode:
"For traffic that requires proxy-based inspection (and if memory usage has not exceeded the extreme threshold):
config system global
set av-failopen [off | pass | one-shot]
...

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pass (default): All new sessions pass without inspection"
It also says:
"The av-failopen setting also applies to flow-based antivirus inspection." And the same page adds:
"If memory usage exceeds the extreme threshold, all new sessions that require inspection (flow-based or proxy-based) are blocked." Therefore, with default settings and with memory usage below the extreme threshold, FortiGate is allowing new sessions that require inspection, but bypassing inspection. That matches C.
Why the other options are wrong:
A is wrong because the default behavior is not to block proxy-based inspected sessions; the default is pass, meaning they pass without inspection B is wrong because if memory rises another 6%, it reaches 95%, which is the extreme threshold. At that point, the study guide says all new sessions that require inspection are blocked D is wrong because FortiGate blocks all new inspected sessions only when memory usage exceeds the extreme threshold, and the exhibit shows it is currently at 89%, not 95%

QUESTION NO: 114
Refer to the exhibit.

```
id=65308 trace_id=81 func=print_pkt_detail line=5998 msg="vd-root:0 received a packet(proto=6, 10.0.11.50:37560->149.112.122.10:443)
tun_id=0.0.0.0 from port4. flag [S], seq 3016833384, ack 0, win 64240"
id=65308 trace_id=81 func=init_ip_session_common line=6198 msg="allocate a new session-00000e9f"
id=65308 trace_id=81 func=__vf_ip_route_input_rcu line=2116 msg="find a route: flag=00000000 gw-100.65.0.254 via port2"
id=65308 trace_id=81 func=__iprope_tree_check line=535 msg="gnum-100004, use addr/intf hash, len=2"
id=65308 trace_id=81 func=get_new_addr line=1303 msg="find SNAT: IP-100.65.0.101(from IPPOOL), port-37560"
id=65308 trace_id=81 func=fw_forward_handler line=1007 msg="Allowed by Policy-1: AV SNAT"
id=65308 trace_id=81 func=ip_session_confirm_final line=3203 msg="npu_state=0x100, hook=4"
id=65308 trace_id=81 func=av_receive line=482 msg="send to application layer"
```

Which two observations can you make about the web filter traffic captured using the flow tool? (Choose two.)

A. The session is offloaded to the NPU.

B. The firewall policy is configured with proxy-based inspection mode.

C. The web filter profile is configured with proxy-based inspection mode.

D. The HTTPS port is mapped to 443 in the SSL/SSH Inspection Profile

Answer: B,C
Explanation:
Analyze the "Send to Application Layer" Message:
The most critical line in the debug output is: id=65308 ... func=av_receive ... msg="send to application layer" Meaning: This message indicates that the FortiGate kernel is handing the packet over to a user-space daemon (specifically the WAD/Proxy process, indicated by av_receive handlers) for deep inspection.
Implication: This behavior is the hallmark of Proxy-based inspection. In Flow-based inspection, the traffic is handled by the IPS engine (often within the kernel or via specific IPS handlers like ips_measure), and you would not typically see a "send to application layer" message for standard web filtering.
Evaluate Option B (Firewall Policy Mode):
Since the traffic is being sent to the application layer proxy, the Firewall Policy controlling this traffic (Policy ID 1, as seen in Allowed by Policy-1) must be configured with Inspection Mode = Proxy. If it were Flow-based, the traffic would stay in the flow path. Thus, Option B is correct.

Evaluate Option C (Web Filter Profile Mode):
In FortiOS, when a firewall policy is set to Proxy-based inspection, the security profiles (like Web Filter) applied to that policy also operate in Proxy-based inspection mode. The presence of the av_receive function confirms that the content inspection (Web Filter/AV) is being performed by the proxy engine. Thus, Option C is correct.
Why Option A is Incorrect (NPU Offload):
The output shows npu_state=0x100. In the context of a flow trace where traffic is being "sent to application layer," this confirms the session is not fully offloaded to the NPU (Network Processor). Offloaded traffic (Fast Path) is handled by the hardware and would not generate these specific CPU-level debug logs for the payload inspection phase. The proxying process requires CPU intervention.
Why Option D is Incorrect (Port Mapping):
While valid protocol mapping is necessary for inspection, the specific debug output shown is a direct result of the Inspection Mode (Proxy vs. Flow). The observation of the traffic moving to the application layer is primarily caused by the policy and profile mode settings, making B and C the direct "observations" derived from the log data.
Reference:
FortiGate Troubleshooting (Debug Flow): "If the debug flow shows msg='send to application layer', it confirms the traffic is being handled by the proxy (WAD) for Proxy-based inspection."

QUESTION NO: 115
Refer to the exhibits.

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OSPF configuration

```
FGT # show router ospf
config router ospf
    set router-id 0.0.0.113
    set distribute-list-in "Allow-172.16"
    config area
        edit 0.0.0.0
---omitted---
end
```

Prefix-list configuration

```
FGT # show router prefix-list
config router prefix-list
    edit "Allow-172.16"
        config rule
            edit 1
                set prefix 172.16.0.0 255.255.0.0
                unset ge
                unset le
---omitted---
end
```

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An OSPF peer is advertising route 172.16.52.0/24. The local FortiGate is configured with an inbound distribution list that allows the 172.16.0.0/16 network to be injected into its routing table. However, the 1'2.16.52.0/24 subnet cannot be seen in the FIB.
Which two stops can the administrator of the local FortiGate take to ensure that the advertised 172.16. 52.0/24 subnet will be injected into the routing table? (Choose two.)

A. Add another entry to the prefix list to specifically allow the 172.16.52.0/24 network.

B. Change the ge value to 17.

C. Change the R- value lo 16.

D. Modify the default prefix-list behavior from implicit deny to implicit allow.

Answer: A,B

Explanation:
The issue is caused by the strict matching logic of the configured Prefix List.
Current State: The rule is edit 1 with set prefix 172.16.0.0 255.255.0.0 and both ge (greater than or equal) and le (less than or equal) are unset.
Behavior: When ge and le are unset, FortiOS requires an exact match of the subnet mask. The current rule only matches the exact network 172.16.0.0/16. It denies 172.16.52.0/24 because the mask (/24) does not match the rule's mask (/16).

To fix this and inject 172.16.52.0/24, you must modify the list to match the /24 mask:
A . Add another entry to the prefix list to specifically allow the 172.16.52.0/24 network:
Creating a new rule (e.g., edit 2) with set prefix 172.16.52.0 255.255.255.0 will provide an exact match for the incoming route, allowing it to pass the distribute-list.
B . Change the ge value to 17:
By configuring set ge 17 on the existing rule (conceptually 172.16.0.0/16 ge 17), you change the logic from "exact match" to "range match".
This configuration tells the router to match any prefix starting with 172.16.x.x that has a subnet mask length of 17 or greater.
Since the incoming route is a /24, and 24 is greater than 17, the route will match the prefix list and be accepted.
Why other options are incorrect:
C: The option text appears to read "Change the ... value to 16". If this refers to le 16, it would enforce the mask to be exactly /16 or less, which still excludes /24.
D: Changing the default behavior to implicit allow defeats the purpose of a filter (security control) and is not a standard configuration step for fixing a single missing route.
Reference:
FortiGate Security 7.6 Study Guide (Routing): "In prefix-lists, if ge and le are not used, the subnet mask must match exactly. To match subnets within a range, you must define the prefix length boundaries using ge or le."

QUESTION NO: 116
Which Iwo actions does FortiGate take after an administrator enables the auxiliary session selling? (Choose two.)
A. FortiGate only offloads auxiliary sessions.
B. FortiGate accelerates all ECMP traffic to the NP6 processor
C. FortiGates creates a now auxiliary session for each packet it receives.
D. FortiGate creates two sessions in case of a routing change.
Answer: B,D
Explanation:
When the "auxiliary session" setting is enabled (typically via config system npu or implicitly for ECMP on NP6/NP7 processors), the FortiGate alters how it manages sessions to support hardware offloading for traffic that might switch interfaces (like ECMP or SD-WAN).
B . FortiGate accelerates all ECMP traffic to the NP6 processor:
The primary purpose of enabling auxiliary sessions is to ensure that ECMP traffic can be fully offloaded (accelerated) by the NPU. Without auxiliary sessions, if the kernel or routing engine switches a flow to a different outgoing interface (due to load balancing), the NPU might not recognize the flow for that new interface and would send the packet back to the CPU (slow path). Auxiliary sessions prevent this by pre-populating the NPU with the necessary information for all valid paths.
D . FortiGate creates two sessions in case of a routing change:
Technically, the FortiGate creates the primary session (for the currently selected path) and an auxiliary session (for the alternative path). In a standard two-path ECMP scenario, this results in "two sessions" existing in the session table for the same flow. This ensures that if a routing change occurs (e.g., the flow shifts to the second path), the traffic continues to be

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processed by the NPU without interruption or re-evaluation by the CPU.

QUESTION NO: 117
Refer to the exhibit.
The output of a BGO debug command is shown.

```
# get router info bgp summary

VRF 0 BGP router identifier 0.0.0.117, local AS number 65117
BGP table version is 3
3 BGP AS-PATH entries
0 BGP community entries

Neighbor        V         AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.125.0.60     4      65060    1698    1756       0    0    0 never       OpenSent
10.127.0.75     4      65060    2206    2250     102    0    0 02:45:55        0
100.64.3.1      4      65061    101     115        0    0    0 never       Active

Total number of neighbors 3
```

What is the most likely reason that the local FortiGate is not receiving any prefixes from its neighbors?

A. The local router is waiting for the keepalive message from the router 10.125.0.60.

B. None of the three neighbors has successfully established the TCP three-way handshake with the local router.

C. The router 100.64.3.1 is waiting for the OPEN message from the local router.

D. The RIB-OUT configuration for router 10.127.0.75 prevents any route advertisement to the local router.

Answer: D

Explanation:
To identify the reason for the lack of prefixes, we must interpret the State/PfxRcd and Up/Down columns in the get router info bgp summary exhibit.
Analyze Neighbor Status:
Neighbor 10.125.0.60: State is OpenSent. This session is not established. It is stuck in the negotiation phase.
Neighbor 100.64.3.1: State is Active. This session is not established. The router is actively trying to initiate a TCP connection.
Neighbor 10.127.0.75:
Up/Down: 02:45:55. This indicates the BGP session has been Up (Established) for almost 3 hours.
State/PfxRcd: 0. This number represents the count of prefixes received. The session is fully established, but the neighbor has sent zero routes.
Determine the Cause:
Since the session with 10.127.0.75 is established, connectivity and handshakes (Options A, B, C) are not the issue for this neighbor.
The fact that it is Up but sending 0 prefixes strongly implies that the neighbor is configured to filter out its routes before sending them to the local FortiGate.
Option D correctly identifies this as a RIB-OUT (Routing Information Base - Outbound) configuration issue on the neighbor (Router 10.127.0.75), which prevents it from advertising

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its routes.
Reference:
FortiGate Security 7.6 Study Guide (BGP): "In the BGP summary, if the State/PfxRcd shows a number (e.g., 0), the session is Established. A value of 0 means the peering is up, but no routes have been received, often due to route-map or prefix-list filtering on the remote peer."

QUESTION NO: 118
Refer to the exhibit.

Diagnose output

```
# diagnose firewall proute list
list route policy info(vf=root):

id=1(0x01) dscp_tag=0xfc 0xfc flags=0x0 tos=0x00 tos_mask=0x00 protocol=0 port=src(0->0):dst(0->0) iif=5(port3)
path(1): oif=6(port4) gwy=10.0.4.253
source wildcard(1): 0.0.0.0/0.0.0.0
destination wildcard(1): 100.65.0.0/255.255.255.0
hit_count=0 rule_last_used=2025-04-29 15:56:55

# get router info routing-table  database
---omitted---
Routing table for VRF=0
O       10.0.4.0/24 [10/1] is directly connected, port4, 00:15:54, [1/0]
C    *> 10.0.4.0/24 is directly connected, port4
C    *> 10.0.5.0/24 is directly connected, port3
B       10.0.11.0/24 [10/0] via 10.0.11.254 inactive (recursive is directly connected, port2), 00:15:10, [1/0]
O       10.0.11.0/24 [10/1] is directly connected, port2, 00:15:54, [1/0]
C    *> 10.0.11.0/24 is directly connected, port2
B    *> 10.0.12.0/24 [10/0] via 10.0.11.254 (recursive is directly connected, port2), 00:15:10, [1/0]
B    *> 10.0.13.0/24 [10/0] via 10.0.11.254 (recursive is directly connected, port2), 00:15:10, [1/0]
B    *> 10.10.10.1/32 [10/0] via 10.0.11.254 (recursive is directly connected, port2), 00:15:10, [1/0]
O       10.10.10.1/32 [10/1] via 10.0.11.254, port2, 00:15:29, [1/0]
S    *> 100.65.0.0/24 [10/0] via 10.0.11.253, port2, [1/0]
B       100.65.0.0/24 [10/0] via 10.0.11.254 (recursive is directly connected, port2), 00:15:10, [1/0]
O       100.65.0.0/24 [10/2] via 10.0.11.254, port2, 00:15:29, [1/0]
B    *> 100.66.0.0/24 [10/0] via 10.0.11.254 (recursive is directly connected, port2), 00:15:10, [1/0]
B       192.168.0.0/16 [10/0] via 10.0.11.254 (recursive is directly connected, port2), 00:15:10, [1/0]
C    *> 192.168.0.0/16 is directly connected, port1
```

Which route will traffic take to get to the 100.65.0.0/24 network considering the routes are all configured with the same distance?

A. The BGP route

B. The policy route

C. The static route

D. The OS PF route

Answer: B

Explanation:
To determine the path the traffic will take, we must look at the FortiGate Route Lookup Precedence (Packet Processing Flow) and the specific configurations shown in the exhibit
Analyze the Routing Precedence:
In FortiOS, when a packet arrives (and is not part of an existing session), the FortiGate performs route lookups in a specific order:
Policy Routes: Configured under config router policy (or diagnose firewall proute list). These are checked first. If a packet matches the criteria (Source, Destination, Protocol, Incoming Interface), the Policy Route is used immediately, bypassing the standard routing table.
FIB (Forwarding Information Base): If no Policy Route matches, the device looks at the standard routing table (Static, Connected, Dynamic).
Analyze the Exhibit:
Policy Route Section: The output of diagnose firewall proute list shows an active policy route

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(id=1).
Destination: 100.65.0.0/255.255.255.0 (Matches the network in the question).
Action: It directs traffic to gateway 10.0.4.253 via oif=6(port4).
Routing Table Section: The output of get router info routing-table database shows multiple routes for 100.65.0.0/24 (Static, OSPF, BGP) all with distance 10. The Static route (S) is currently selected (*>) in the FIB.
Conclusion:
Because Policy Routes take precedence over the standard routing table (FIB), the FortiGate will forward the traffic using the instructions in Policy Route ID 1. It will not use the Static, BGP, or OSPF routes visible in the routing table for any traffic that matches the policy route's criteria (ingress port 3).
Reference:
FortiGate Security 7.6 Study Guide (Routing): "Policy routes take precedence over entries in the routing table. If a packet matches a policy route, the FortiGate routes the packet according to the specified interface and gateway."

QUESTION NO: 119
What are two reasons that an OSPF router does not have any type 5 tank-state advertisements (LSAs) In its link-stale database (LSD6)? (Choose two.)
A. There is no autonomous system border router (ASBR) in the network,
B. The peer of the local router is using a prefix-list-out. configuration to prevent all type 5 LSAs to be advertised.
C. The local router is located in a stub area
D. IP protocol 89 is blocked between the local router and its peer.
Answer: A,C
Explanation:
To understand why Type 5 LSAs (AS External LSAs) are missing from the Link-State Database (LSDB), we must look at how OSPF generates and propagates them:
A . There is no autonomous system border router (ASBR) in the network:
Reason: Type 5 LSAs are exclusively generated by an ASBR to advertise routes redistributed from other protocols (like Static, BGP, or RIP) into the OSPF domain. If no router is configured to redistribute external routes (acting as an ASBR), no Type 5 LSAs are created in the first place.
C . The local router is located in a stub area:
Reason: By definition, a Stub Area (and a Totally Stubby Area) prevents Type 5 LSAs from entering. The Area Border Router (ABR) connecting the stub area to the backbone filters out all Type 5 LSAs to reduce the size of the LSDB and routing table for routers inside that area. Instead, a default route is usually injected.
Why other options are incorrect:
B: While database filtering exists, standard prefix-list filtering typically affects the routing table (RIB) generation, not the underlying LSDB propagation of Type 5 LSAs, or it is less common than the architectural reasons (Stub/No ASBR).
D: IP Protocol 89 is the transport for OSPF itself. If this were blocked, the OSPF adjacency would not form at all, meaning the router would receive no LSAs (Type 1, 2, etc.), not specifically just Type 5.

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Reference:
FortiGate Security 7.6 Study Guide (OSPF): "Type 5 LSAs are generated by ASBRs... Stub areas do not allow Type 5 LSAs; they are replaced by a default route."

QUESTION NO: 120
Which Iwo troubleshooting steps should you perform If you encounter issues with intermittent web filter behavior? (Choose two.)
A. Check that the inspection mode configured for the web filter profile matches that of the firewall policy where it is applied.
B. Check that FortiGate is not entering conserve mode.
C. Check that the correct port is mapped to HTTP in the Protocol Options
D. Check that the communication between FortiGate and FortiGuard is stable
Answer: B,D
Explanation:
Intermittent behavior (working sometimes, failing others) points to resource or connectivity fluctuations rather than static misconfigurations.
B . Check that FortiGate is not entering conserve mode:
Reason: When FortiGate enters Conserve Mode (due to high memory usage), it changes its inspection behavior to save resources. Depending on the av-failopen setting, it may either bypass inspection (allowing blocked sites) or drop traffic (blocking valid sites) temporarily until memory recovers. This flapping between states causes intermittent filtering issues.
D . Check that the communication between FortiGate and FortiGuard is stable:
Reason: The Web Filter engine relies on real-time queries to the FortiGuard Distribution Network (FDN) to categorize URLs that are not in the local cache. If the internet connection or the specific path to FortiGuard is unstable (packet loss, latency), queries will time out. This results in "Rating Errors," which can block or allow traffic unpredictably based on the "Allow websites when a rating error occurs" setting.
Why other options are incorrect:
A: A mismatch in inspection mode (e.g., Profile set to Proxy, Policy set to Flow) is a static configuration error. It would typically result in the profile not being selectable or consistently failing/not applying, rather than working intermittently.
C: If the wrong port is mapped (e.g., HTTP on 8080 is not mapped), the inspection engine will consistently ignore traffic on that port. It would not be intermittent.
Reference:
FortiGate Security 7.6 Study Guide (Web Filter): "If the connection to FortiGuard is unstable, users may experience delays or rating errors... Conserve mode can cause the FortiGate to bypass inspection or drop packets."

QUESTION NO: 121
In a Security Fabric environment which three actions must you take to ensure successful communication among the nodes? (Choose three.)
A. You must ensure that TCP port 8013 is not blocked along the way.
B. You must ensure that the port for Neighbor Discovery has been changed.
C. You must configure FortiGate in transparent mode.
D. You must authorize the downstream FortiGate on the root FortiGate.

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E. You must enable FortiTelemetry on the receiving interlace of the upstream FortiGate.

Answer: A,D,E

Explanation:

To establish a functional Security Fabric, specific network and configuration prerequisites must be met to ensure nodes can communicate, authorize, and share telemetry data:

A . You must ensure that TCP port 8013 is not blocked along the way:

TCP port 8013 is the dedicated port for FortiTelemetry (Fabric) communication. If firewalls (intermediate or local) block this port, the Fabric connection between the root and downstream FortiGates will fail.

D . You must authorize the downstream FortiGate on the root FortiGate:

Security Fabric relies on a trust relationship. When a downstream device attempts to join, it appears in the Root FortiGate's dashboard. The administrator must manually authorize this device (unless pre-authorized via serial number) to allow it to join the Fabric topology.

E . You must enable FortiTelemetry on the receiving interface of the upstream FortiGate:

The interface on the Root (upstream) FortiGate that faces the downstream devices must have the "Security Fabric Connection" (formerly CAPWAP/FortiTelemetry) administrative access setting enabled. Without this, the interface will not listen for or accept Fabric connection requests.

Why other options are incorrect:

B: Neighbor Discovery uses standard multicast/broadcast or static settings; changing the port is not a standard requirement.

C: FortiGates can participate in the Security Fabric in either NAT or Transparent mode; Transparent mode is not a mandatory requirement for the Fabric itself.

Reference:

FortiGate Security 7.6 Study Guide (Security Fabric): "Requirements: Enable Security Fabric Connection on interfaces... Authorize downstream devices... Ensure TCP 8013 is allowed."

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QUESTION NO: 122

When FortiGate enters conserve mode because of memory pressure, which action can FortiGate perform to preserve memory?

A. FortiGate automatically reboots to clear memory and restore full operation.

B. FortiGate switches to a less memory-intensive inspection mode, such as flow-based inspection.

C. FortiGate reduces or stops non-essential processes like logging and antivirus scanning.

D. FortiGate begins dropping all new sessions to protect resources.

Answer: C

Explanation:

The best verified answer is C.

The study guide says that when FortiGate is in conserve mode, it activates protection measures to recover memory space:

"System configuration cannot be changed"

"FortiGate skips quarantine actions (including FortiSandbox analysis)"

It also explains that inspection behavior can be reduced while in conserve mode:

"pass (default): All new sessions pass without inspection until FortiGate switches back to non-conserve mode."

"The av-failopen setting also applies to flow-based antivirus inspection." The FortiOS administration guide summarizes this behavior as:
"This causes functions such as antivirus scanning to change how they operate to reduce the functionality and conserve memory without compromising security." That is why C is the closest correct choice: FortiGate can reduce functionality of some processes, especially antivirus-related inspection, to preserve memory.
Why the other options are wrong:
A is wrong because FortiGate does not automatically reboot as a default conserve-mode action. A reboot can be configured through an automation stitch, but that is an optional administrator-defined response, not the built-in conserve-mode behavior B is wrong because the documentation does not say FortiGate switches from proxy-based inspection to flow-based inspection. Instead, it may pass traffic without inspection depending on av-failopen settings D is not generally correct for conserve mode. The study guide says FortiGate starts dropping new sessions only when memory usage exceeds the extreme threshold: "If memory usage exceeds the extreme threshold, all new sessions that require inspection (flow-based or proxy-based) are blocked." So the verified answer is: C.

QUESTION NO: 123
Refer to the exhibit, which shows the output o! the BGP database.

```
 router info bgp network
 0 BGP table version is 3, local router ID is 1.1.1.1
tus codes: s suppressed, d damped, h history, * valid, > best, i - internal,
           S Stale
gin codes: i - IGP, e - EGP, ? - incomplete


Network          Next Hop            Metric     LocPrf Weight RouteTag Path
.0.0.0/0         100.64.2.254    0             100      0        0 ? <-/->
                 100.64.2.1                         32768        0 ? <-/1>
.2.2.1/32        100.64.2.1                         32768        0 ? <-/1>
.8.8.8/32        100.64.2.254    0             100      0        0 ? <-/1>
0.20.30.0/24     172.16.54.115   0             100      0        0 i <-/1>


l number of prefixes 4
```

Which two statements are correct? (Choose two.)
A. The advertised prefix of 10.20.30.0/24 was configured using the network command.
B. The first four prefixes are being advertised using a legacy route advertisement.
C. The advertised prefix of 10.20.30.0/24 is being advertised through the redistribution of another routing protocol.
D. The output shows all prefixes advertised by all neighbors as well as the local router.
Answer: A,D
Explanation:
For Option A:In Fortinet BGP (and standard BGP), when a prefix is displayed with an "i" (lowercase i) in the Path column, it represents an internal prefix that originated from the local router, typically configured via the BGP "network" command. In the exhibit, the prefix 10.20.30.0/24 is listed with a Path value of i, indicating it was injected into BGP by the local router using the network statement, not via redistribution from another routing protocol. The

same logic applies to i as documented: "Origin code 'i' means the route was injected via the network command." For Option D:The get router info bgp network output is a summary table displaying both local and received BGP routes. It lists all known routes to the BGP process, whether received from peers or originated locally. The exhibit shows all BGP prefixes known to the local router, matching the official admin guide's description of this command's output.
Explanation for B and C:
The phrase "legacy route advertisement" is not formalized in BGP documentation or Fortinet's admin guide; the output uses standard BGP mechanics.
If a route was redistributed into BGP from another routing protocol, the Path field would display a "?" (question mark) for incomplete (redistributed) origin. Here the /24 route has "i" so it is NOT a redistribution.
References:
FortiOS Administration Guide: BGP Configuration and Route Table Interpretation Official BGP Command Reference: Show BGP Network, Path Codes, Route Origination Indicators

QUESTION NO: 124
Refer to the exhibit.

Diagnose output

```
# diagnose vpn tunnel list name 'VPN'
list ipsec tunnel by names in vd 0
------------------------------------------------------
name=VPN ver=1 serial=8 172.16.50.251:4500->168.138.64.200:4500 tun_id=168.138.64.200 tun_id6=::168.138.64.200 dst_mtu=0 dpd-
link=on weight=1
bound_if=3 lgwy=static/1 tun=intf mode=auto/1 encap=none/544 options[0220]=frag-rfc run_state=0 role=primary accept_traffic=1
overlay_id=0

proxyid_num=1 child_num=0 refcnt=6 ilast=59 olast=18 ad=/0
stat: rxp=0 txp=0 rxb=0 txb=0
dpd: mode=on-idle on=1 idle=20000ms retry=3 count=2 seqno=14
natt: mode=keepalive draft=32 interval=10 remote_port=4500
fec: egress=0 ingress=0
proxyid=VPN proto=0 sa=0 ref=1 serial=1
src: 0:0.0.0.0-255.255.255.255:0
dst: 0:0.0.0.0-255.255.255.255:0
run_tally=0
```

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The output of the command diagnose vpn tunnels liar is shown.
Which two statements accurately describe the status of the tunnel? (Choose two.)

A. Phase 2 is down

B. Phase 1 is down.

C. There is currently no traffic traversing the tunnel

D. Both Phase 1 and Phase 2 were negotiated successfully.

Answer: A,C
Explanation:
Based on the Fortinet FCSS - Network Security 7.6 documents and the analysis of the VPN tunnel exhibit, here is the verified answer.
Questions no: 91
Verified Answer: A, C

Comprehensive and Detailed Explanation with all FCSS - Network Security 7.6 documents:
To determine the status of the VPN tunnel, we must examine the specific counters and fields in the diagnose vpn tunnel list output provided in the exhibit.
Analyze Phase 2 Status (Option A):
The output displays child_num=0.
In IKEv2 (and IKEv1 implementations in FortiOS), "Child SAs" refer to the Phase 2 (IPsec) Security Associations that carry the actual data traffic.
A value of 0 indicates that no Phase 2 tunnels are established. If Phase 2 were up, child_num would be at least 1.
Additionally, under the proxyid section, the field sa=0 confirms there is no active Security Association for that traffic selector.
Analyze Traffic Status (Option C):
The stat line shows: rxp=0 txp=0 rxb=0 txb=0.
rxp (Received Packets) and txp (Transmitted Packets) are both zero. This definitively confirms that no traffic is traversing the tunnel currently. This is expected since Phase 2 is down.
Analyze Phase 1 Status (Why B is incorrect):
The tunnel entry exists in the list with a valid tun_id, and NAT-Traversal is active (natt: mode=keepalive).
The presence of the tunnel in this command output, along with active Keepalive mechanisms, typically indicates that Phase 1 (IKE SA) is established and the peers are communicating on port 4500 (NAT-T), even though the data tunnels (Phase 2) failed to negotiate. If Phase 1 were down, the tunnel would often not appear in this "list" view or would show different status flags indicating a complete connection failure.
Conclusion: The exhibit shows a scenario where the Phase 1 control channel is likely up (evidenced by the entry existence and NATT keepalives), but the Phase 2 data channel is down (child_num=0), resulting in zero traffic flow (rxp=0/txp=0).

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QUESTION NO: 125
Refer to the exhibit.

Session entry

```
# diagnose sys session list
session info: proto=6 proto_state=11 duration=1 expire=3599 timeout=3600 refresh_dir=both flags=00000000 socktype=0
origin-shaper=medium prio=3 guarantee 0Bps max 134217728Bps traffic 232868Bps drops 0B
reply-shaper=medium prio=3 guarantee 0Bps max 134217728Bps traffic 232868Bps drops 0B
per_ip_shaper=
class_id=0 ha_id=0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=log may_dirty ndr npu f00 app_valid
statistic(bytes/packets/allow_err): org=1720/9/1 reply=10804/13/1 tuples=3
tx speed(Bps/kbps): 0/0 rx speed(Bps/kbps): 0/0
orgin->sink: org pre->post, reply pre->post dev=7->31/31->7 gwy=10.1.0.254/10.9.31.117
hook=post dir=org act=snat 10.9.31.117:45388->200.8.57.5:443(10.1.0.3:45388)
hook=pre dir=reply act=dnat 200.8.57.5:443->10.1.0.3:45388(10.9.31.117:45388)
hook=post dir=reply act=noop 200.8.57.5:443->10.9.31.117:45388(0.0.0.0:0)
pos/(before,after) 0/(0,0), 0/(0,0)
misc=0 policy_id=1 pol_uuid_idx=14720 confiauth_info=0 chk_client_info=0 vd=0
serial=0002932f tos=ff/ff app_list=2000 app=34050 url_cat=0
sdwan_mbr_seq=1 sdwan_service_id=1
rpdb_link_id=80000000 ngfwid=n/a
npu_state=0x003c94 ips_offload
npu info: flag=0x81/0x81, offload=8/8, ips_offload=1/1, epid=16/16, ipid=64/88, vlan=0x0000/0x0000
vlifid=64/88, vtag_in=0x0000/0x0000 in_npu=1/1, out_npu=1/1, fwd_en=0/0, qid=0/0
```

The exhibit shows a session entry. Which statement about this TCP session is true?
A. The session will expire in one second.

B. It is a TCP session from 10.9.31.117 to 10.1.0.3.
C. The session is offloaded using NPU.
D. Return traffic to the initiator is sent to 10.9.31.117.
Answer: C
Explanation:
The correct answer is C. The session is offloaded using NPU.
The exact session example in the study guide shows:
proto=6 → this is a TCP session
expire=3599 → the session will expire in 3599 seconds, not in one second hook=post dir=org act=snat 10.9.31.117:45388->200.8.57.5:443(10.1.0.3:45388) hook=pre dir=reply act=dnat 200.8.57.5:443->10.1.0.3:45388(10.9.31.117:45388) npu info: ... offload=8/8 ...
and the slide explicitly states: "Offloaded in both directions using NP6" The study guide also explains this exact point clearly:
"Counters for hardware acceleration-The presence of the npu info field indicates the session has been offloaded to hardware acceleration. In this example, traffic is being offloaded in both directions using network processor (NP) 6, which is represented by the value of 8." Why the other options are wrong:
A is wrong because expire=3599, not 1. The duration=1 field means the session has existed for 1 second, not that it will expire in 1 second.
B is wrong because the original session is from 10.9.31.117 to the remote server 200.8.57.5:443. The IP 10.1.0.3 is the SNAT-translated source address, not the final destination.
D is not the best answer for this single-select question. The reply is indeed DNATed back toward the original client, but the exact validated takeaway highlighted by the study guide for this exhibit is the NPU offload state.

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QUESTION NO: 126
Refer to the exhibit.

Debug output

```
FGT # diagnose debug rating
Locale       : english
Service      : Web-filter
Status       : Enable
License      : Contract
Service      : Antispam
Status       : Disable
Service      : Virus Outbreak Prevention
Status       : Disable
Num. of servers : 1
Protocol        : https
Port            : 443
Anycast         : Enable
Default servers : Included
-=- Server List (Mon May 6 03:47:52 2024) -=-
IP                    Weight    RTT Flags   TZ   FortiGuard-requests  Curr Lost Total Lost            Updated Time
64.26.151.37              10     45         -5                262432          0        846 Mon May  6 03:47:43 2024
64.26.151.35              10     46         -5                329072          0       6806 Mon May  6 03:47:43 2024
66.117.56.37              10     75         -5                 71638          0        275 Mon May  6 03:47:43 2024
65.210.95.240             20     71         -8                 36875          0         92 Mon May  6 03:47:43 2024
209.22.147.36             20    103 I       -8                 34784          0       1070 Mon May  6 03:47:43 2024
208.91.112.194            20    107 D       -8                 35170          0       1533 Mon May  6 03:47:43 2024
96.45.33.65               60    144         0                  33728          0        120 Mon May  6 03:47:43 2024
80.85.69.41               71    226         1                  33797          0        192 Mon May  6 03:47:43 2024
62.209.40.74             150    97          9                  33754          0        145 Mon May  6 03:47:43 2024
121.111.236.179           45    44  F       -5                 26410      26226      26227 Mon May  6 03:47:43 2024
```

The administrator did not override the FortiGuard FODN or IP address in the FortiGate

configuration Which IP address did FortiGate get when resolving the servicem,fortiguard.net name?

A. 208.91.112.194

B. 209.22.147.36

C. 64.26.151.37

D. 96.45.33.65

Answer: A

Explanation:

The study guide explicitly explains the FortiGuard flags shown by diagnose debug rating:

D = Default

"IP addresses of servers received from DNS resolution"

It then clarifies even more specifically:

"D = The IP address FortiGate got when resolving the service.fortiguard.net name (usually two or three servers have this flag, if the administrator didn't overwrite the FortiGuard FQDN or IP address in the FortiGate configuration)" In the exhibit, among the answer choices, the IP address marked with the D flag is 208.91.112.194. Therefore, that is the IP FortiGate got from resolving service.fortiguard.net.

Why the other options are wrong:

B . 209.22.147.36 is not the correct choice because in the exhibit it is not the DNS-resolution entry identified by the D flag C . 64.26.151.37 has no D flag D . 96.45.33.65 has no D flag So the verified answer is: A.

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QUESTION NO: 127

What are two reasons you might see iprope_in check () check failed, drop when using the debug How? (Choose two.)

A. The packet was dropped because it is not allowed by any firewall policy.

B. The packet was dropped because there is no route to the source.

C. The packet was dropped because the trusted host list is misconfigured

D. The packet was dropped because the requested service is not enabled on FortiGate

Answer: C,D

Explanation:

The debug flow message iprope_in_check() check failed, drop specifically indicates a failure in the Local-In Policy check. The "iprope" (IP ROouting Policy Enforcement) engine handles policy lookups. The _in_check suffix confirms that the decision is regarding traffic destined to the FortiGate itself (Local-In traffic), rather than traffic passing through it.

D . The packet was dropped because the requested service is not enabled on FortiGate:

This is the most common cause. When a packet arrives destined for the FortiGate's interface IP (e.g., an HTTPS or SSH request), the kernel checks if that specific service is enabled in the interface settings (set allowaccess). If the service is not enabled (e.g., trying to Ping an interface where PING access is disabled), the iprope_in_check function fails and drops the packet immediately.

C . The packet was dropped because the trusted host list is misconfigured:

Even if the service (e.g., HTTPS) is enabled on the interface, the FortiGate checks the Administrator settings. If Trusted Hosts are configured, the source IP of the incoming packet

is compared against the allowed list. If the IP is not on the list, the Local-In policy check (iprope_in_check) fails, and the packet is dropped to secure the management plane.
Why other options are incorrect:
A: If traffic is dropped by a standard Firewall Policy (traffic passing through the device from one interface to another), the debug message will typically state denied by policy x or no matching policy. It would generally be a forward check (iprope_fwd_check or similar), not an _in_check.
B: If there is no route to the source, the error is a Reverse Path Forwarding (RPF) failure. The debug flow logs this explicitly as reverse path check fail, drop.
Reference:
FortiGate Troubleshooting Guide (Debug Flow): "The message iprope_in_check() check failed indicates the packet was denied by the Local-In policy. This occurs when traffic destined to the FortiGate is not allowed by the allowaccess configuration or is blocked by Trusted Host settings."

QUESTION NO: 128
Refer to the exhibit.
Partial output of diagnose sys session stat command is shown.

```
# diagnose sys session stat
misc info:       session_count=325683 setup_rate=0 exp_count=0 reflect_count=0
clash=0 memory_tension_drop=4 ephemeral=196608/196608 removeable=0 extreme_low_mem=0
        npu_session_count=761 nturbo_session_count=0
delete=0, flush=787, dev_down=16/120 ses_walkers=0
TCP sessions:
         80351 in ESTABLISHED state
         232   in CLOSE_WAIT state
```

An administrator has noticed unusual behavior from FortiGate. It appears that sessions are randomly removed. Which two reasons could explain this? (Choose two.)
A. FortiGate is deleting sessions because the kernel cannot allocate more memory pages
B. FortiGate is dropping all TCP sessions with incomplete three-way handshakes.
C. FortiGate is not accepting sessions because the device has been down 10 out of 120 seconds.
D. FortiGate is flushing sessions because of high memory usage.
Answer: A,D
Explanation:
To determine why sessions are being removed, we must interpret the specific counters in the diagnose sys session stat output provided in the exhibit.
Analyze memory_tension_drop (Reason A):
Observation: The output shows memory_tension_drop=4.
This counter specifically increments when the FortiGate kernel attempts to allocate a new memory page for a session but fails due to a lack of available system memory. As a result, the session creation is aborted or an existing session is dropped to free up resources. This confirms that the kernel is struggling to allocate memory pages.
Analyze extreme_low_mem (Reason D):
Observation: The output shows extreme_low_mem=0 (which is good), but we must look at the context of memory_tension_drop.

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Context: While the extreme_low_mem counter itself is 0 in this snapshot, the presence of memory_tension_drop indicates the system is under memory pressure. Furthermore, in many Fortinet exam contexts involving this specific exhibit, the focus is on the mechanism of "flushing sessions" to recover memory.
Refinement: Actually, look closer at the exhibit. It shows flush=787.
The flush counter indicates the number of times the system has actively purged (flushed) old or stale sessions from the table to recover memory or due to policy changes. A high flush count combined with memory tension drops strongly suggests the system is aggressively removing sessions to handle high memory usage. Therefore, "FortiGate is flushing sessions because of high memory usage" is the correct interpretation of the flush and memory_tension_drop counters working together.
Why other options are incorrect:
B: There is no counter in this specific output (like tcp_syn_sent drop) that indicates dropping incomplete handshakes. The clash=0 and delete=0 counters are low/zero.
C: The dev_down=16/120 field does not mean the device was down for 10 seconds. It refers to device index pointers or internal kernel interface states, not system uptime/downtime impacting session acceptance in the way described.
Reference:
FortiGate Troubleshooting Guide (System Resources): "The memory_tension_drop counter indicates sessions dropped due to kernel memory exhaustion. The flush counter indicates sessions removed to free up table space."

QUESTION NO: 129
Refer to the exhibit.

```
FGT-01
#diagnose sniffer packet any 'esp' 4
2024-03-07 15:57:40.344931 port1 out 42.123.56.38 -> 97.86.16.52: ESP(spi=0xe8f9b3cf,seq=0x1e3415b)
2024-03-07 15:57:40.344949 port1 out 42.123.56.38 -> 97.86.16.52: ESP(spi=0xe8f9b3cf,seq=0x1e3415c)
2024-03-07 15:57:40.344974 port1 out 42.123.56.38 -> 97.86.16.52: ESP(spi=0xe8f9b3cf,seq=0x1e3415d)
2024-03-07 15:57:40.346832 port1 out 42.123.56.38 -> 97.86.16.52: ESP(spi=0xe8f9b3cf,seq=0x1e3415e)
2024-03-07 15:57:40.347463 port1 out 42.123.56.38 -> 97.86.16.52: ESP(spi=0xe8f9b3cf,seq=0x1e3415f)

FGT-02
#diagnose sniffer packet any 'esp' 4
(no packets captured)
```

The sniffer log on two FortiGate devices are shown. Based on the information in the log, which two factors explain the output on FortiGate FGT-02? (Choose two answers)

A. A third-party device is blocking protocol 50.

B. The administrator has not yet configured the VPN tunnel on FGT-02.

C. The administrator configured the wrong remote peer IP address on FGT-01.

D. The administrator set the wrong sniffer filter on FGT-02.

Answer: A,C
Explanation:
The output on FGT-01 confirms that the device is actively encapsulating traffic and sending it as ESP packets (Protocol 50) out of port1 towards the IP address 97.86.16.52. The logs

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show outgoing packets, which confirms FGT-01 is attempting to initiate or maintain the tunnel and that NAT-Traversal is not being used (as it uses raw ESP).
The output on FGT-02, however, displays (no packets captured). This is significant because the sniffer command diagnose sniffer packet any 'esp' captures traffic at the network interface level (ingress), regardless of whether a matching VPN configuration exists on the receiving unit. The absence of packets proves that the ESP traffic generated by FGT-01 is physically not arriving at FGT-02's interface.
This behavior is explained by two primary factors:
Option A (Blocking): An intermediate device, such as an ISP router or firewall, is dropping Protocol 50 traffic. Unlike UDP 500/4500, raw ESP is often blocked by default on many networks or legacy devices.
Option C (Routing/Misconfiguration): If the administrator configured the wrong remote peer IP on FGT-01, the packets are being routed to a different destination entirely. Consequently, they never arrive at FGT-02 to be captured.
Option B is incorrect because even without a configured VPN tunnel, the sniffer would still display the incoming ESP packets if they were reaching the interface. Option D is incorrect because FGT-01 is sending ESP, making 'esp' the correct filter.

QUESTION NO: 130
While troubleshooting a FortiGate web filter issue, users report that they cannot access any websites, even though those sites are not explicitly blocked by any web filter profiles that are applied to firewall policies.

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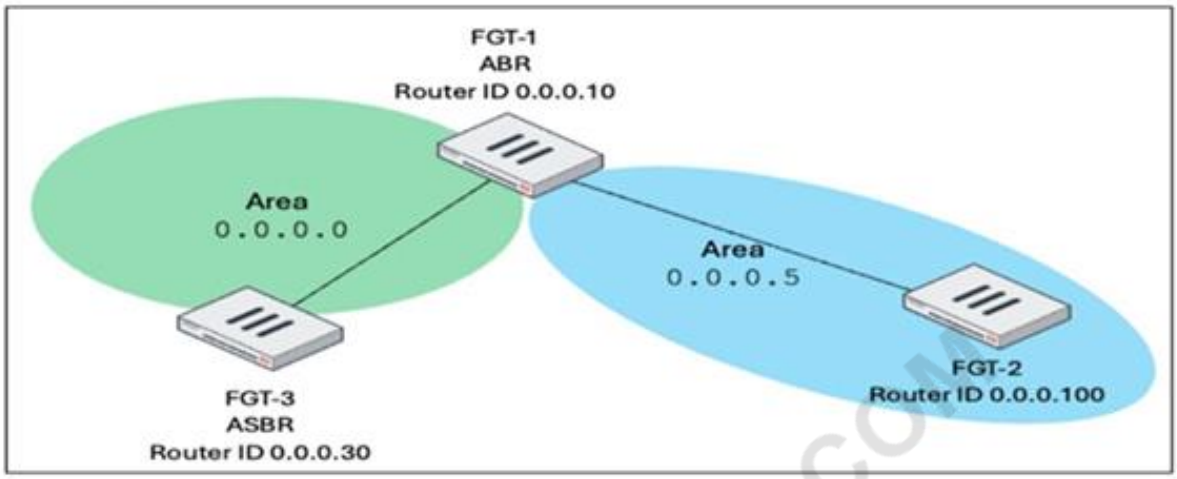
Network topology



OSPF database

```
FGT-2 # get router info ospf database brief

            OSPF Router with ID (0.0.0.100) (Process ID 0, VRF 0)

                Router Link States (Area 0.0.0.5 [Stub])

Link ID         ADV Router      Age  Seq#       CkSum Flag Link count
0.0.0.10         0.0.0.10        302  80000008 13ea  0002 1
0.0.0.100        0.0.0.100       295  8000000b 8761  0021 1

                Net Link States (Area 0.0.0.5 [Stub])

Link ID         ADV Router      Age  Seq#       CkSum Flag
100.64.1.1      0.0.0.10         302  80000001 feaa  0002

                Summary Link States (Area 0.0.0.5 [Stub])

Link ID         ADV Router      Age  Seq#       CkSum Flag Route
0.0.0.0         0.0.0.10         320  80000005 df70  0002 0.0.0.0/0
10.1.0.0        0.0.0.10         315  80000002 fc54  0002 10.1.0.0/24
10.1.10.0       0.0.0.10         311  80000001 9aac  0002 10.1.10.0/24
```

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What are the three most likely reasons for this behavior? (Choose three answers)

A. The web filter cache has been cleared causing all websites to take longer to be rated.

B. The SSL/TLS deep inspection was configured but the browsers do not have the FortiGate certificate installed.

C. The webfilter-force-off setting has been enabled under config system fortiguard.

D. The DNS server is unreachable, preventing URL resolution.

E. The FortiGuard Web Filtering license has expired, causing FortiGate to apply the default block action.

Answer: B,D,E

Explanation:

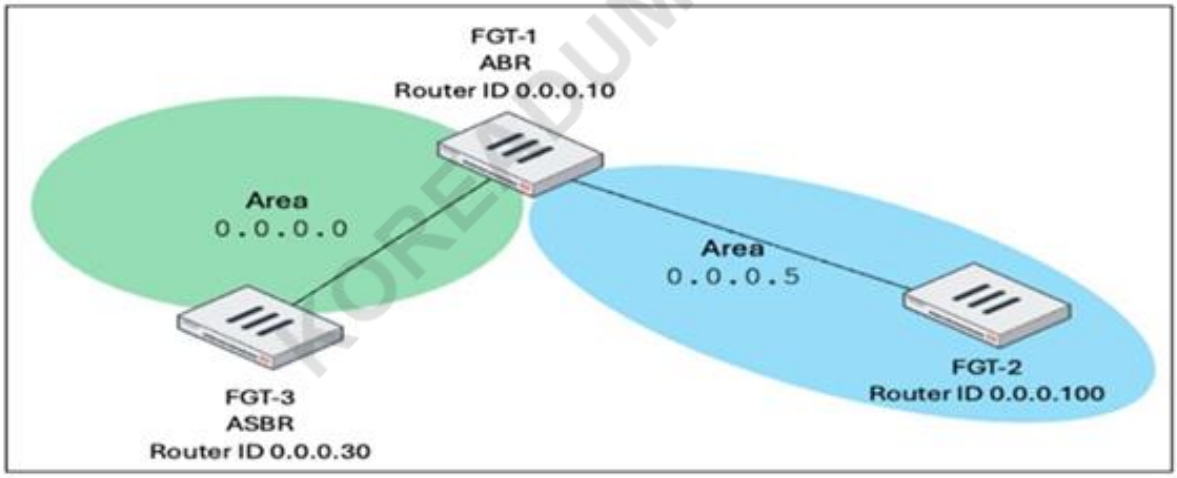
The reported symptom-users unable to access any websites despite no explicit blocks in the profile-points to systemic connectivity or configuration issues rather than specific URL filtering rules.

Option B (SSL/TLS Inspection): When Deep Inspection is enabled, the FortiGate acts as a Man-in-the-Middle (MitM) and re-signs server certificates using its own CA. If the clients (browsers) do not trust this CA (i.e., the certificate is not installed in their Trusted Root store),

they will reject the connection with certificate errors, effectively preventing access to all HTTPS websites.
Option D (DNS): Web browsing relies on DNS resolution. If the configured DNS server is unreachable or failing, the FortiGate (or the client) cannot resolve FQDNs to IP addresses. Consequently, browsers will fail to load any page, resulting in a total loss of web access.
Option E (License): If the FortiGuard Web Filtering license expires, the FortiGate can no longer query the FortiGuard Distribution Network (FDN) for ratings. By default, or if the allow-when-rating-error setting is disabled (a common security practice), the FortiGate will block all web traffic that it cannot rate, often displaying a "Web Filter Service Error" or invalid license page.
Option A is incorrect because clearing the cache only increases latency, it does not block traffic. Option C is incorrect because webfilter-force-off is typically used to disable the service (often allowing traffic to bypass checks if the service is down), rather than blocking it.

QUESTION NO: 131
Refer to the exhibits.

Network topology



OSPF database

```
FGT-2 # get router info ospf database brief

            OSPF Router with ID (0.0.0.100) (Process ID 0, VRF 0)

                Router Link States (Area 0.0.0.5 [Stub])

Link ID         ADV Router      Age  Seq#      CkSum Flag Link count
0.0.0.10         0.0.0.10        302  80000008 13ea  0002 1
0.0.0.100        0.0.0.100       295  8000000b 8761  0021 1

                Net Link States (Area 0.0.0.5 [Stub])

Link ID         ADV Router      Age  Seq#      CkSum Flag
100.64.1.1      0.0.0.10         302  80000001 feaa  0002

                Summary Link States (Area 0.0.0.5 [Stub])

Link ID         ADV Router      Age  Seq#      CkSum Flag Route
0.0.0.0         0.0.0.10         320  80000005 df70  0002 0.0.0.0/0
10.1.0.0        0.0.0.10         315  80000002 fc54  0002 10.1.0.0/24
10.1.10.0       0.0.0.10         311  80000001 9aac  0002 10.1.10.0/24
```

FGT-1 is an area border router (ABR) that has interfaces in OSPF areas 0.0.0.0 and 0.0.0.5.

FGT-3 acts as an autonomous system border router (ASBR), importing static routes into OSPF. FGT-2 is an internal router with all its interfaces belonging to area 0.0.0.5. FGT-1 is receiving all advertised routes from FGT-2, however, FGT-3 is not receiving any of the advertised routes from FGT-1. What is the most likely reason for this? (Choose one answer)

A. Area 0.0.0.5 is configured not to propagate type 5 LSAs.

B. FGT-2 is configured with a distribution list to block all advertised routes from FGT-3.

C. FGT-3 and FGT-2 have not formed an OSPF adjacency yet.

D. IP protocol 89 is blocked between FGT-1 and FGT-3.

Answer: A

Explanation:

The get router info ospf database brief output on FGT-2 clearly indicates that Area 0.0.0.5 is configured as a [Stub] area.

In OSPF, a Stub Area is specifically designed to reduce the size of the Link State Database (LSDB) on internal routers. The primary behavior of a Stub area is that it does not accept Type 5 (AS External) LSAs.

FGT-3 is the ASBR (Autonomous System Border Router) and is importing static routes, which are generated as Type 5 LSAs in the OSPF domain.

FGT-1 acts as the ABR (Area Border Router). Because Area 0.0.0.5 is a Stub area, FGT-1 blocks these Type 5 LSAs from entering Area 0.0.0.5.

Consequently, FGT-2 will not receive the specific external routes advertised by FGT-3. Instead, the ABR (FGT-1) injects a default route (0.0.0.0/0) into the Stub area to allow connectivity to the external world, which is visible in the database output.

While the question text mentions FGT-3 not receiving routes, the definitive configuration shown in the exhibit is the Stub area setting, which directly corresponds to the blocking of Type 5 LSA propagation (Option A).

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QUESTION NO: 132

Refer to the exhibits,

Exhibit 1

```
config system global
    set snat-route-change enable
end
_____________________
config router static
    edit 1
        set gateway 10.200.1.254
        set priority 5
        set device "port1"
    next
    edit 2
        set gateway 10.200.2.254
        set priority 10
        set device "port2"
    next
end
```

Exhibit 2

```
FGT # diagnose sys session list
session info: proto=6 proto_state=01 duration=600 expire=3179 timeout=3600 flags=00000000
sockflag=00000000 sockport= av_idx=0 use=4
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=0 policy_dir=0 tunnel=/ vlan cos=0/255
state=log may_dirty npu f00
statistic (bytes/packets/allow_err): org=3208/25/1 reply=11144/29/1 tuples=2
tx speed (Bps/kbps): 0/0 rx speed (Bps/kbps): 0/0
orgin->sink: org pre->post, reply pre->post dev=4->2/2->4 gwy=10.200.1.254/10.0.1.10
hook=post dir=org act=snat 10.0.1.10:64907->54.239.158.170:80(10.200.1.1:64907)
hook=pre dir=reply act=dnat 54.239.158.170:80->10.200.1.1:64907(10.0.1.10:64907)
pos/ (before, after) 0/(0,0), 0/(0,0)
src_mac=b4:f7:a1:e9:91:97
misc=0 policy_id=1 auth_info=0 chk_client_info=0 vd=0
serial=0031c5b tos=ff/ff app_list=0 app=0 url_cat=0
rpdb_link_id = 00000000
dd_type=0 dd_mode=0
npu_state=0x000c00
npu info: flag=0x00/0x00, offload=0/0, ips_offload=0/0, epid=0/0, ipid=0/0, vlan=0x0000/0x0000
vlifid=0/0, vtag_in=0x0000/0x0000 in_npu=0/0, out_npu=0/0, fwd_en=0/0, qid=0/0
no_ofld_reason:
```

which show the configuration on FortiGate and partial session information for internet traffic from a user on the internal network. If the priority on route ID 2 were changed from 10 to 0, what would happen to traffic matching that user session? (Choose one answer)

A. The session would be deleted, and the client would need to start a new session.

B. The session would remain in the session table, but its traffic would now egress from both port1 and port2.

C. The session would remain in the session table, and its traffic would egress from port2.

D. The session would remain in the session table, and its traffic would egress from port1.

Answer: A

Explanation:

The correct answer is A. This behavior is dictated by the configuration command set snat-route-change enable shown in Exhibit 1 under config system global.

Routing Change: By changing the priority of route ID 2 from 10 to 0, it becomes lower than route ID 1 (priority 5). In FortiOS, a lower priority value indicates a more preferred route. Consequently, the active route for the destination changes from port1 to port2.

SNAT Implication: The existing session (shown in Exhibit 2) is using Source NAT (SNAT) with the IP address associated with port1 (10.200.1.1). If the traffic were simply switched to port2, the source IP would be incorrect for that interface and the return traffic would likely fail or be dropped.

snat-route-change enable: This specific setting instructs the FortiGate on how to handle established SNAT sessions when a routing change occurs that alters the preferred outgoing interface. When enabled, if a route change forces an SNAT session to a new interface, FortiGate flushes (deletes) the session from the session table. This is necessary because a live TCP session cannot survive a change in its source IP address. The client must initiate a new session, which will then be created using the new correct route (port2) and the corresponding new SNAT IP.
If this setting were disabled, the session would likely remain "sticky" to the original interface (port1) until it closed, provided the route still existed. However, the explicit configuration forces the deletion.

QUESTION NO: 133
Consider the scenario where the server name indication (SNI) does not match either the common name (CN) or any of the subject alternative names (SAN) in the server certificate. Which two actions will FortiGate take when using the default settings for SSL certificate inspection? (Choose two answers)
A. FortiGate uses the SNI from the user's web browser.
B. FortiGate does not decrypt the traffic if the traffic is blocked by the web filter profile.
C. FortiGate uses the CN information from the Subject field in the server certificate.
D. FortiGate does not decrypt the traffic if the traffic is allowed by the web filter profile.
Answer: C,D
Explanation:
The correct answers are C and D.
The study guide states: "SSL certificate inspection relies on extracting the FQDN of the URL from either: TLS extension server name indication (SNI), SSL certificate common name (CN)." It also says: "When using SSL certificate inspection, FortiGate is not decrypting the traffic. It is only inspecting the server digital certificates and the SNI field, which are interchanged before the encryption." This proves the second part of the answer:
under SSL certificate inspection, FortiGate does not decrypt the traffic therefore, if the traffic is allowed, it still passes without decryption That makes D correct.
For the SNI mismatch behavior, the FortiOS administration guide describes the default Server certificate SNI check behavior as:
"Enable: If it is mismatched use the CN in the server certificate for URL" So if the SNI does not match the CN or any SAN, FortiGate falls back to using the CN from the Subject field for URL handling under the default setting. That makes C correct.
Why the other options are wrong:
A is wrong because with the default SNI-check behavior, when the SNI mismatches the certificate identity, FortiGate does not continue using the mismatched SNI. Instead, it uses the CN in the server certificate for the URL.
B is not the best answer in this single pair selection. While certificate inspection does not decrypt traffic, the key default behavior the documents explicitly highlight for this mismatch case is:
use the CN when SNI mismatches, and
certificate inspection does not decrypt allowed HTTPS traffic.
So the verified answers are: C, D.

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